

第一章 半导体器件

1. 半导体基础知识

纯净：本征半导体

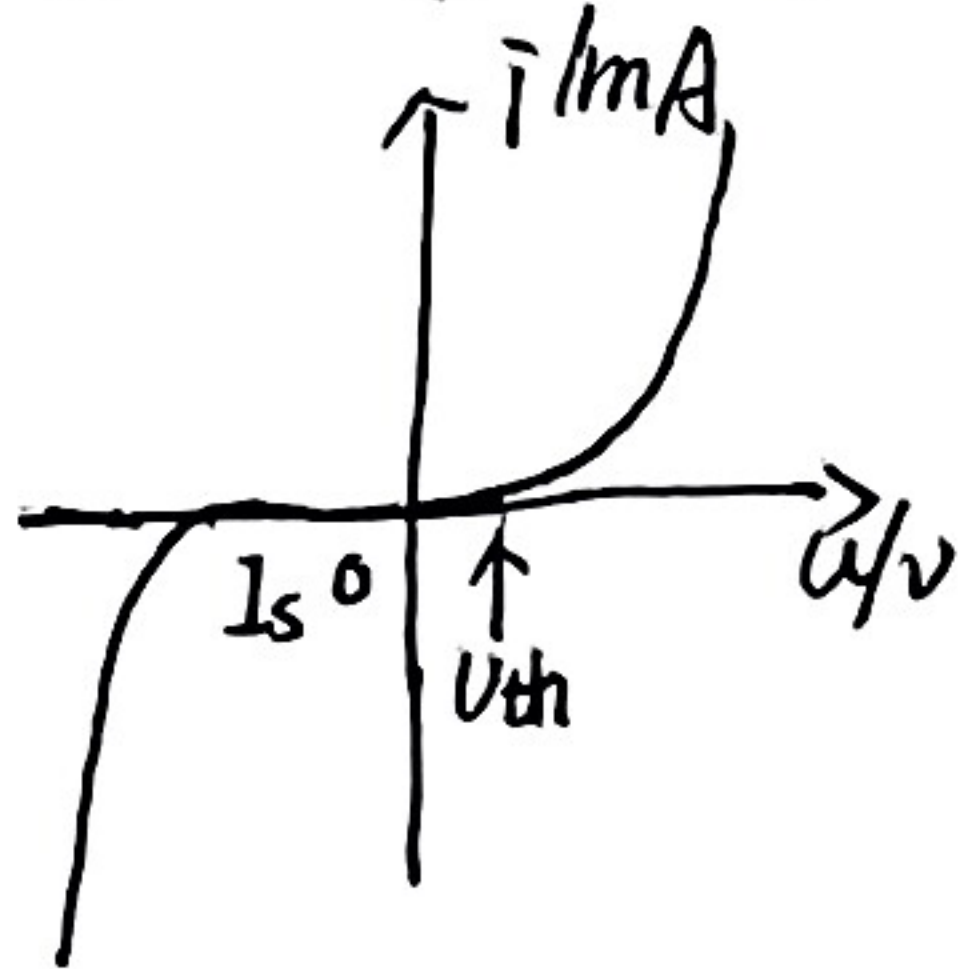
掺入杂质：掺杂半导体

三个特性 { 光敏
热敏
掺杂

运动 { 扩散运动：浓度差
漂移运动：电场

2. PN结与二极管

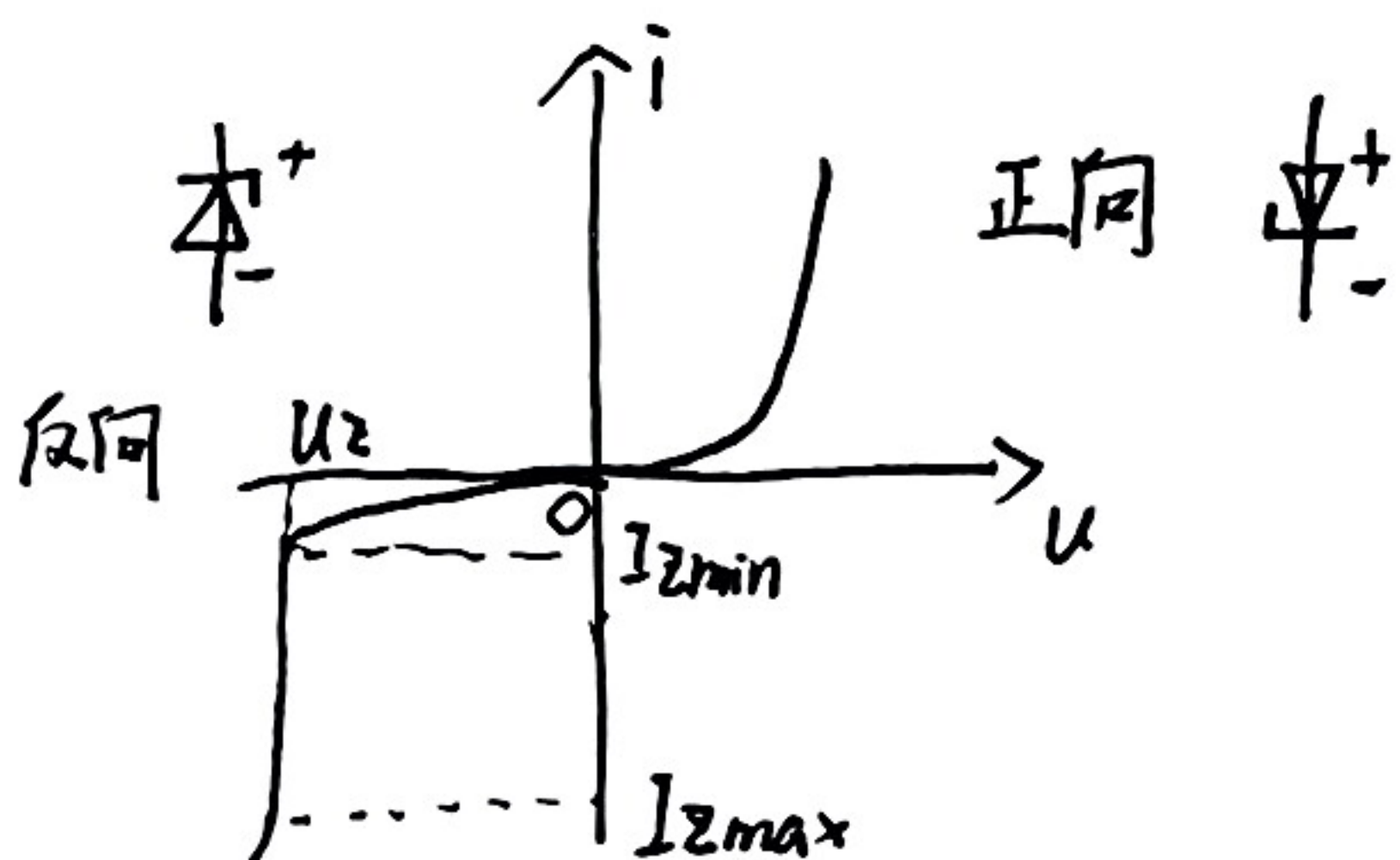
二极管伏安特性曲线：



U_{th} ：导通电压，死区电压 { 硅：0.7V
锗：0.2V } 直流电阻 $R_D = \frac{U_a}{I_a}$
 I_S ：反向饱和电流 交流电阻 $r_d = \frac{du}{di} = \frac{26mV}{I_a}$
 $I_S(T_2) = I_S(T_1) \cdot 2^{\frac{T_2-T_1}{T_0}}$
表达式： $i = I_S (e^{\frac{u}{U_T}} - 1)$ $U_T = 26mV$

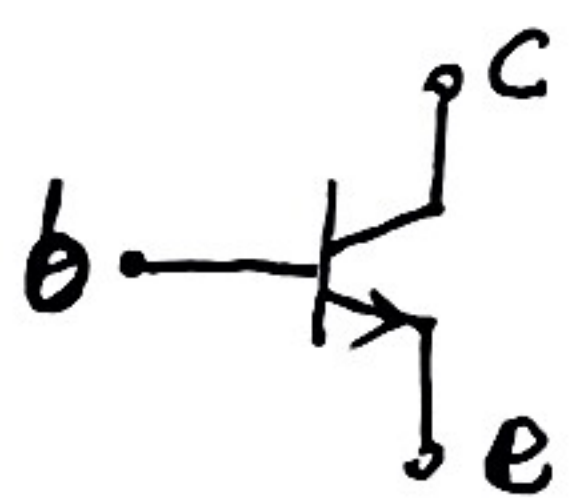
应用 { 整流 给图画波形
门电路 求 U_b ，压差大的先导通
限幅 注意：二极管方向，反向都不通

3. 特殊二极管



电压达到 U_Z 时有稳压效果 (反向电压 $< U_Z$ 无稳压作用)
电流要在 $I_{Zmin} < I < I_{Zmax}$ 之间
正常工作电压为 U_Z

4. 三极管



e (发射区) 多子浓度高，b (基区) 浓度低且薄，c (集电区) 面积大

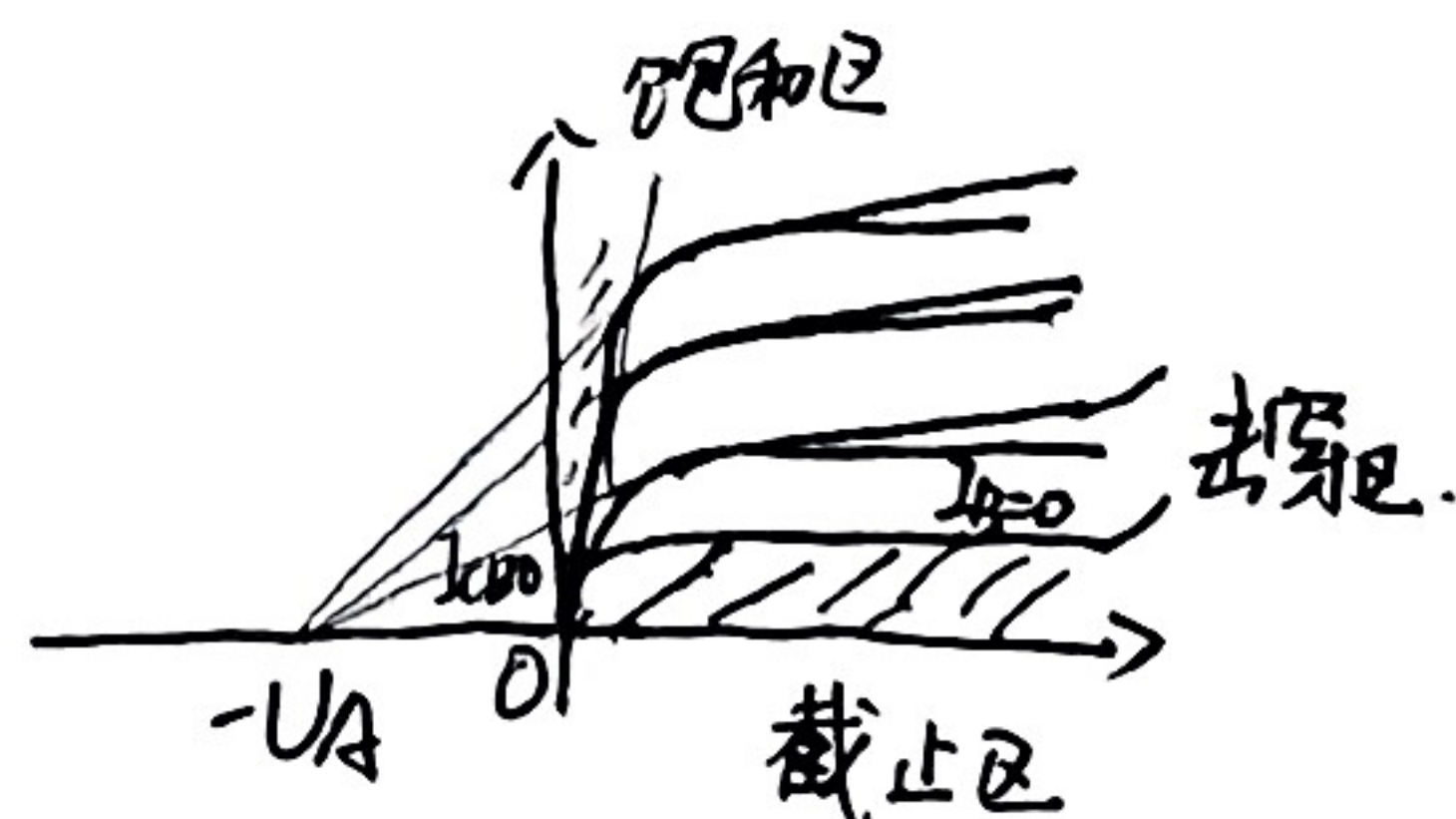
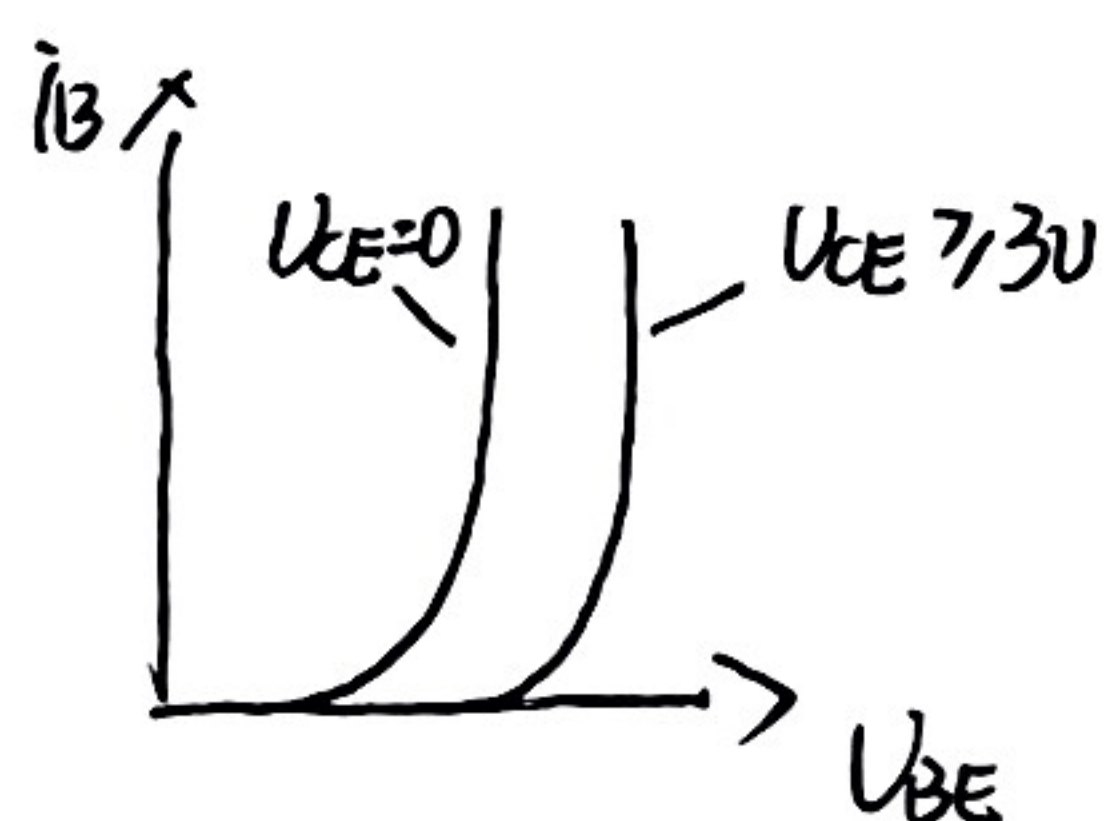
放大区：发射结正偏，集电结反偏。 电位沿箭头方向减小 $\phi_c > \phi_b > \phi_e$

NPN 扩散 复合 漂移
 $I_E = I_B + I_C$
↓
很小

$$\beta = \frac{I_{BC}}{I_B} \quad \alpha = \frac{I_C}{I_E}$$

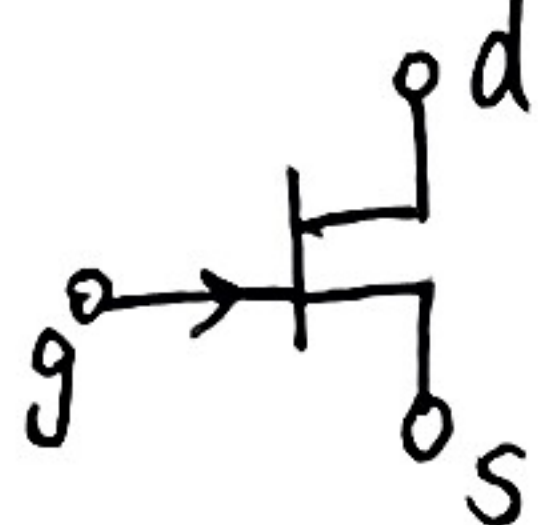
输入特性曲线 $I_B = f(U_{BE}, U_{CE}) | U_{CE} \text{ 为常数}$

输出特性曲线 $I_C = f(I_B, U_{CE}) | I_B \text{ 为常数}$



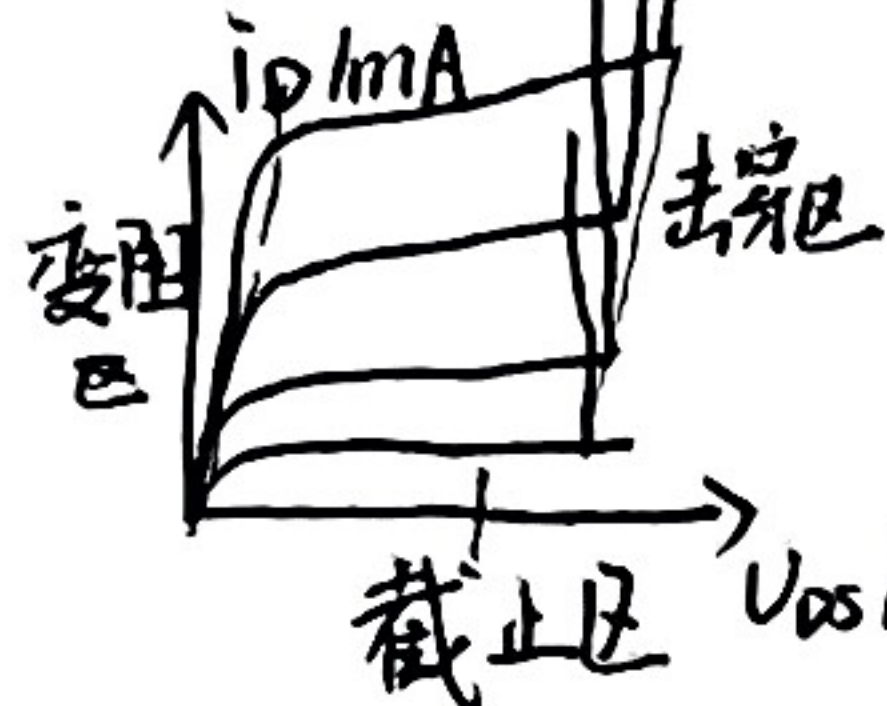
5. 场效应管

结型场效应管



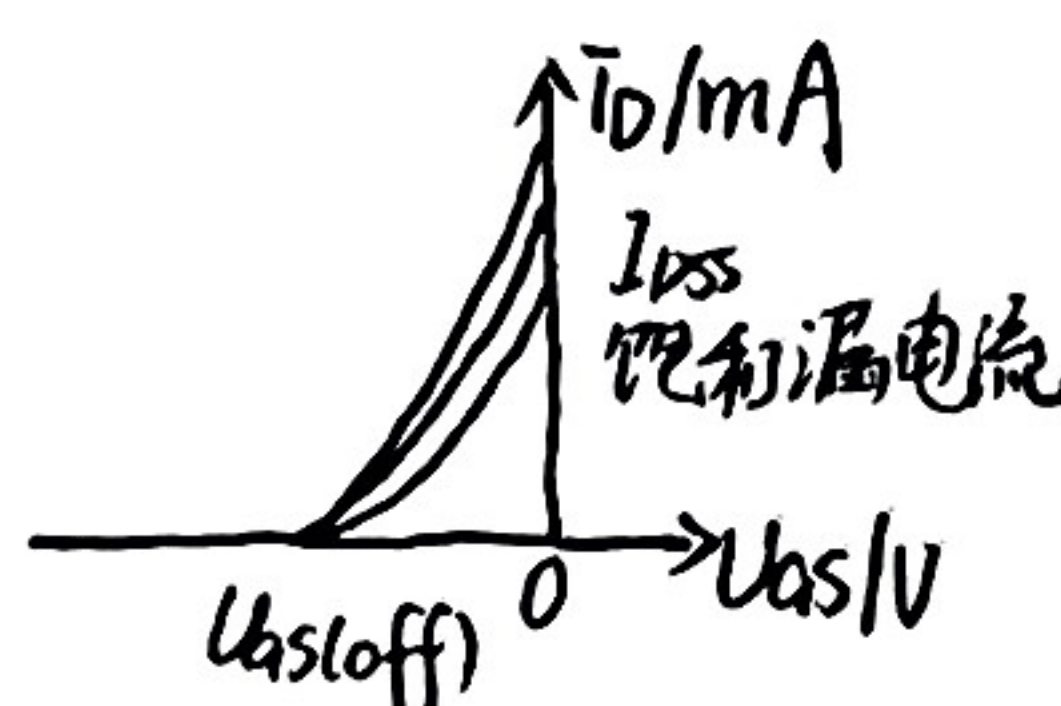
预夹断: U_{a0}
夹断: $U_{as(off)}$

输出特性曲线



可变电阻区:
预夹断之前 $U_{DS} < U_{GS} - U_{as(off)}$
放大区:
预夹断之后 $U_{DS} > U_{GS} - U_{as(off)}$

转移特性曲线



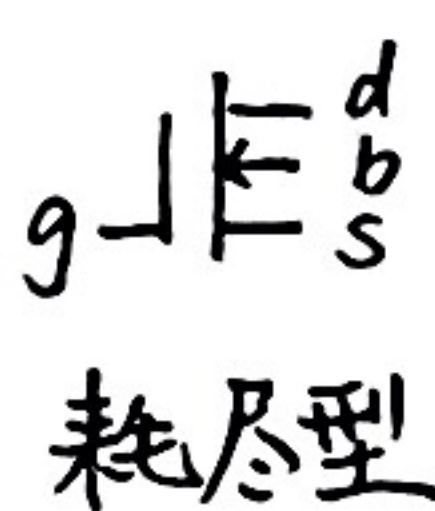
$$i_D = I_{DSS} \left(1 - \frac{U_{GS}}{U_{GS(off)}}\right)^2$$

$i_D - U_{GS}$ 的导数——跨导

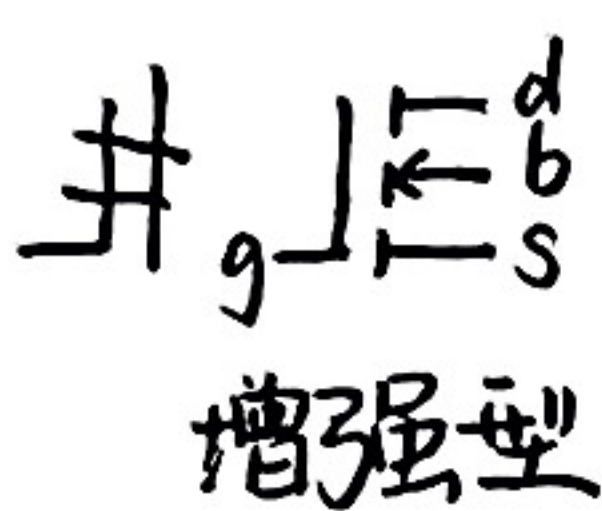
$$g_m = \frac{\partial i_D}{\partial U_{GS}} = -\frac{2I_{DSS}}{U_{GS(off)}} \left(1 - \frac{U_{GS}}{U_{GS(off)}}\right)$$

$$i_D = g_m U_{GS}$$

绝缘栅型场效应管

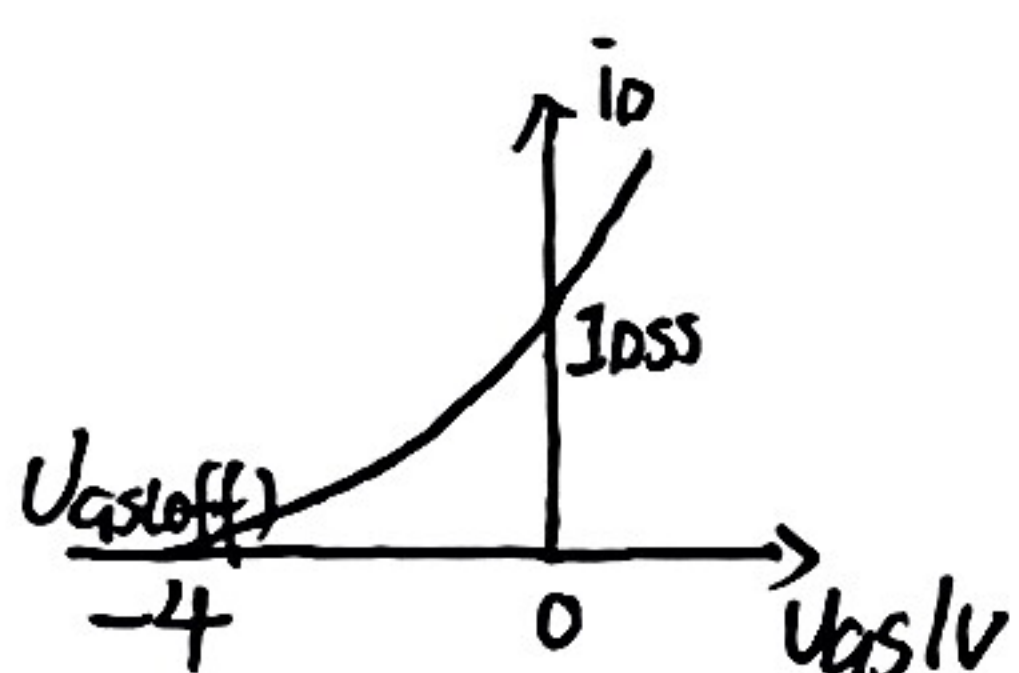


耗尽型

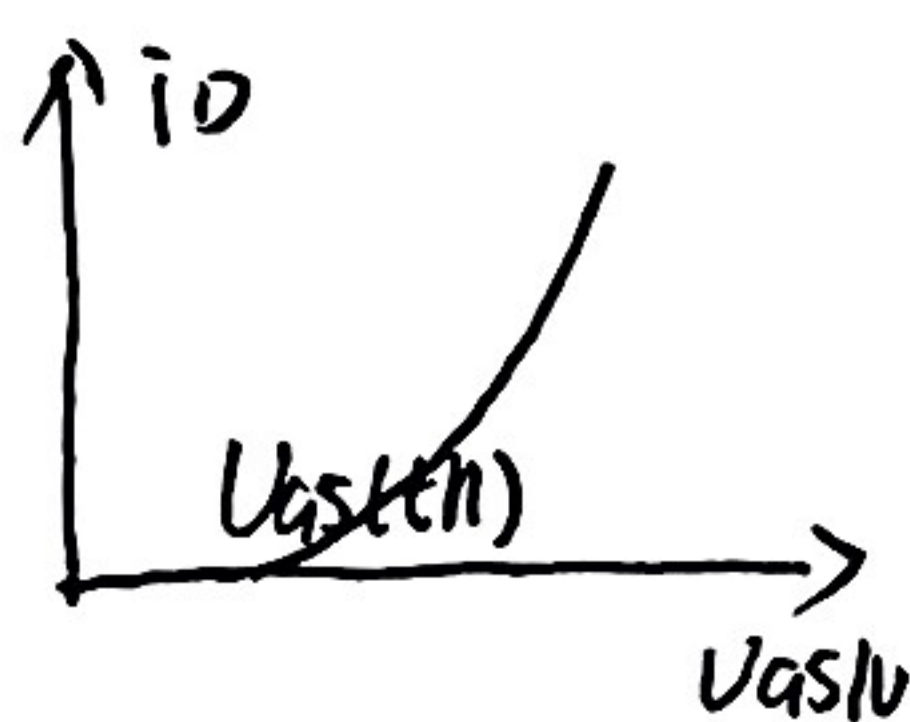


增强型

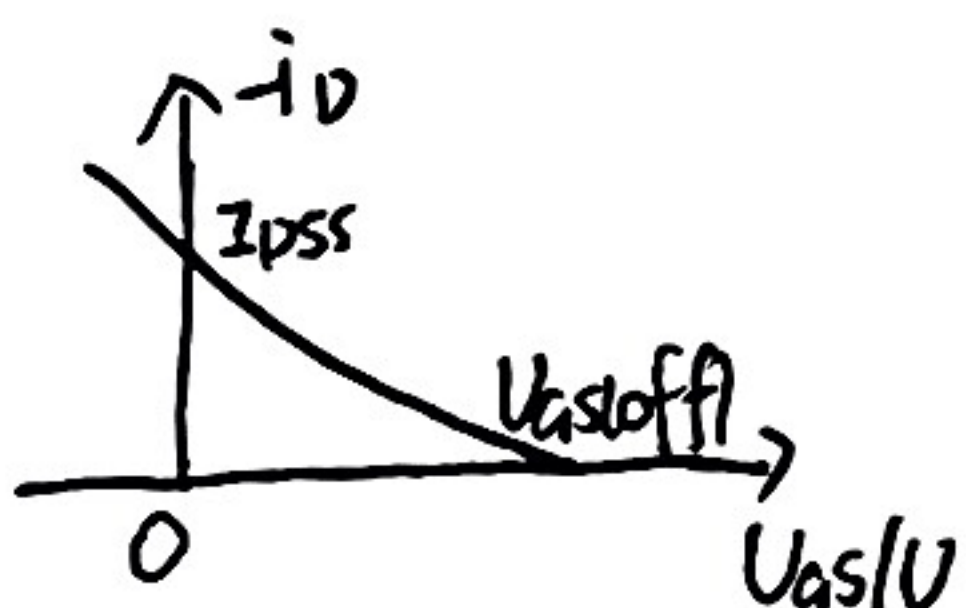
耗尽型直线, 增强型虚线.
一般源极和衬底相连, 对外仍三个口



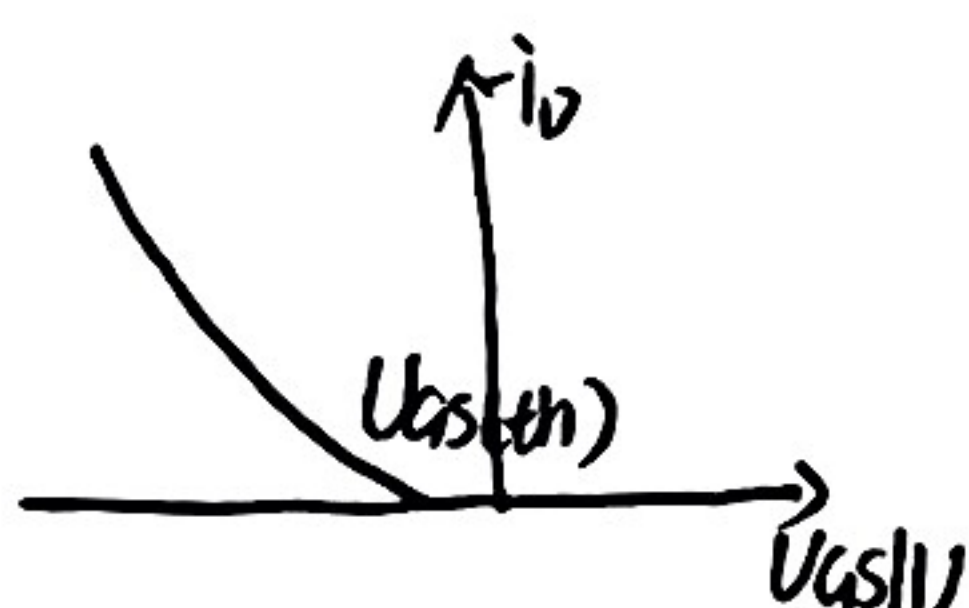
N 耗尽



N 增强



P 耗尽



P 增强

耗尽型与坐标轴有两个交点, 增强型只有一个
耗尽型随栅极电压的减小, 预夹断→夹断

夹断 $U_{GS(off)}$

增强型随栅极电压的增大产生沟道

开启电压: $U_{GS(th)}$

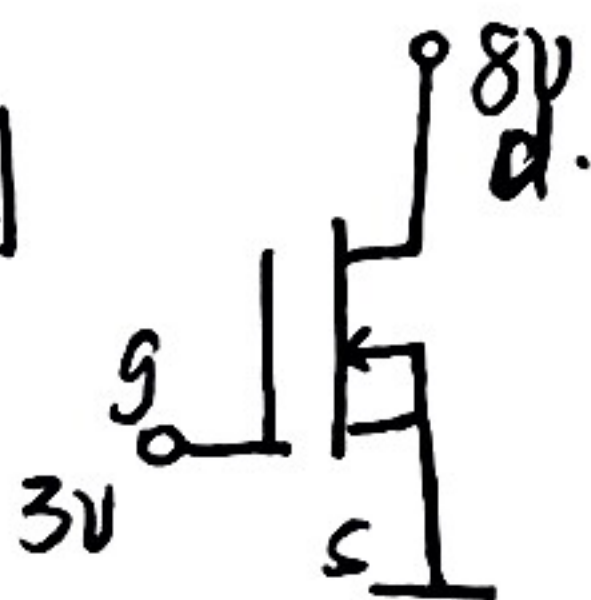
N 沟道单增 P 沟道单减

判断场效应管的工作状态

①. 是否导通: N型 $U_{GS} > U_{GS(off)} (U_{GS(th)})$ P型 $U_{GS} < U_{GS(off)} (U_{GS(th)})$

②. 是否在放大区 N型 $U_{DS} > U_{GS} - U_{GS(off)} (U_{GS(th)})$ P型 $U_{DS} < U_{GS} - U_{GS(off)} (U_{GS(th)})$

例: 已知 $|U_{GS(off)}| = 2V$



①. 是否导通 $U_{GS} = 3 > U_{GS(off)}$ 导通

②. 是否在放大区 $U_{DS} = 8 > U_{GS} - U_{GS(off)}$ 放大区

三极管 $U_{BE} > 0.7 (0.2)$

② 假设放大区 $U_{BE} > 0.7$

第=章 放大器基础 (Part 1)

1. 放大器 概述

u 瞬时值 U : 有效值

u_{be} 交流电压 U_{be} : 电压

U_{BE} : 电压瞬时值

U_{BE} 电压有效值 (直流)

u_{be} : 交流电压瞬时值

U_{be} : 交流电压的有效值

U_{bem} : U_{be} 的峰值 (振幅)

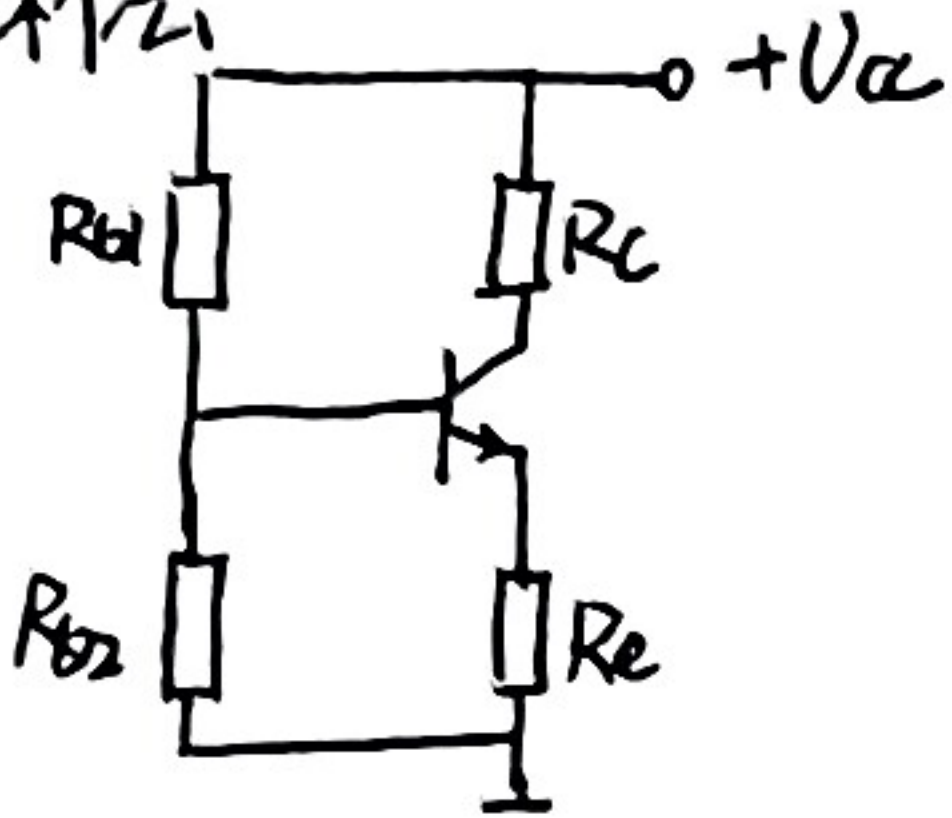
输入为电压源时, 输入电阻越大越好; 输出电阻越小越好.

主要性能指标: R_i , R_o , A_u , $A_{us} = \frac{U_o}{U_s}$ A_i

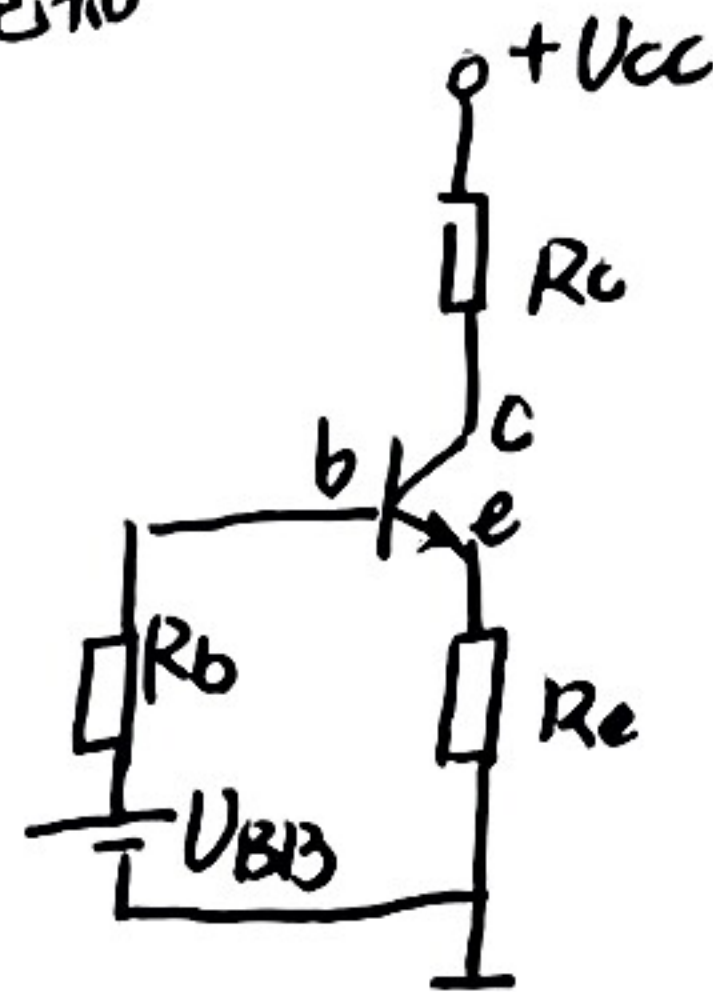
$$A_{us} = \frac{R_i}{R_i + R_s} A_u$$

2. 静态分析: 求静态工作点 (I_{BQ} , U_{BEQ} , I_{CQ} , U_{CEQ}) 直流通路: 电容断路.

解析法



① 等效



$$R_b = R_{b1} \parallel R_{b2}$$

$$U_{BB} = \frac{R_{b2}}{R_{b1} + R_{b2}} U_{cc}$$

② 求 I_{BQ} .

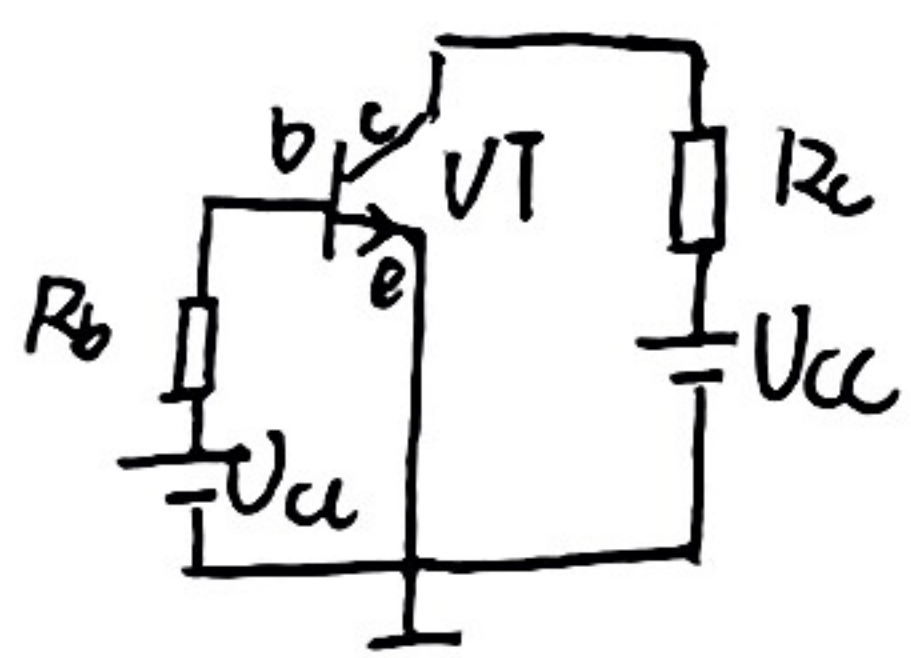
$$U_{BB} - 0.7 = I_{BQ} R_b + (1 + \beta) I_{BQ} R_e$$

③ 求 $I_{CQ} = \beta I_{BQ}$.

④ 求 $U_{CEQ} = U_{cc} - I_{CQ} R_c - R_e I_{EQ}$.

分析三极管的工作状态: 假设三极管工作在放大区. 求静态工作点, ($U_{CEQ} < 0$ 饱和区)

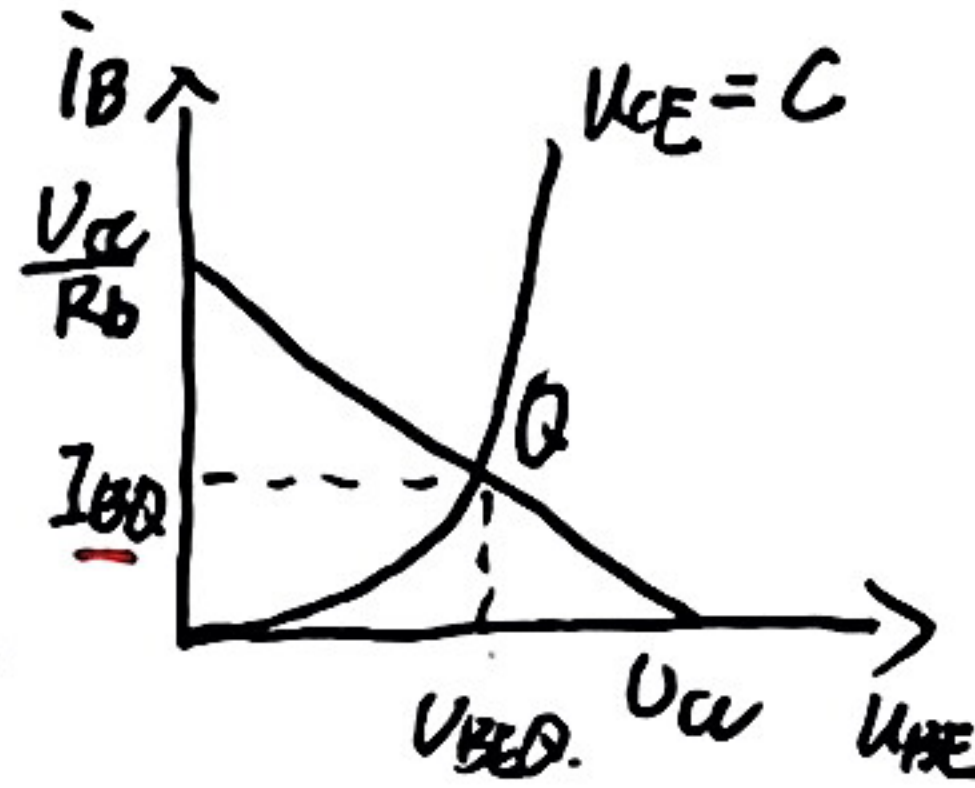
图解法.



输入回路:

$$U_{BE} = U_{cc} - I_{BQ} R_b$$

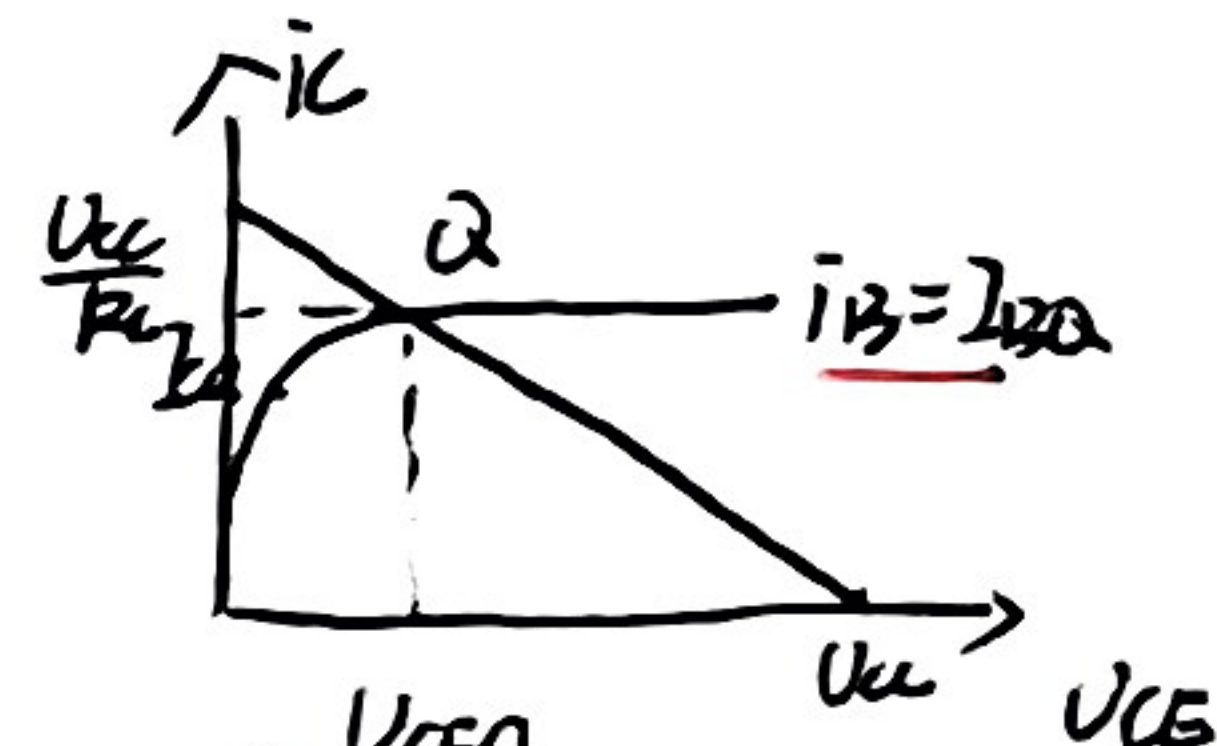
$$I_{BQ} = -\frac{U_{BE}}{R_b} + \frac{U_{cc}}{R_b}$$



输出回路

$$U_{CE} = U_{cc} - I_{CQ} R_c$$

$$I_{CQ} = -\frac{U_{CE}}{R_c} + \frac{U_{cc}}{R_c}$$



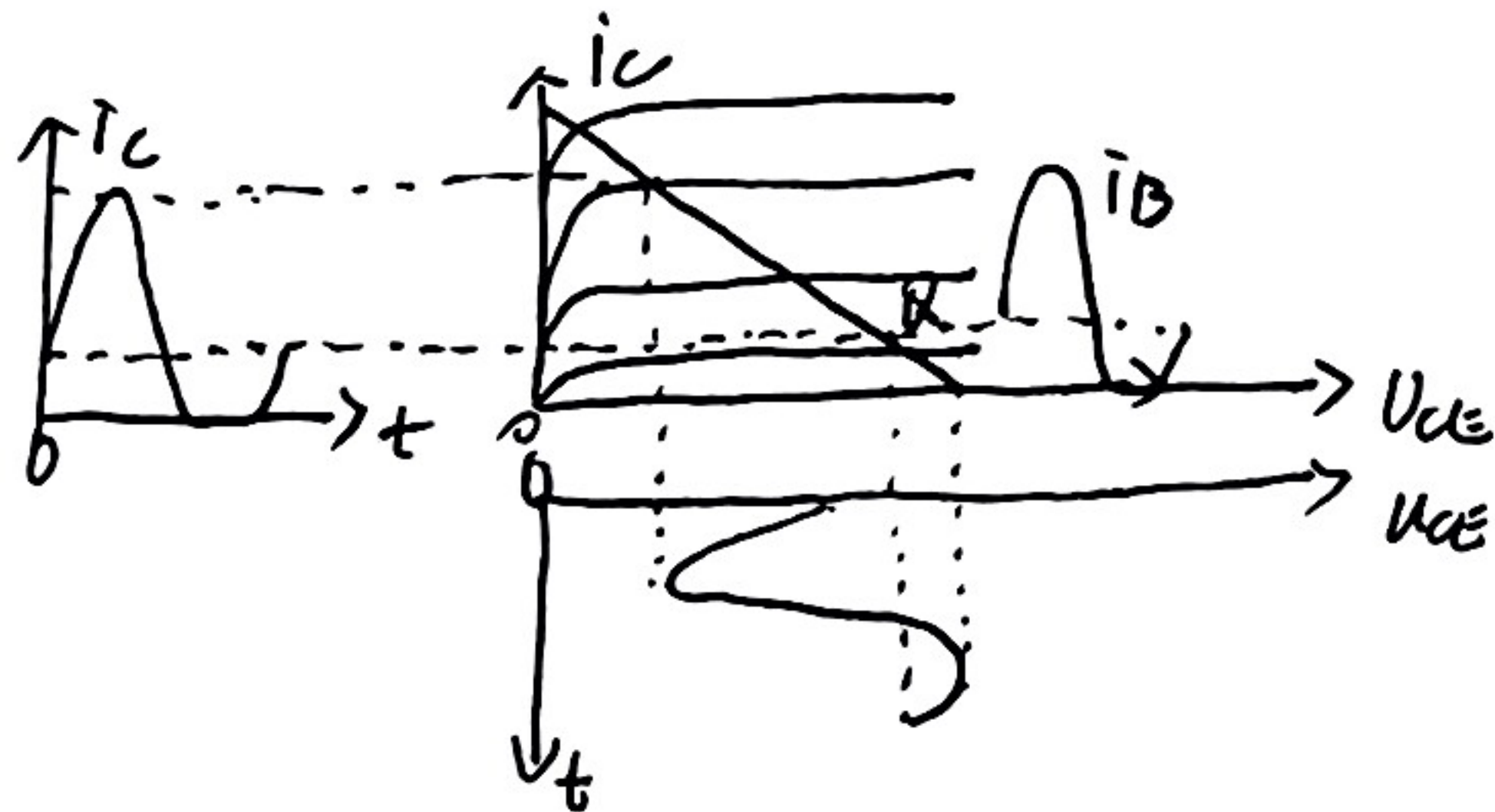
3. 动态分析: 求主要性能指标.

交流通路: 电容短路. 直流源置零 (接地)

调整静态工作点

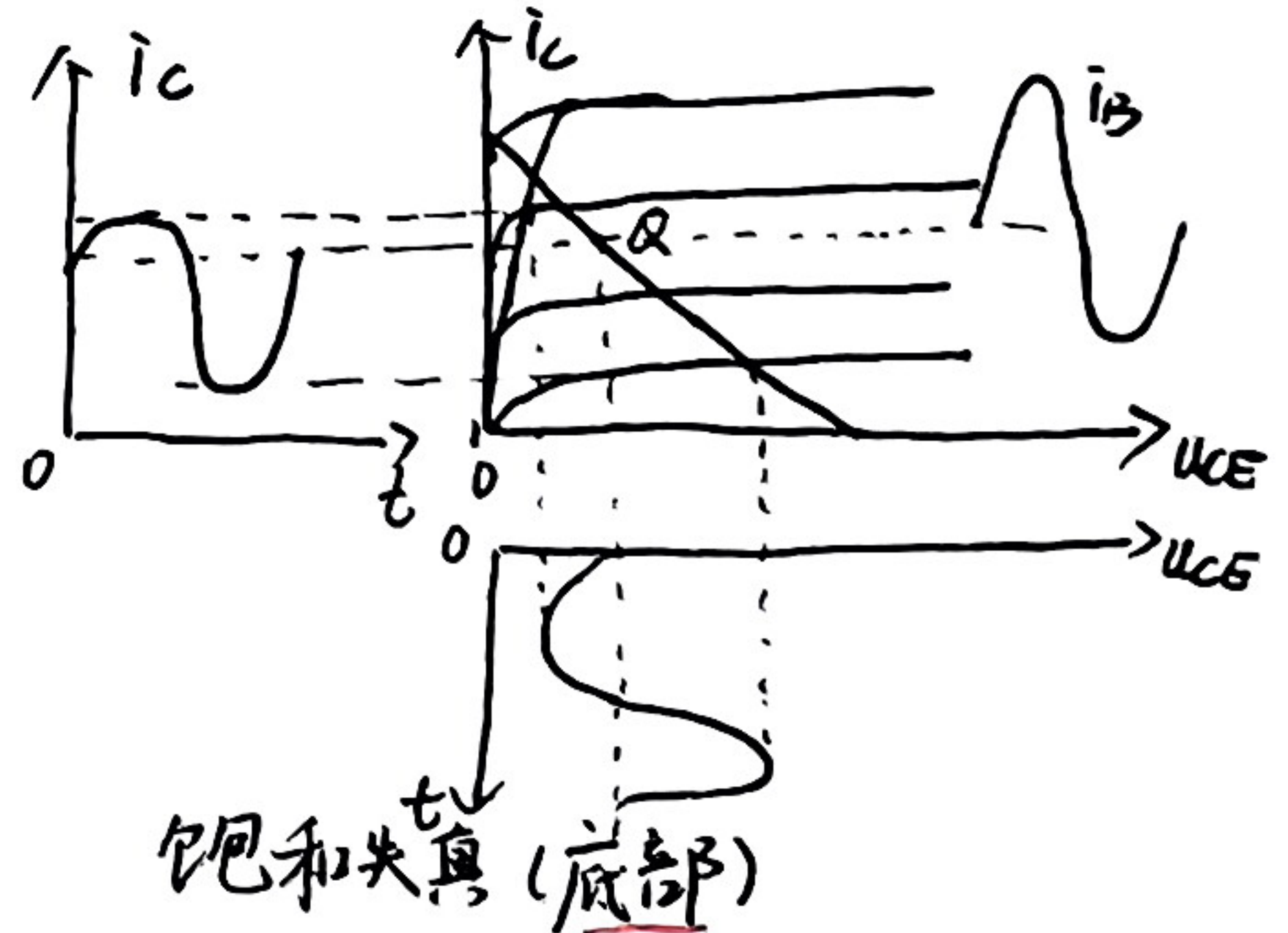
① 减 R_b , $I_B \uparrow$ $Q \uparrow$

图解法: 求最大不失真输出电压



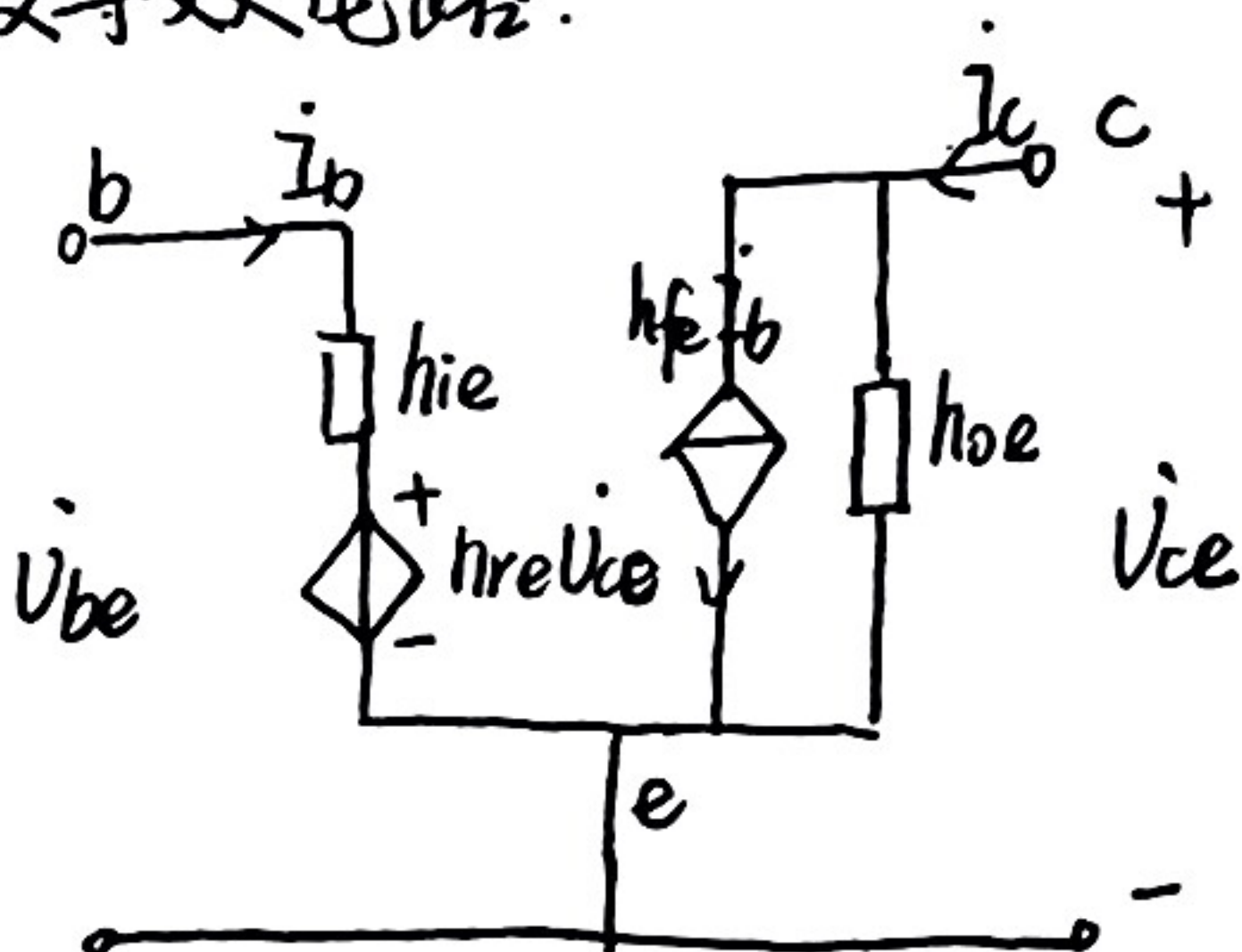
截止失真 (顶部)

对于 U_{CE} 而言.

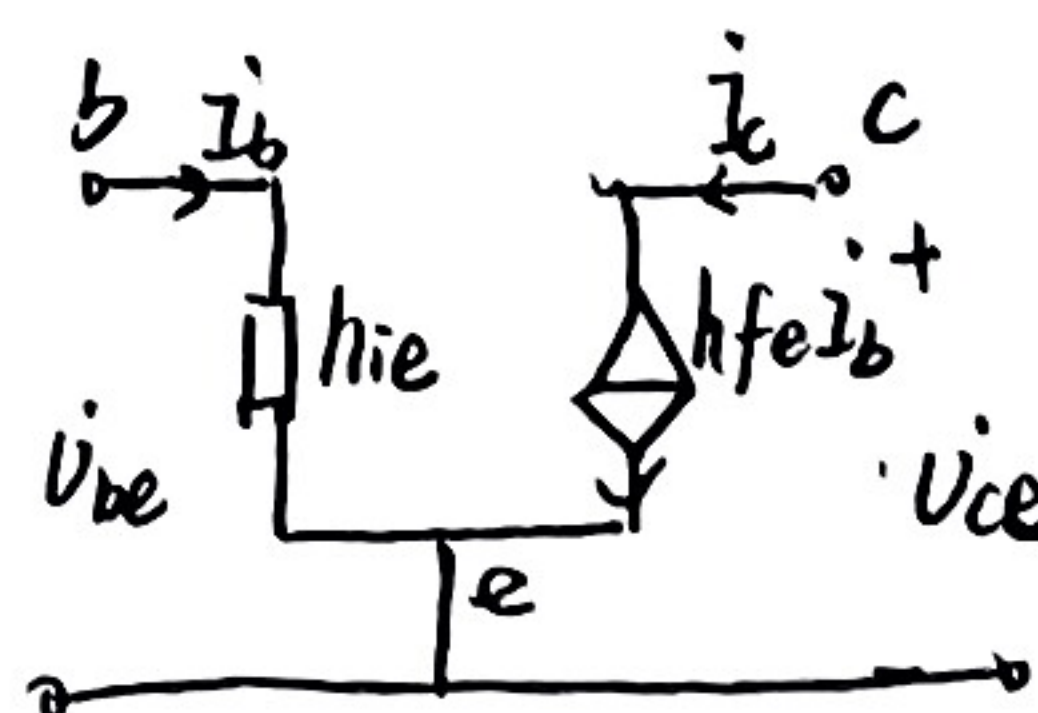


饱和失真 (底部)

微变等效电路.



简化



$$h_{ie} \quad r_{oe} = r_{bb'} + (1 + \beta) r_e$$

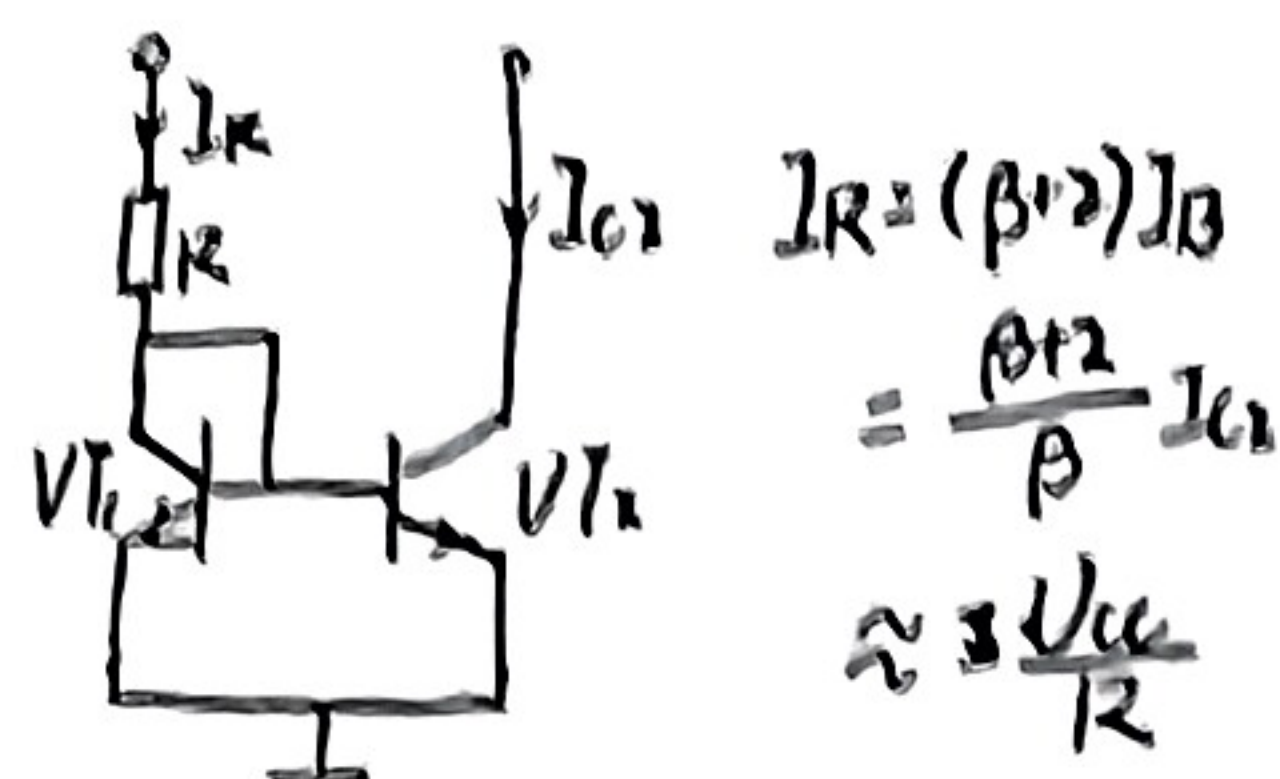
$$= r_{bb'} + (1 + \beta) \frac{26}{I_{EQ}}$$

$$\frac{26}{I_{EQ}} = \beta \frac{26}{I_{CQ}}$$

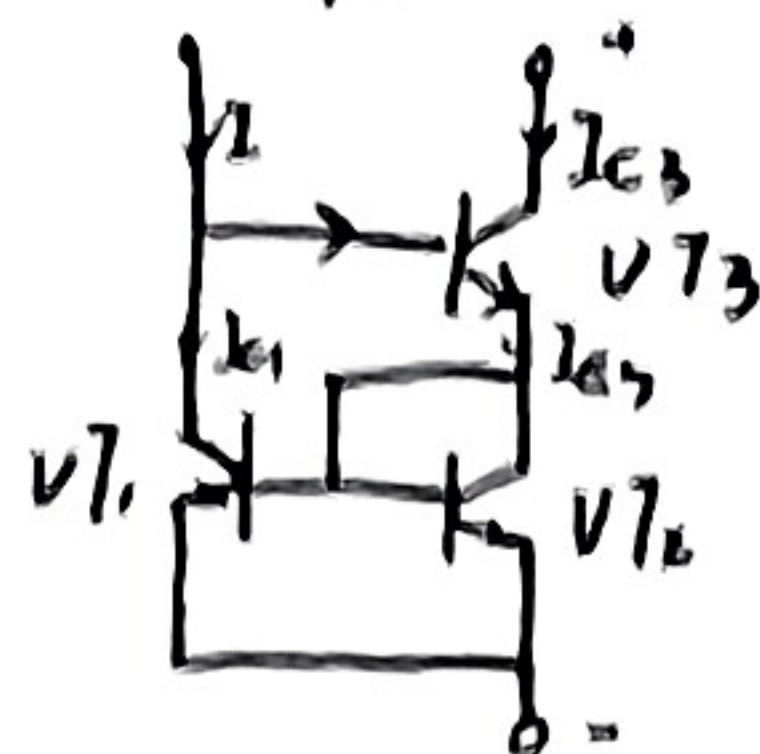
I_b 与 $h_{fe} I_b$ 同流入/流出射级 $h_{fe} = \beta$

4. 电流源偏置电路

基本电流源

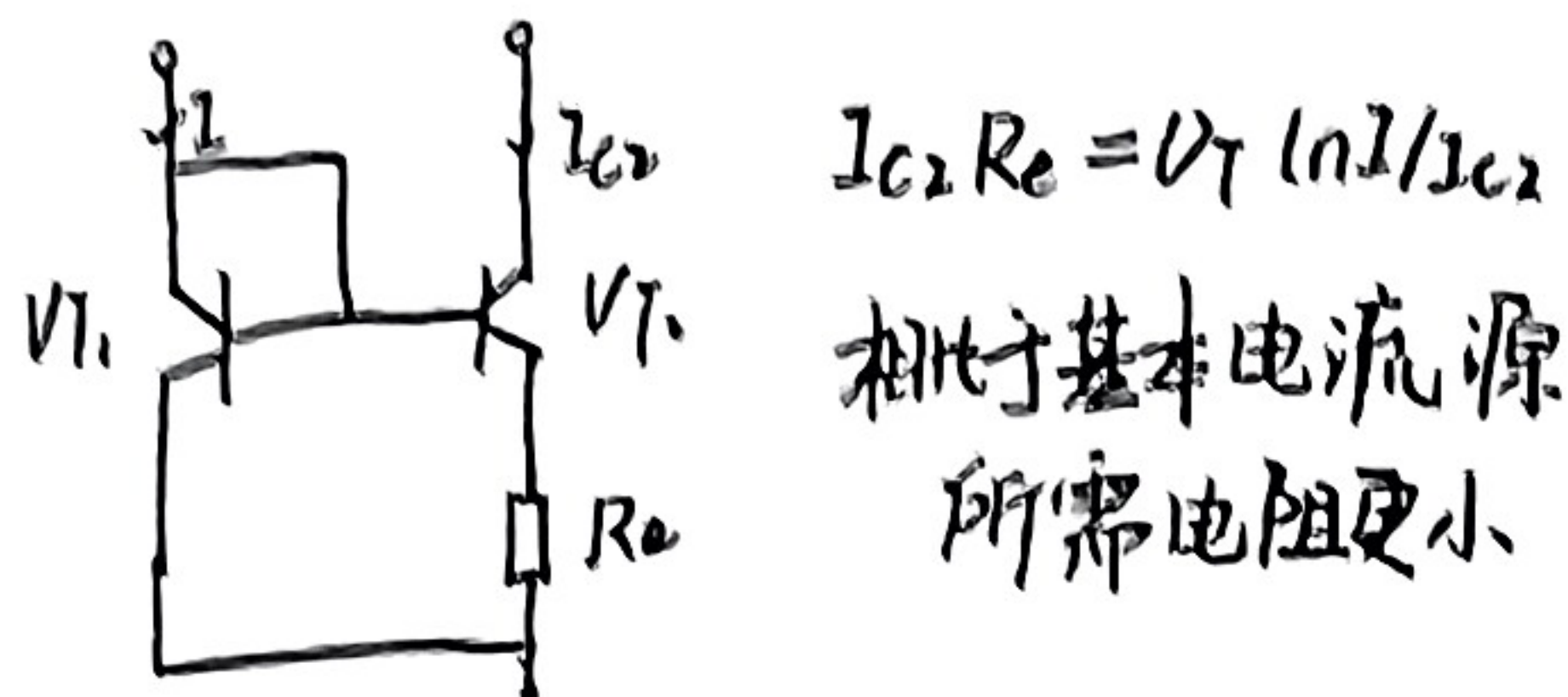


威尔逊电流源

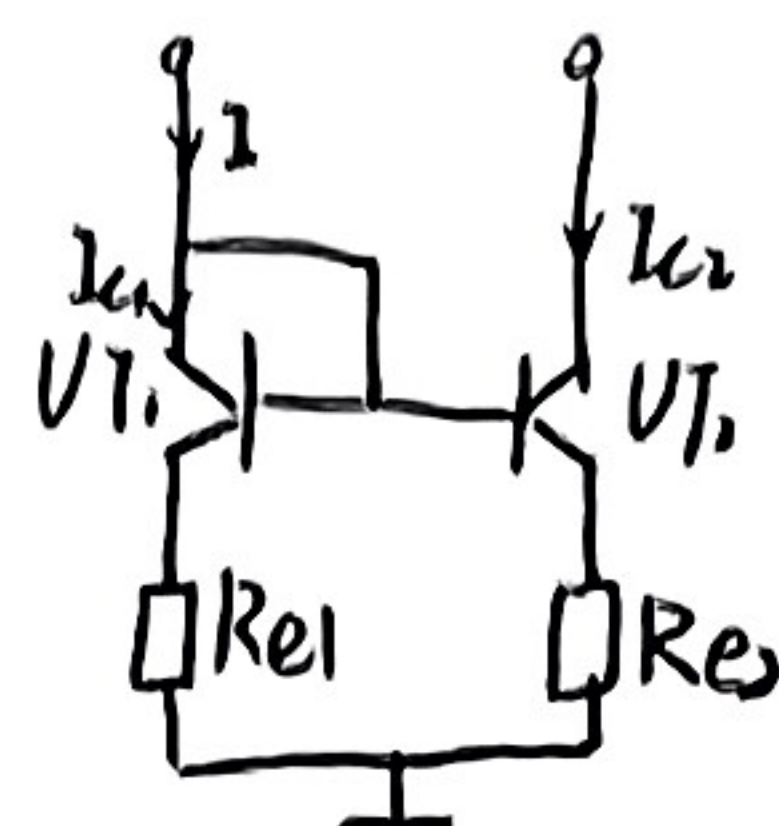


相比于基本电流源
 I_{C3} 更加近似等于 I_{C1}

微电流电流源

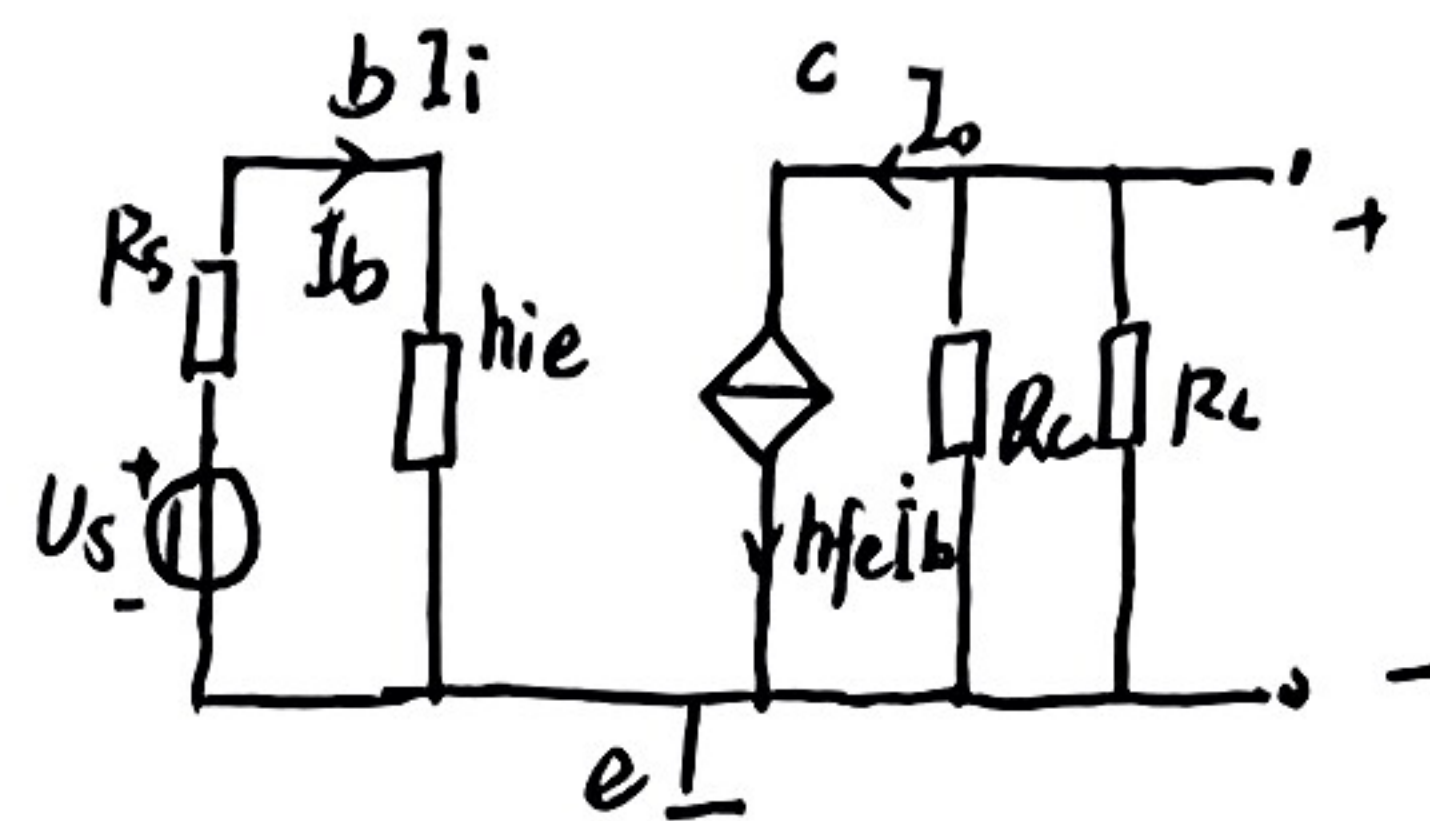
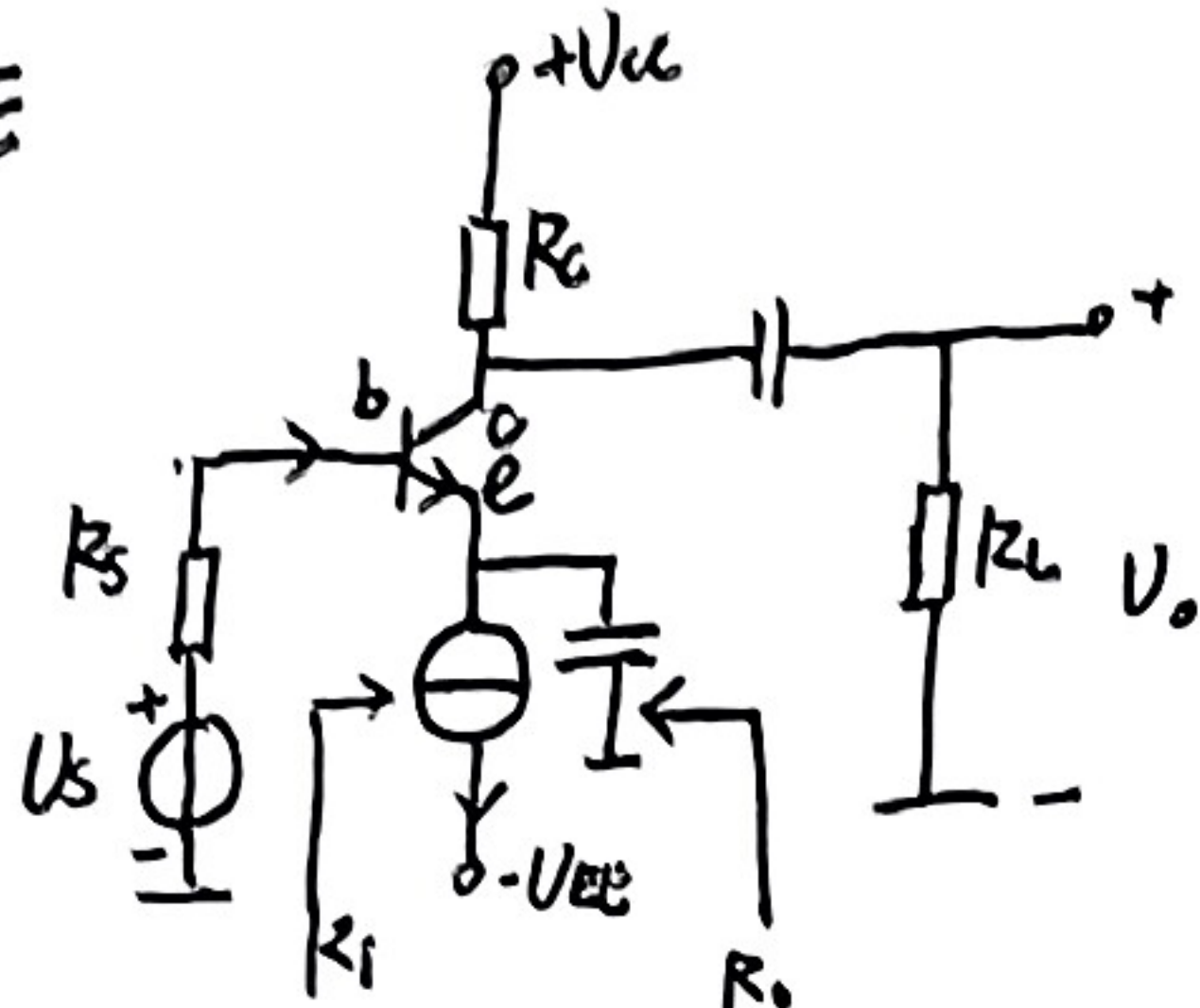


比例电流源



5. 晶体管放大器

CE



$$A_i = \frac{i_o}{i_i} = \frac{h_{fe} i_b}{i_b} = h_{fe}$$

$$R_i = \frac{V_i}{i_i} = \frac{h_{ie} i_b}{i_b} = h_{ie}$$

$$A_{um} = \frac{V_o}{V_i} = \frac{-h_{fe} i_b (R_c || R_L)}{h_{ie} i_b} = -\frac{h_{fe} (R_c || R_L)}{h_{ie}}$$

$$A_{usm} = \frac{V_o}{V_s} = \frac{-h_{fe} i_b (R_c || R_L)}{(h_{ie} + R_s) i_b} = -\frac{h_{fe} (R_c || R_L)}{h_{ie} + R_s}$$

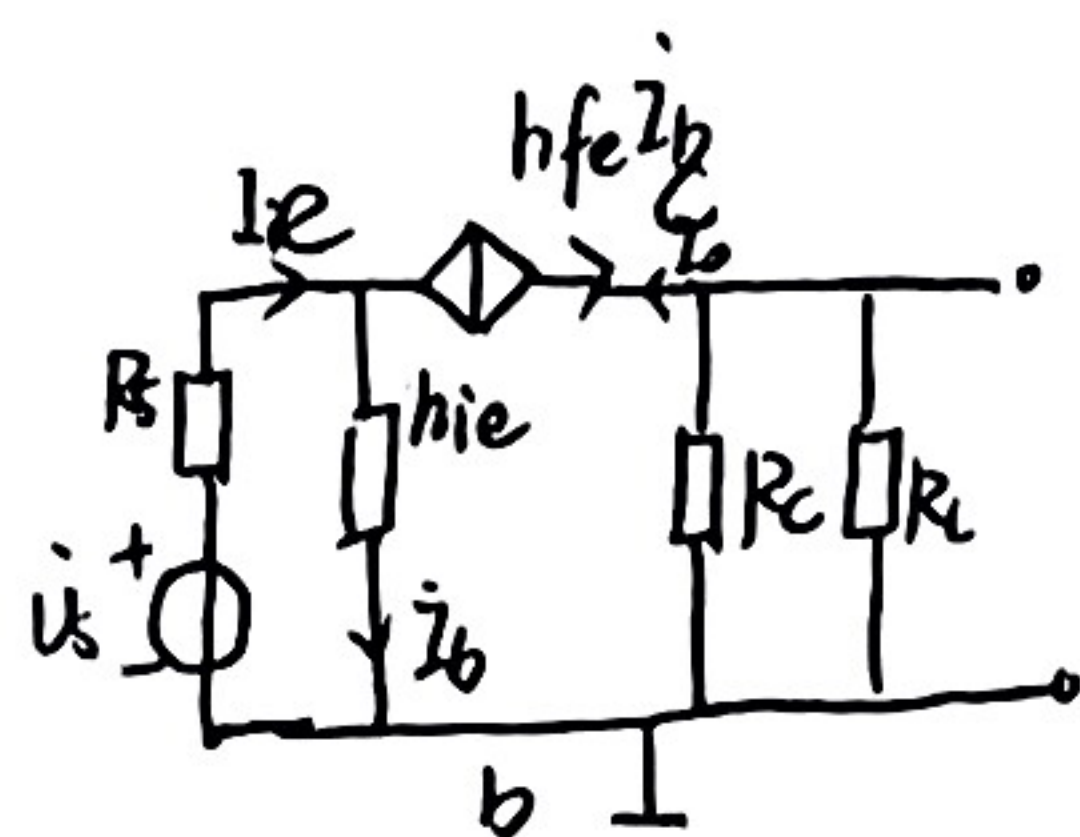
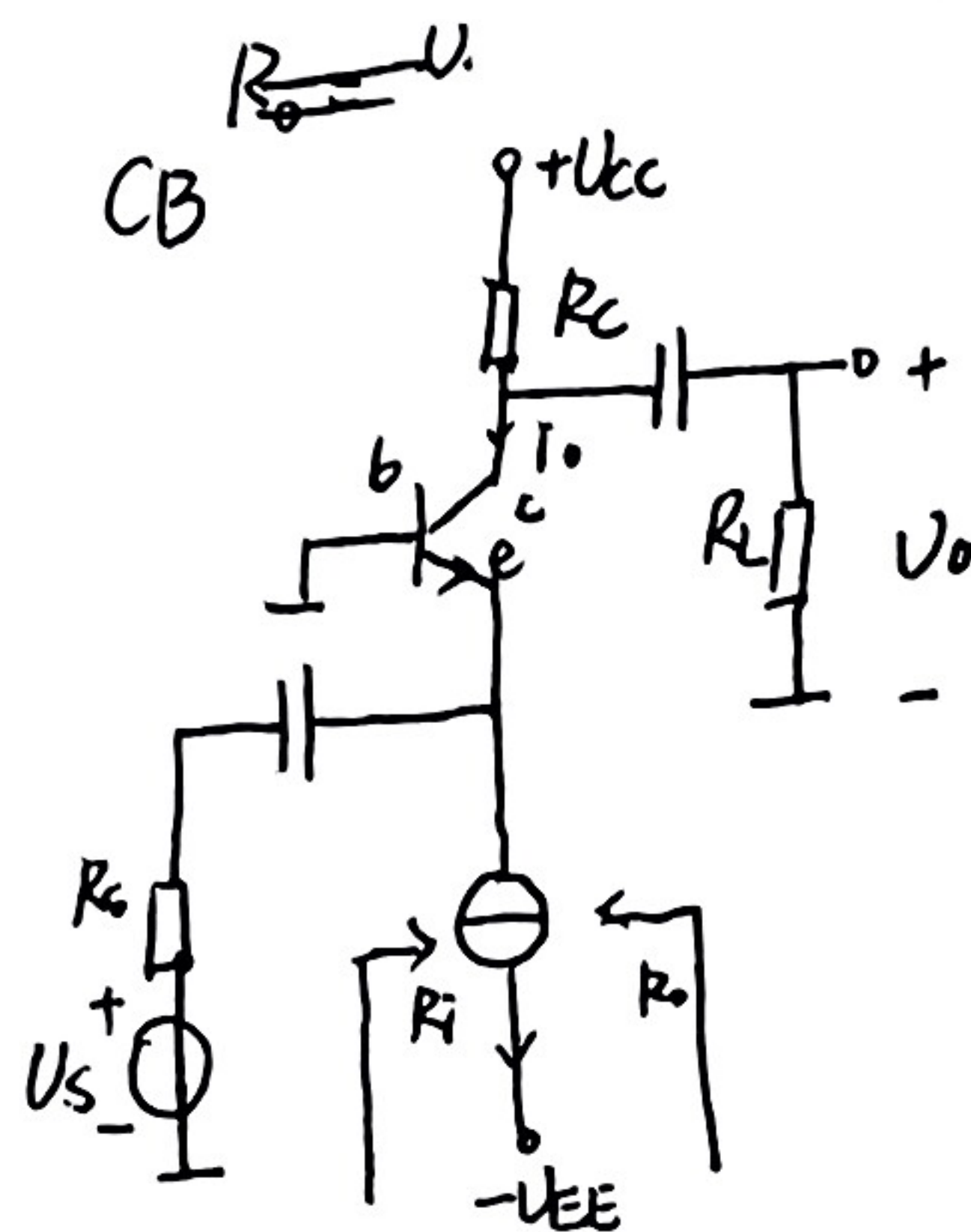
R_o : 戴维南等效.

电源置零 $U_s = 0$ 短路.

$i_b = 0 \rightarrow h_{fe} i_b = 0$ 断.

外加电源
 $R_o = R_c$

CB



$$A_i = \frac{i_o}{i_i} = \frac{-h_{fe} i_b}{i_b + h_{fe} i_b} = \frac{-h_{fe}}{1 + h_{fe}} \approx -1$$

$$R_i = \frac{V_i}{i_i} = \frac{h_{ie} i_b}{i_b + h_{fe} i_b} = \frac{h_{ie}}{1 + h_{fe}}$$

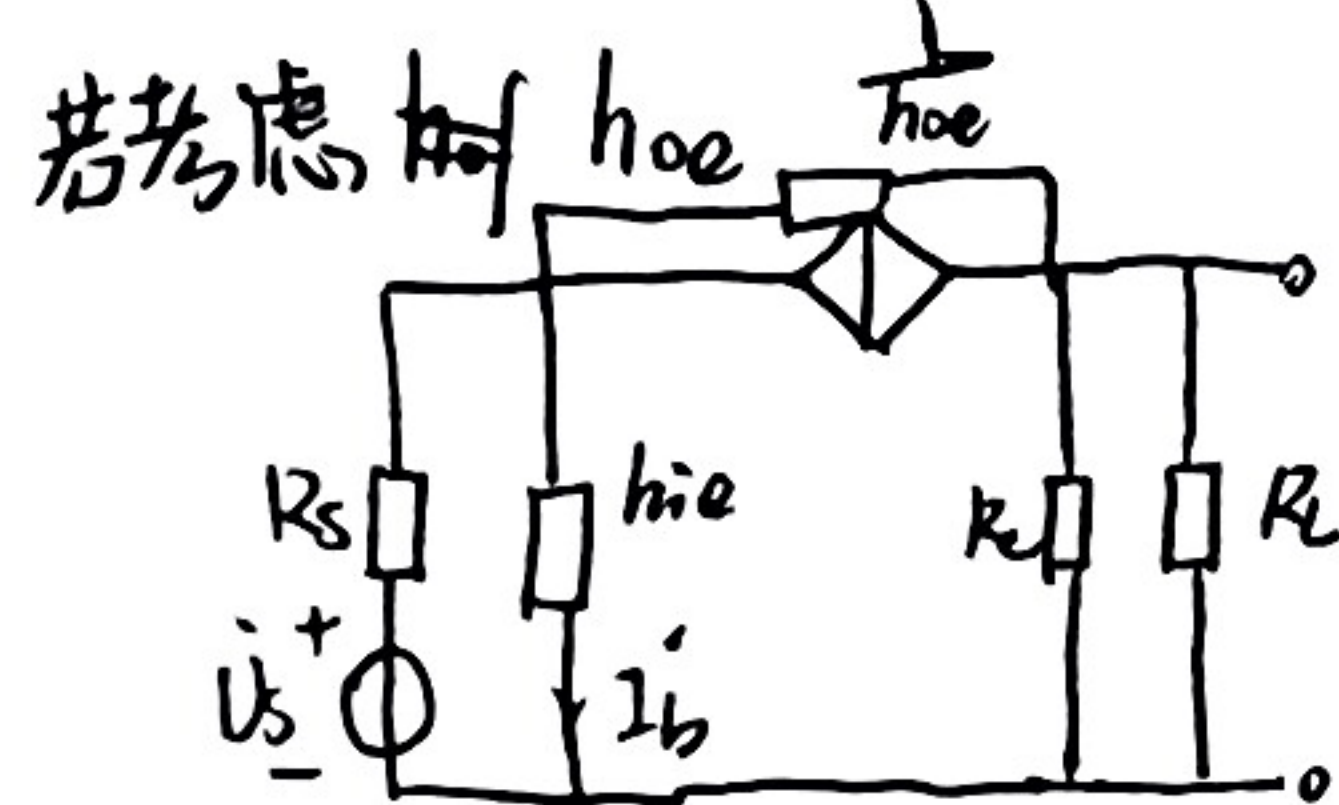
$$A_{um} = \frac{V_o}{V_i} = \frac{h_{fe} i_b (R_c || R_L)}{h_{ie} i_b} = \frac{h_{fe} (R_c || R_L)}{h_{ie}}$$

$$A_{usm} = \frac{V_o}{V_s} = \frac{h_{fe} i_b (R_c || R_L)}{h_{ie} i_b + (i_b + h_{fe} i_b) R_s} = \frac{h_{fe} (R_c || R_L)}{h_{ie} + (1 + h_{fe}) R_s}$$

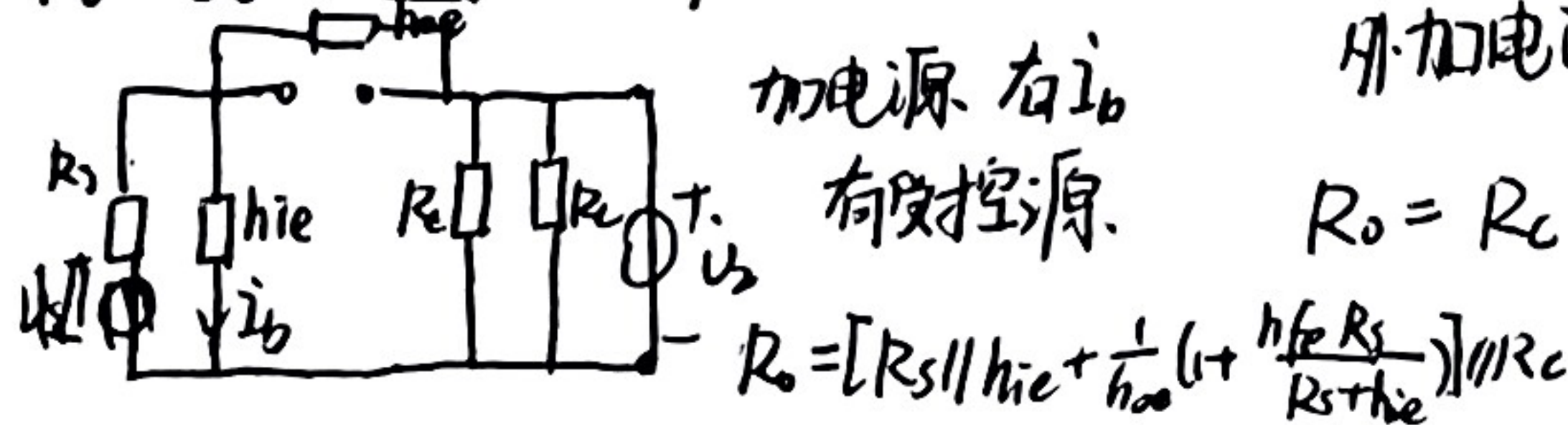
R_o : $U_s = 0 \rightarrow i_b = 0 \rightarrow h_{fe} i_b = 0$

外加电源 仍无 i_b

$R_o = R_c$

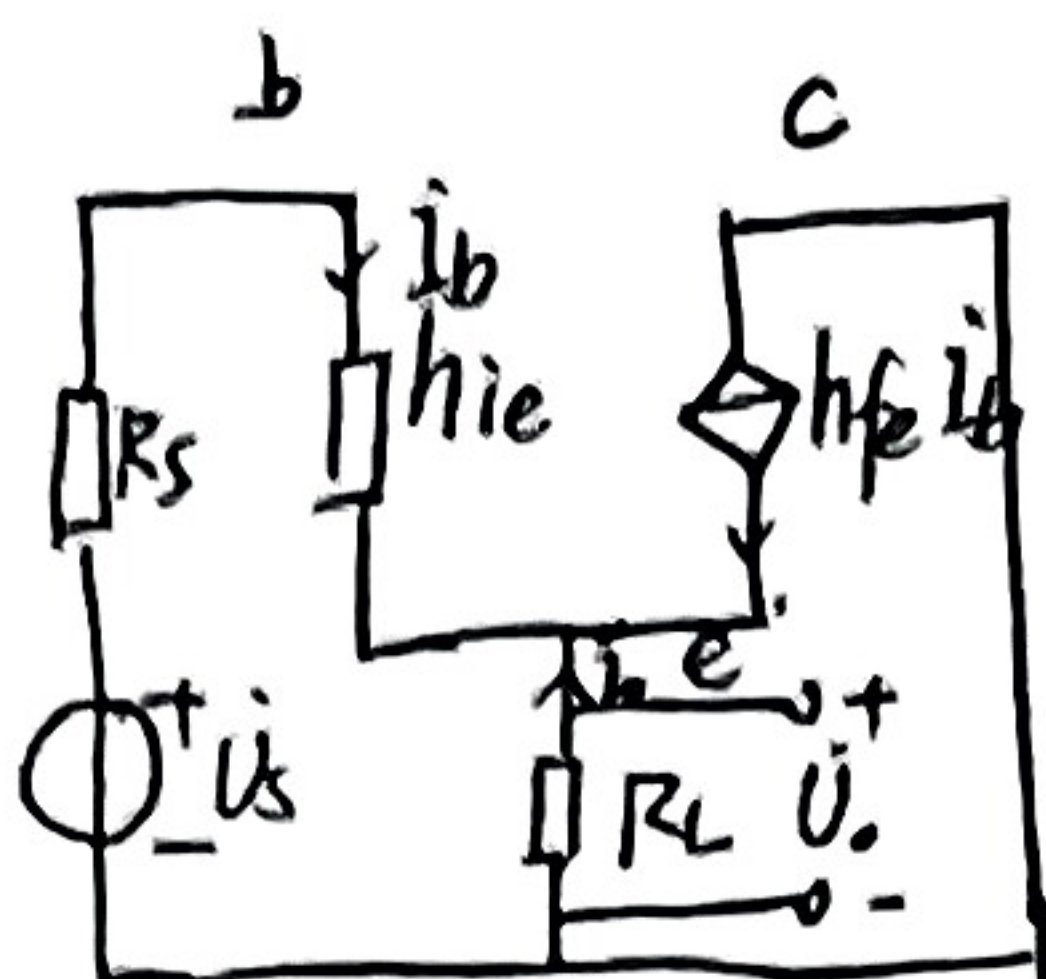
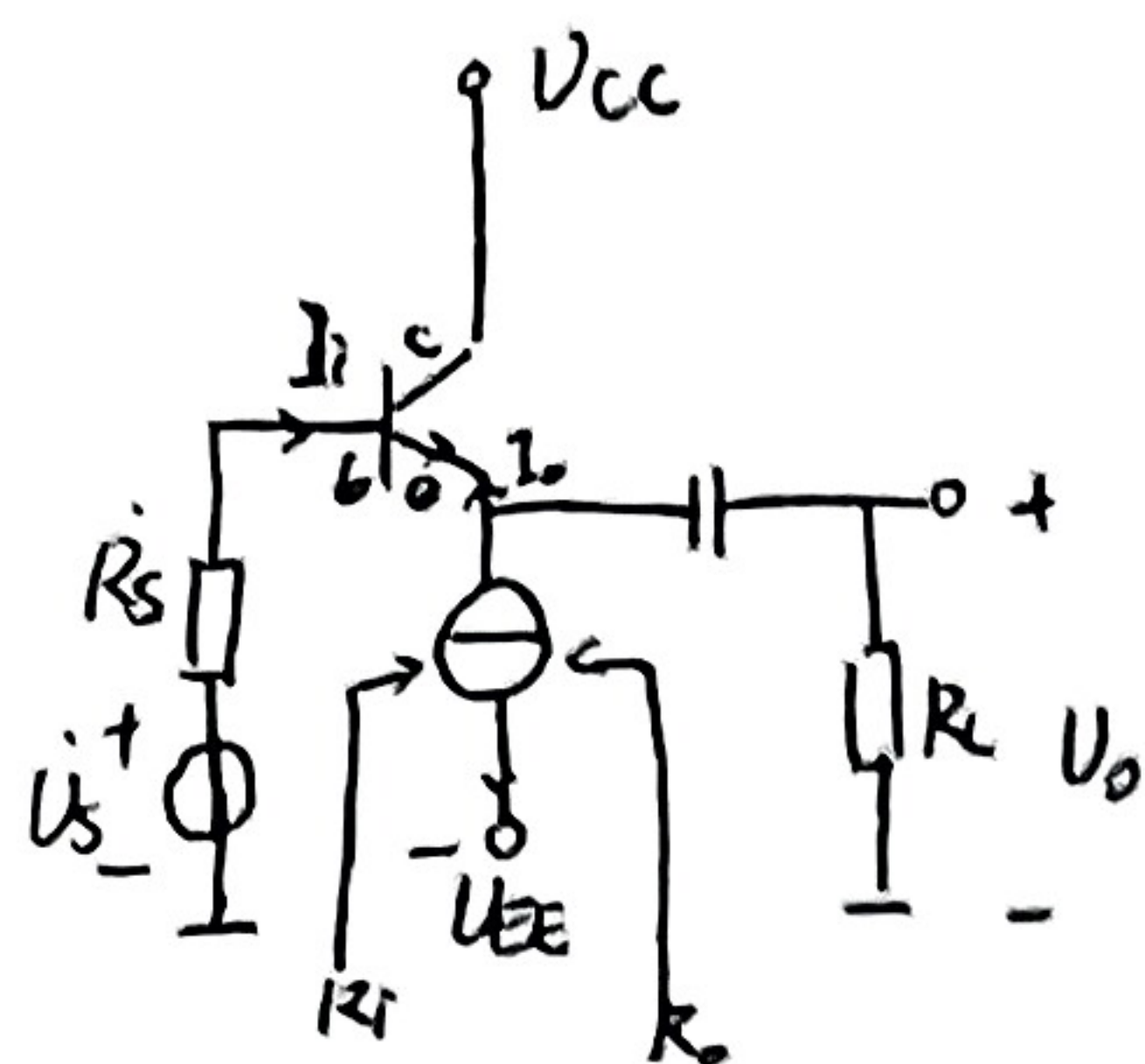


R_o : $U_s = 0, i_b = 0, h_{fe} i_b = 0$



加电源 有 i_b
有受控源.

CC



$$A_i = \frac{i_o}{i_i} = \frac{-(h_{fe} i_b + i_b)}{i_b} = -(h_{fe} + 1) = -(1 + \beta)$$

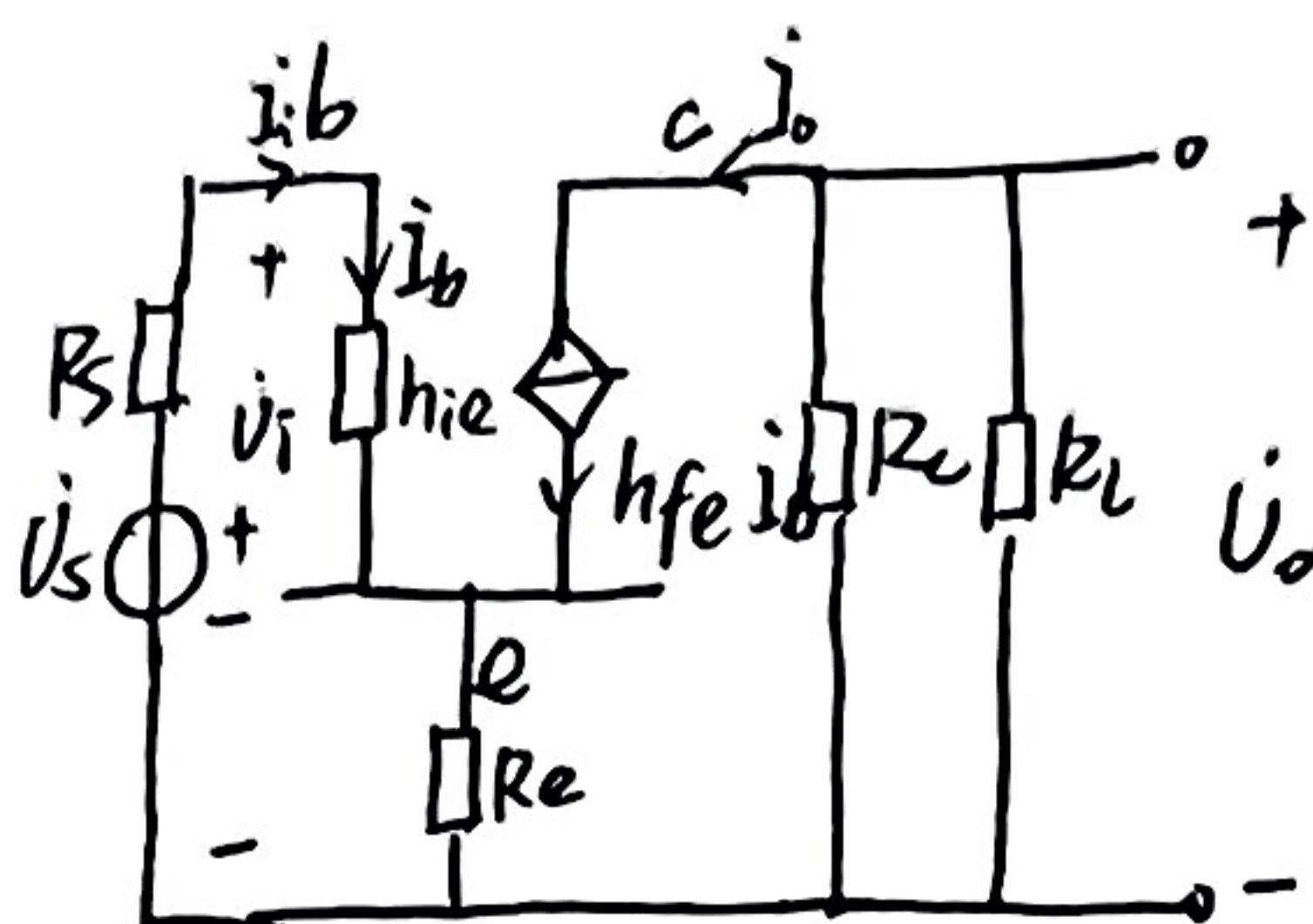
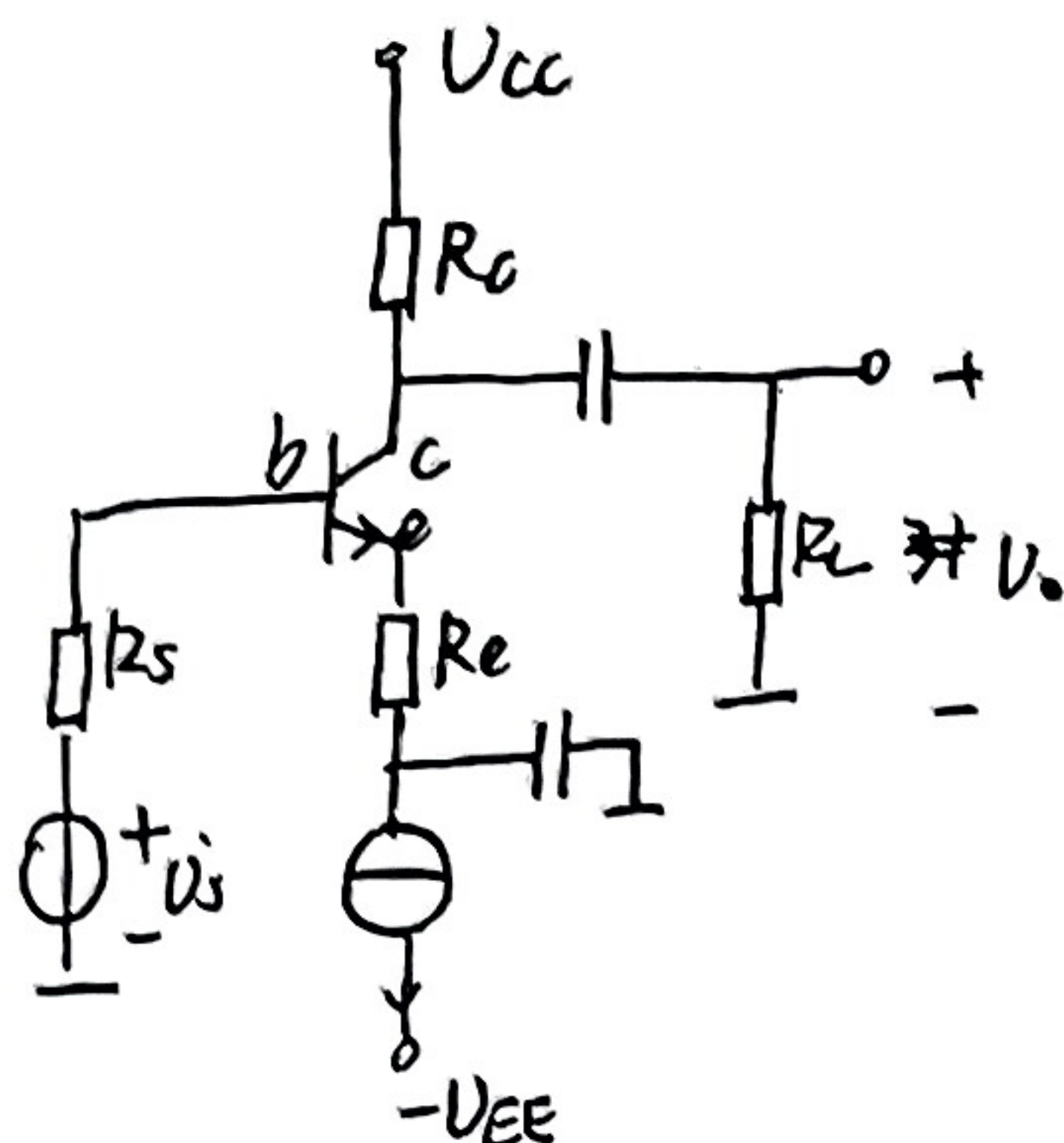
$$R_i = \frac{U_i}{I_i} = \frac{h_{ie} i_b + (i_b + h_{fe} i_b) R_E}{i_b} = h_{ie} + (1 + h_{fe}) R_E$$

$$A_{um} = \frac{U_o}{U_i} = \frac{(h_{fe} i_b + i_b) R_E}{h_{ie} i_b + (i_b + h_{fe} i_b) R_E} = \frac{(1 + h_{fe}) R_E}{h_{ie} + (1 + h_{fe}) R_E}$$

$$A_{usm} = \frac{U_o}{U_s} = \frac{(1 + h_{fe}) R_E}{R_S + h_{ie} + (1 + h_{fe}) R_E}$$

$$R_o = \frac{R_S + h_{ie}}{1 + h_{fe}}$$

射极带电阻的CE



$$A_i = \frac{i_o}{i_i} = \frac{h_{fe} i_b}{i_b} = h_{fe}$$

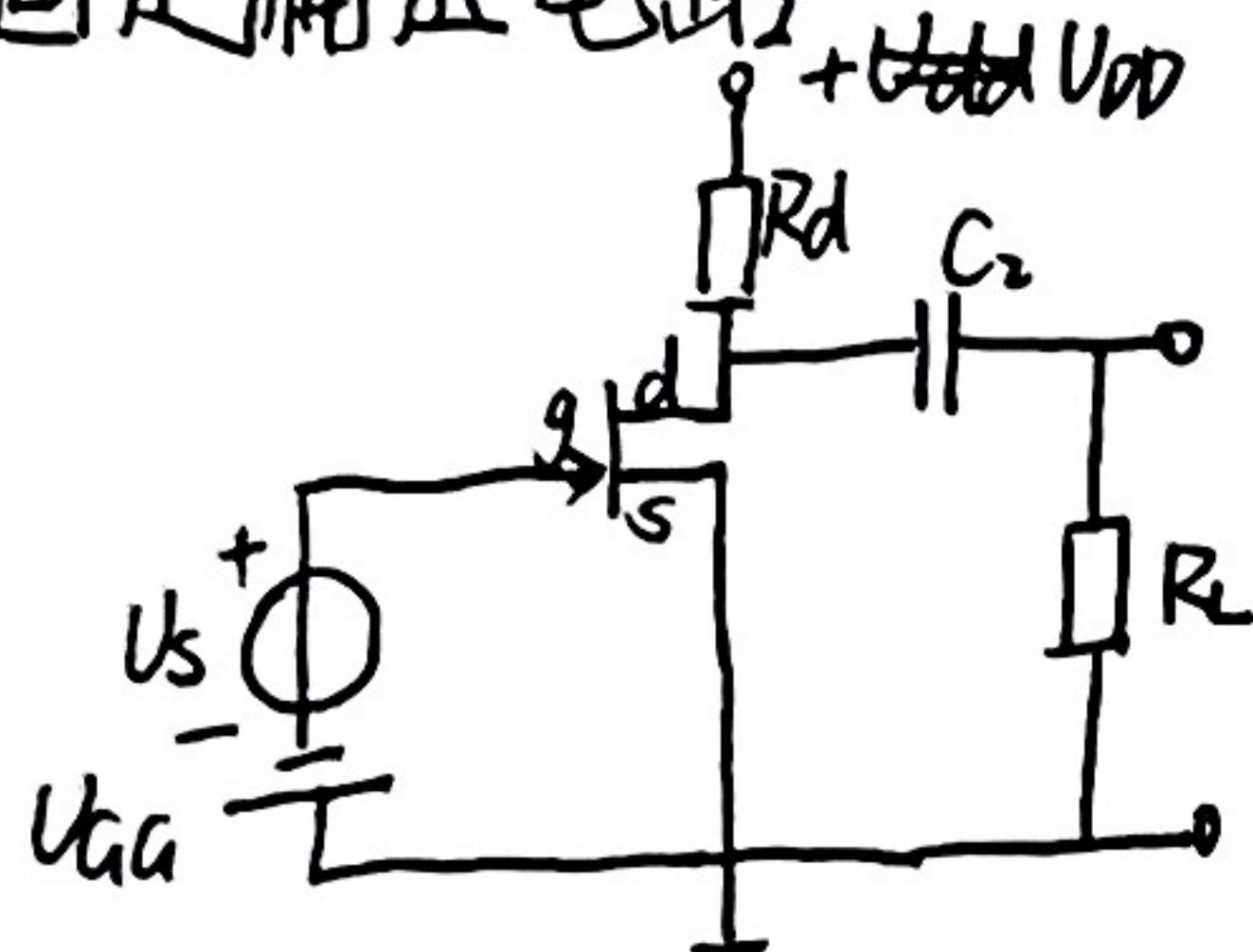
$$R_i = \frac{U_i}{I_i} = \frac{h_{ie} i_b + (i_b + h_{fe} i_b) R_E}{i_b} = h_{ie} + (1 + h_{fe}) R_E$$

$$R_o = R_C$$

$$A_{um} = \frac{U_o}{U_i} = \frac{-h_{fe} i_b (R_C \parallel R_L)}{h_{ie} i_b + (i_b + h_{fe} i_b) R_E} = - \frac{h_{fe} (R_C \parallel R_L)}{h_{ie} + (1 + h_{fe}) R_E}$$

6. 场效应管放大器 直流偏置电路

固定偏压电路

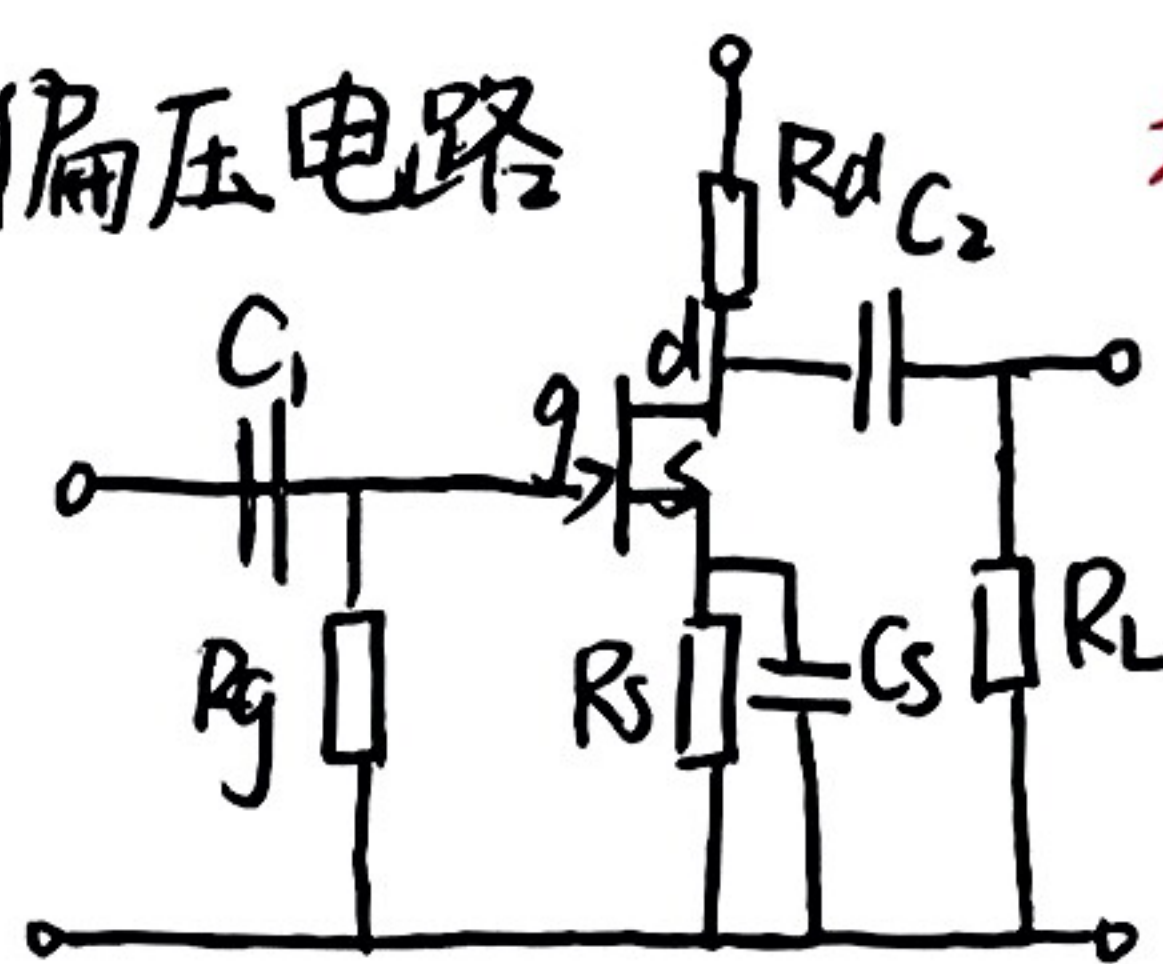


$$U_{GS} = -U_{GQ}$$

$$I_{DQ} = I_{DSS} \left(1 - \frac{U_{GSQ}}{U_{GS(off)}}\right)^2$$

$$U_{DSQ} = U_{DD} - I_{DQ} R_d$$

自偏压电路



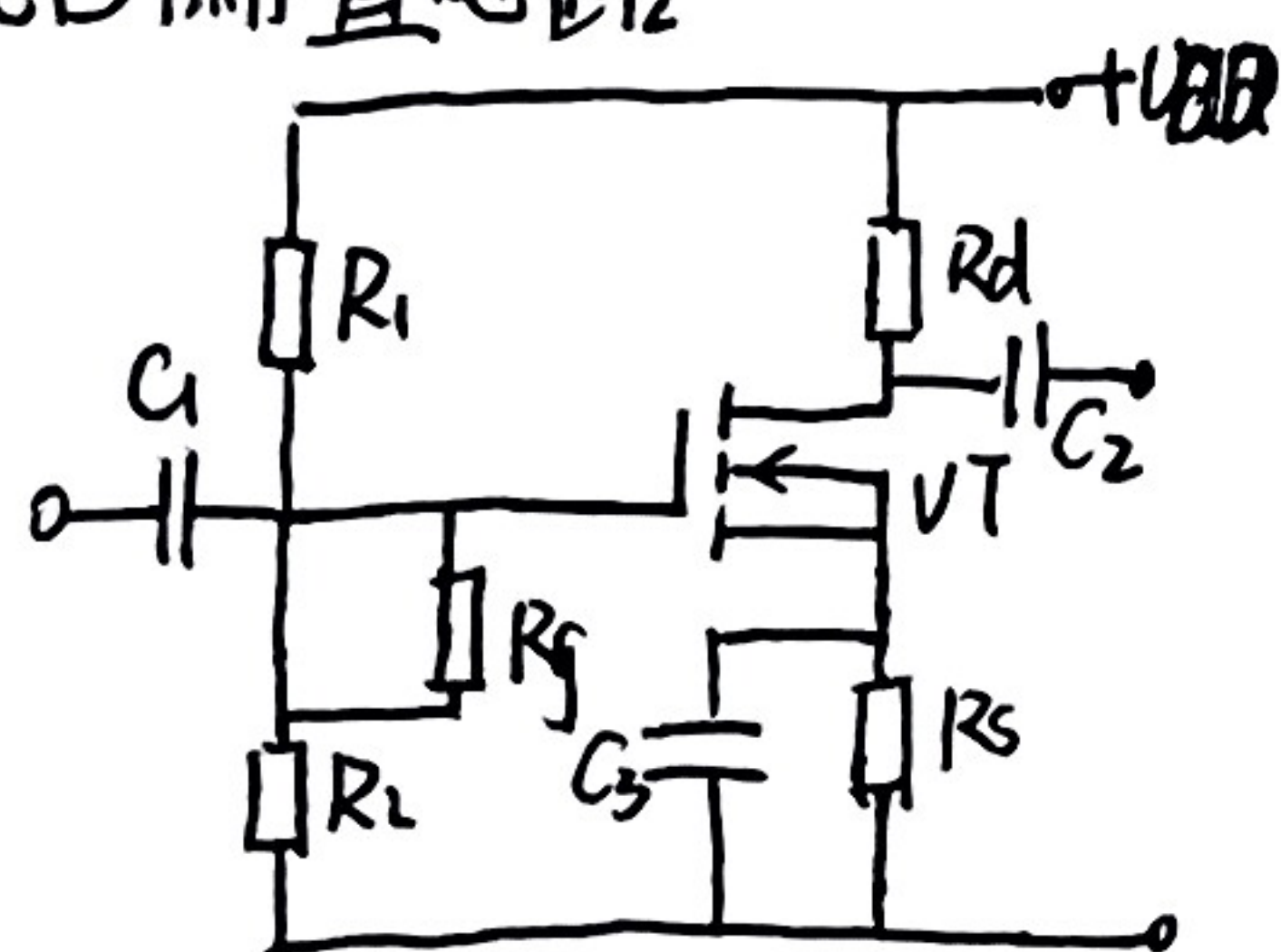
增强型不可用

$$U_{GS} = -I_D R_s$$

$$I_D = I_{DSS} \left(1 - \frac{U_{GS}}{U_{GS(off)}}\right)^2$$

$$U_{DSQ} = U_{DD} - I_{DQ} (R_d + R_s)$$

混合偏置电路



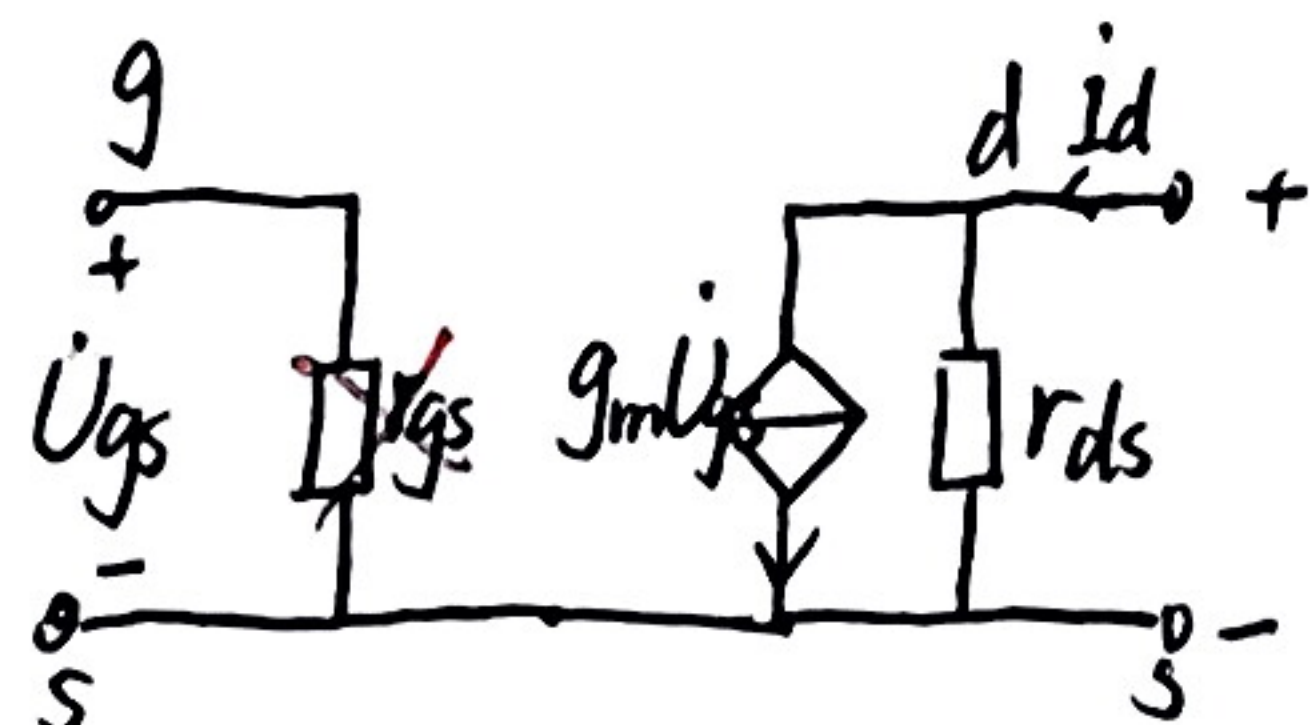
$$U_{GS} = U_{GQ} - I_D R_s$$

$$U_{GQ} = \frac{R_2}{R_1 + R_2} U_{DD}$$

恒流源电路

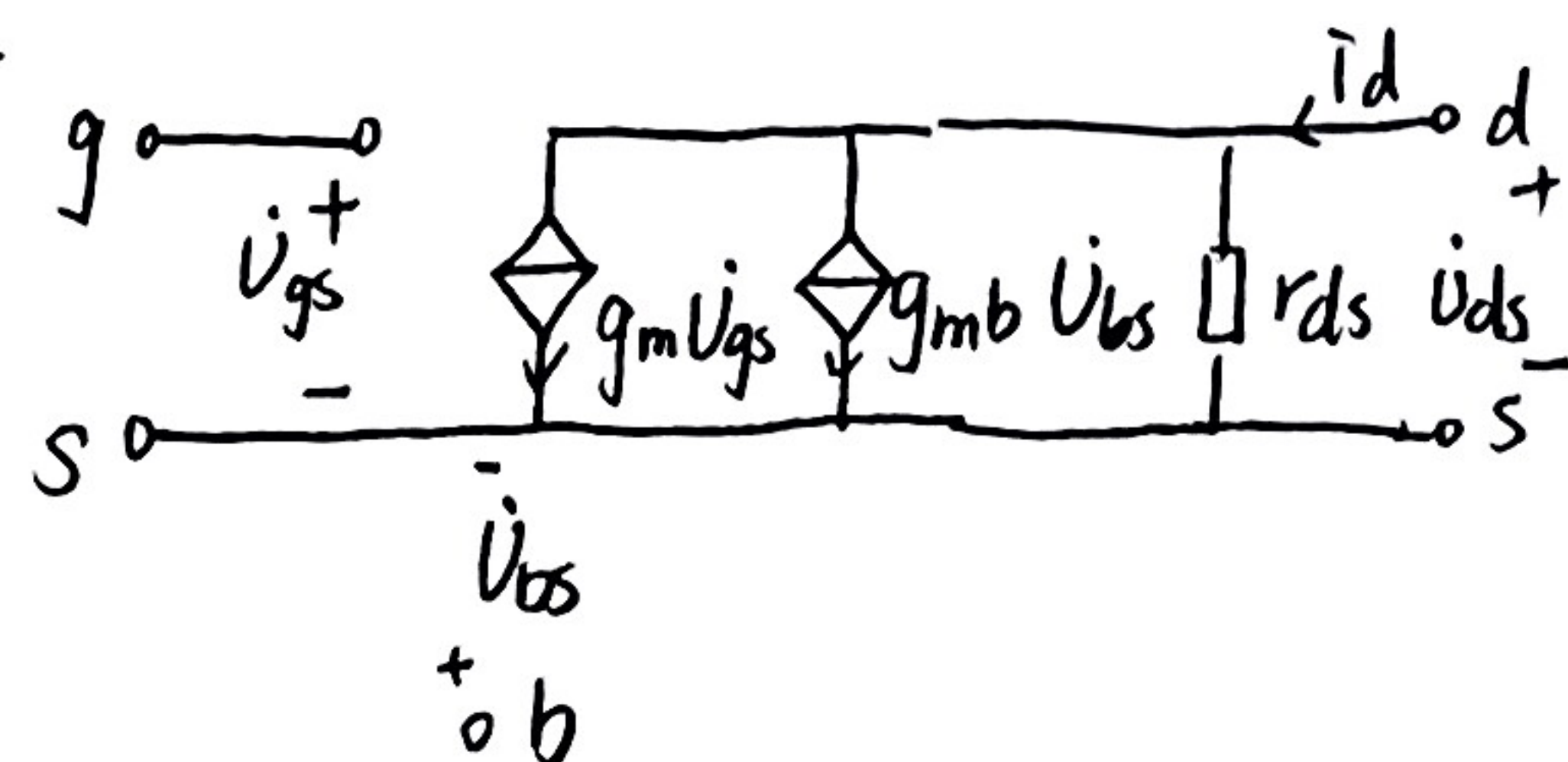
7. 动态分析

JFET

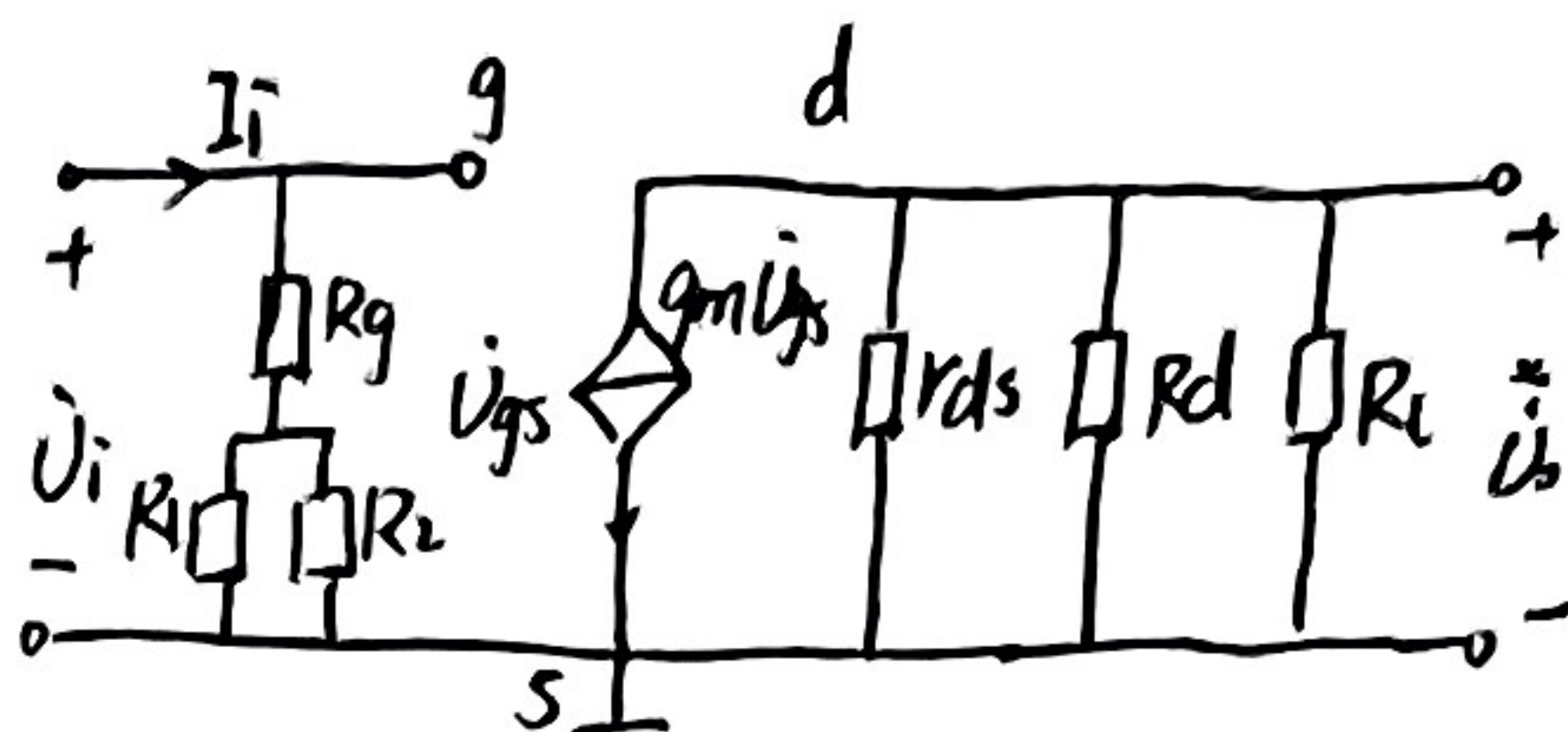
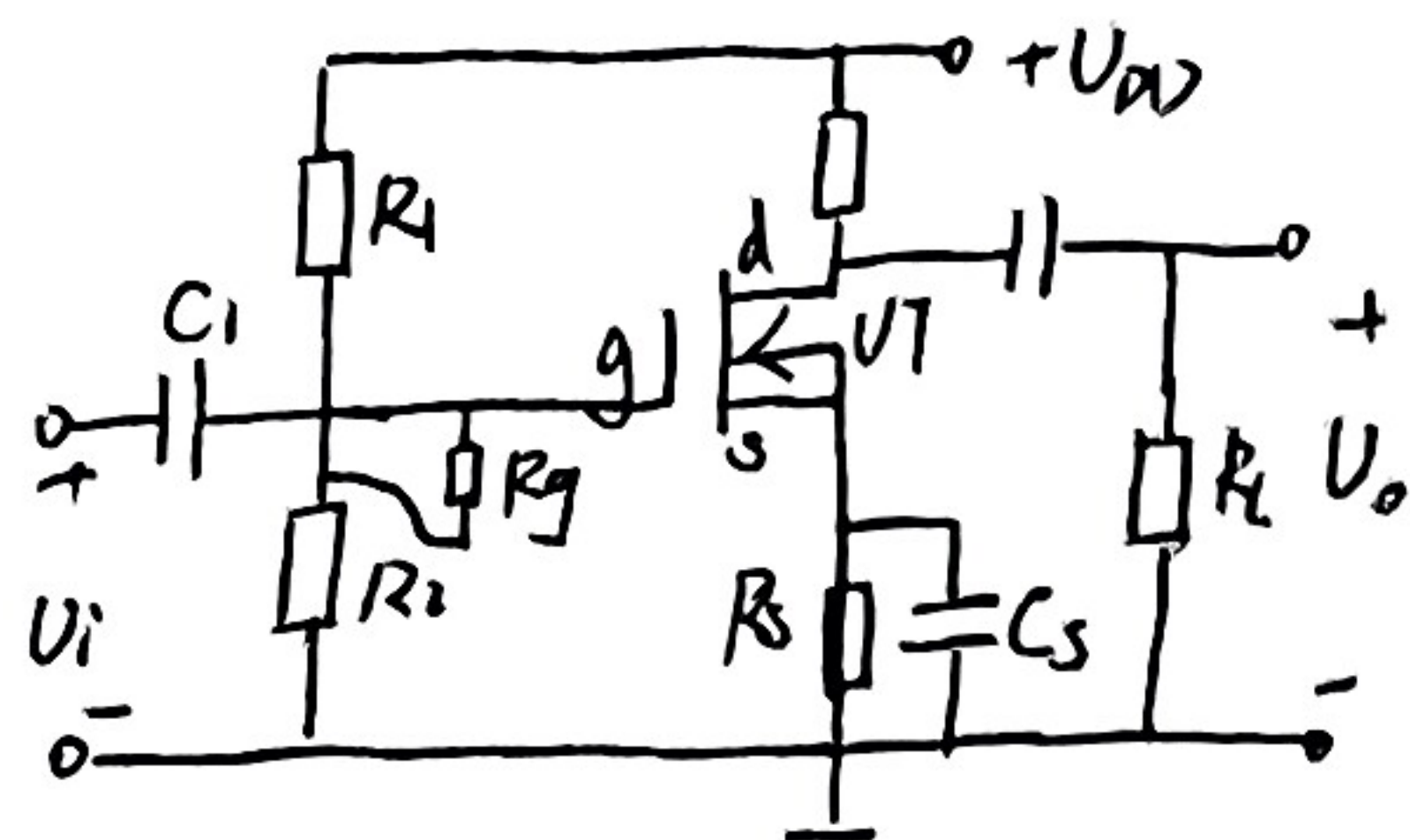


一般不考 \$r_{gs}\$

MOSFET



共源



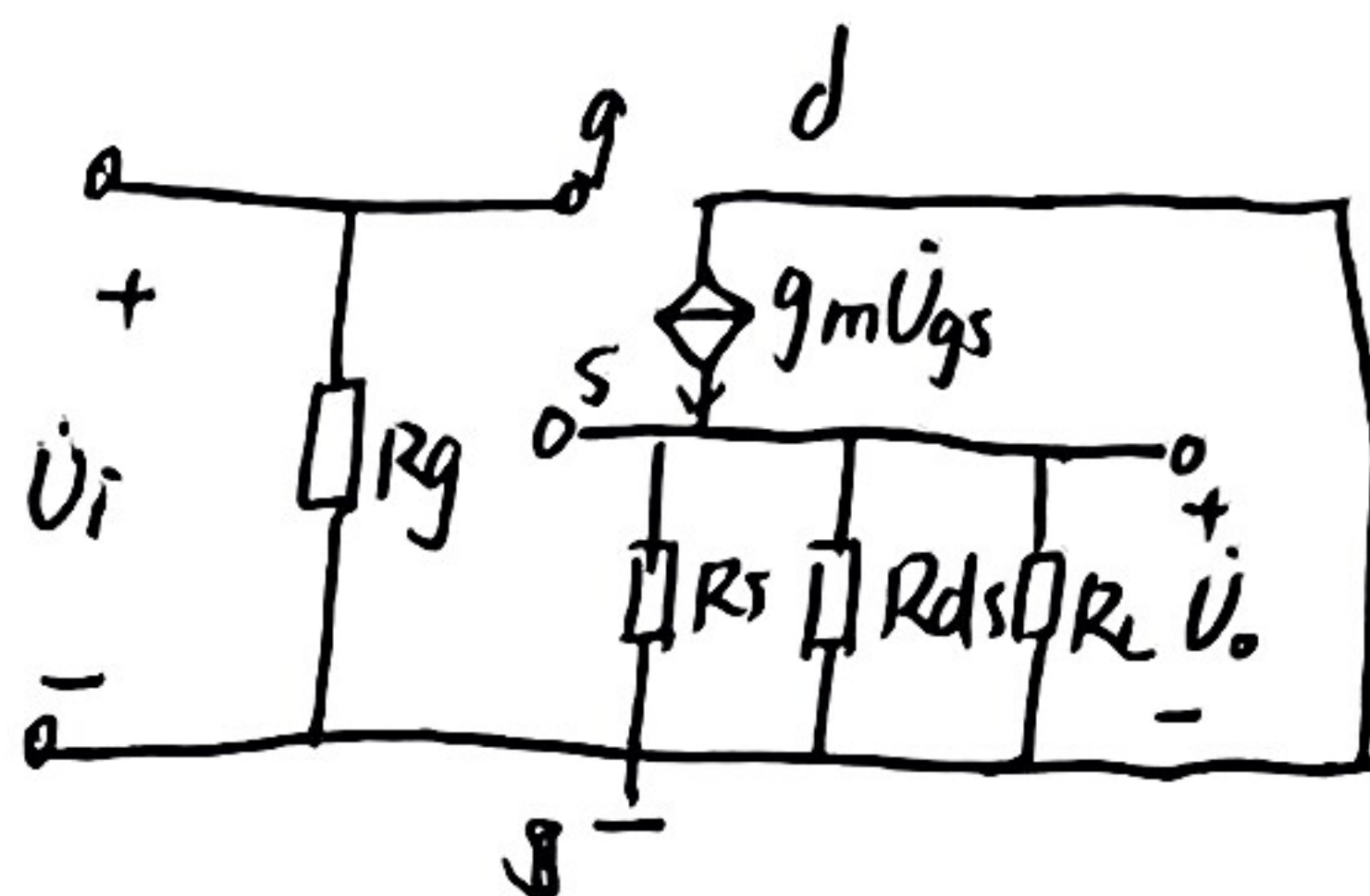
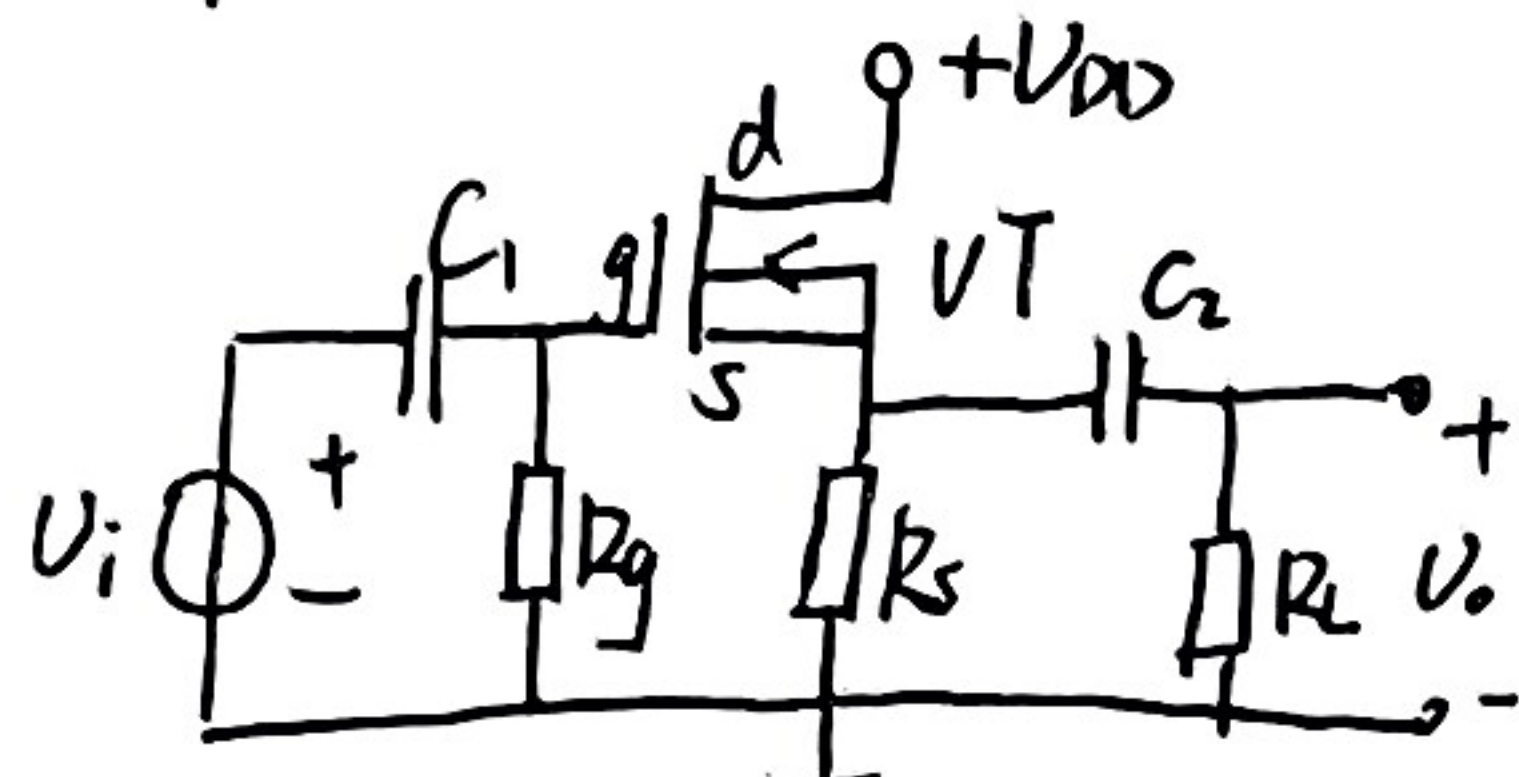
$$A_{um} = \frac{U_o}{U_i} = \frac{-g_m U_{gs} (R_d || R_L || r_{ds})}{U_{gs}}$$

$$= -g_m (R_d || R_L || r_{ds})$$

$$R_i = \frac{U_i}{I_i} = R_g + R_1 || R_2$$

$$R_o = r_{ds} || R_d$$

共漏 (源极跟随器)



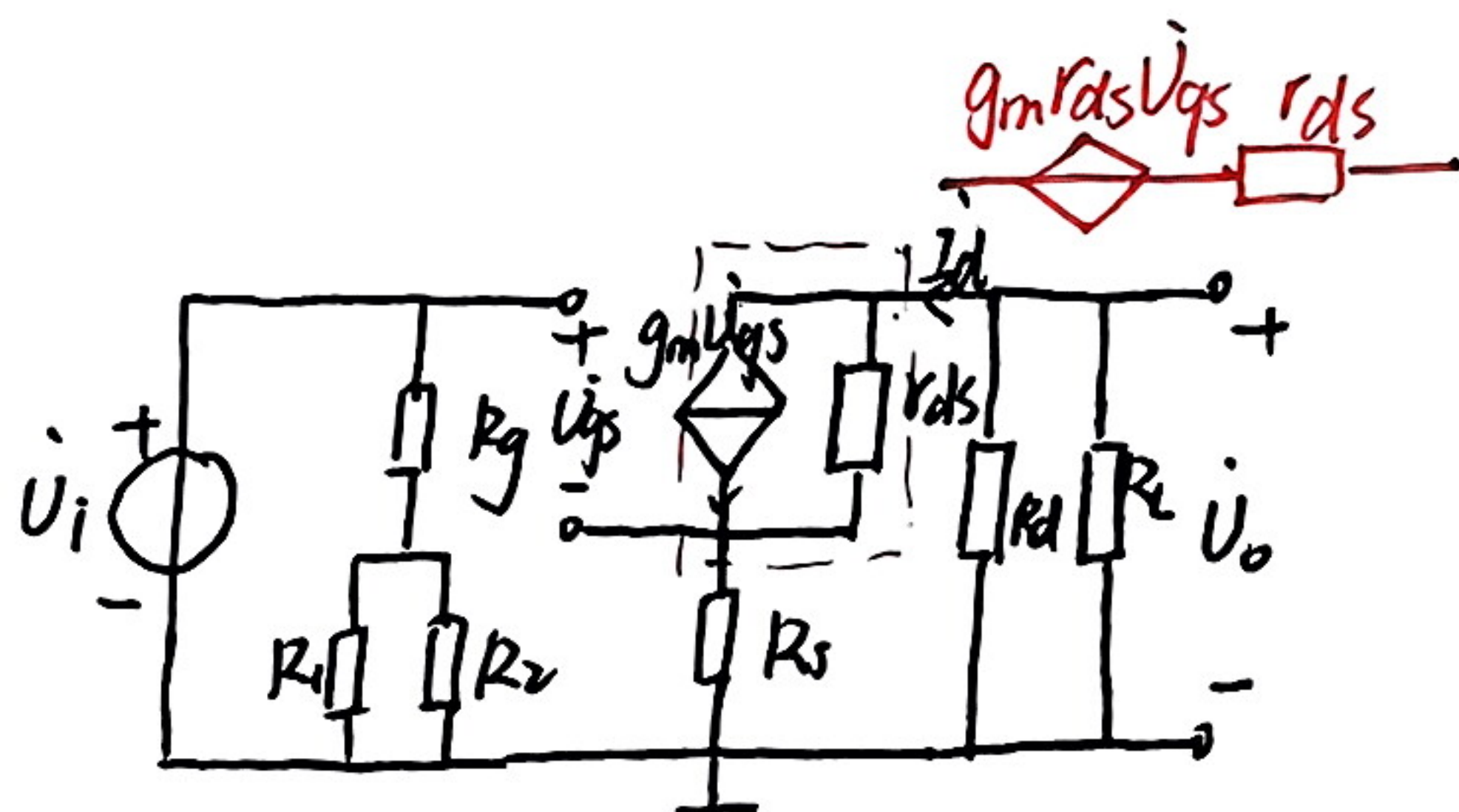
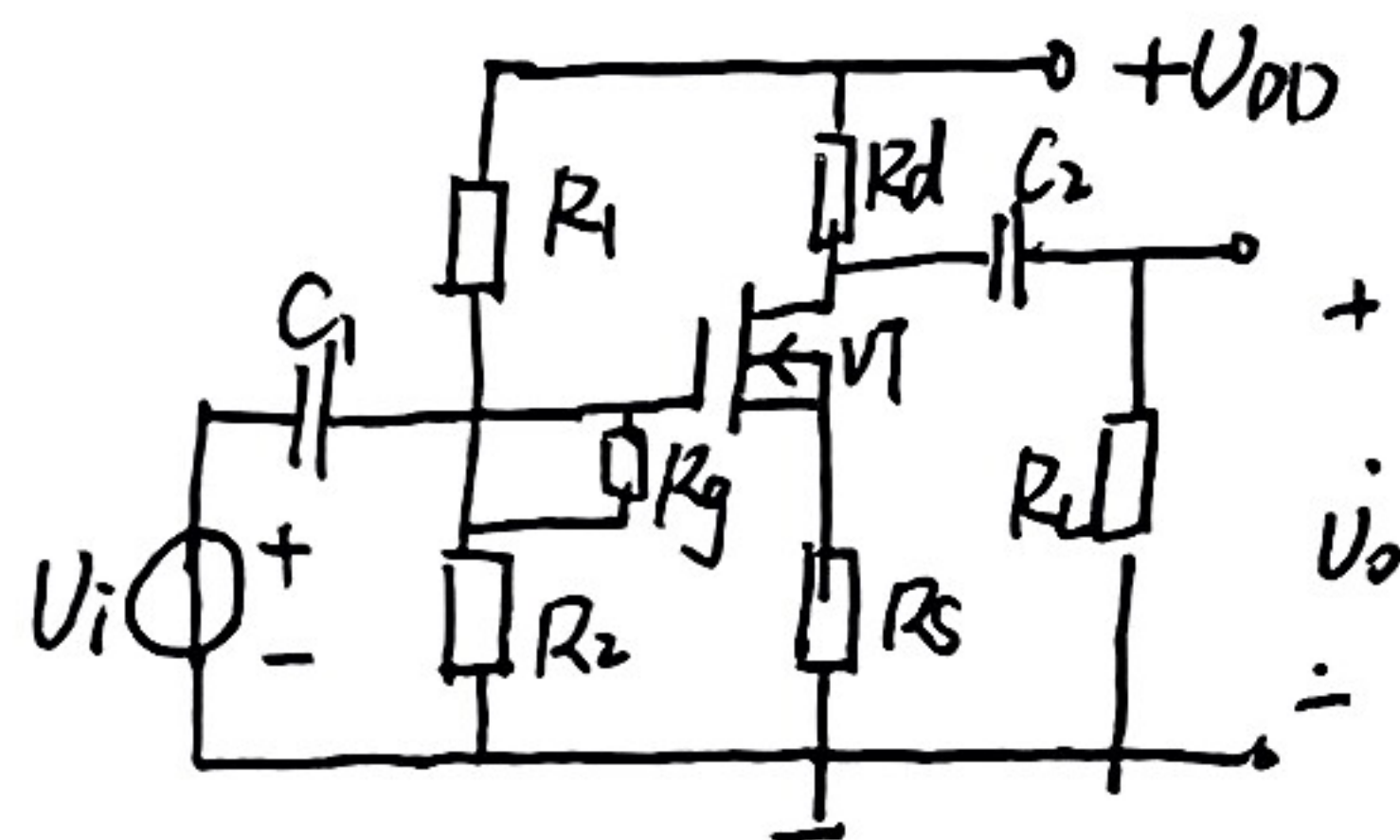
$$A_{um} = \frac{U_o}{U_i} = \frac{g_m U_{gs} (R_s || R_{ds} || r_{ds})}{U_{gs} + g_m U_{gs} (R_s || R_{ds} || r_{ds})}$$

$$= \frac{g_m (R_s || R_{ds} || r_{ds})}{1 + g_m (R_s || R_{ds} || r_{ds})}$$

$$R_i = R_g$$

$$R_o = \frac{U_o}{I_o} \Big|_{U_i=0} = r_{ds} || R_s || \frac{1}{g_m}$$

源极接电阻的共源



$$I_d = \frac{g_m r_{ds} U_{gs}}{r_{ds} + R_s + R_d || R_L}$$

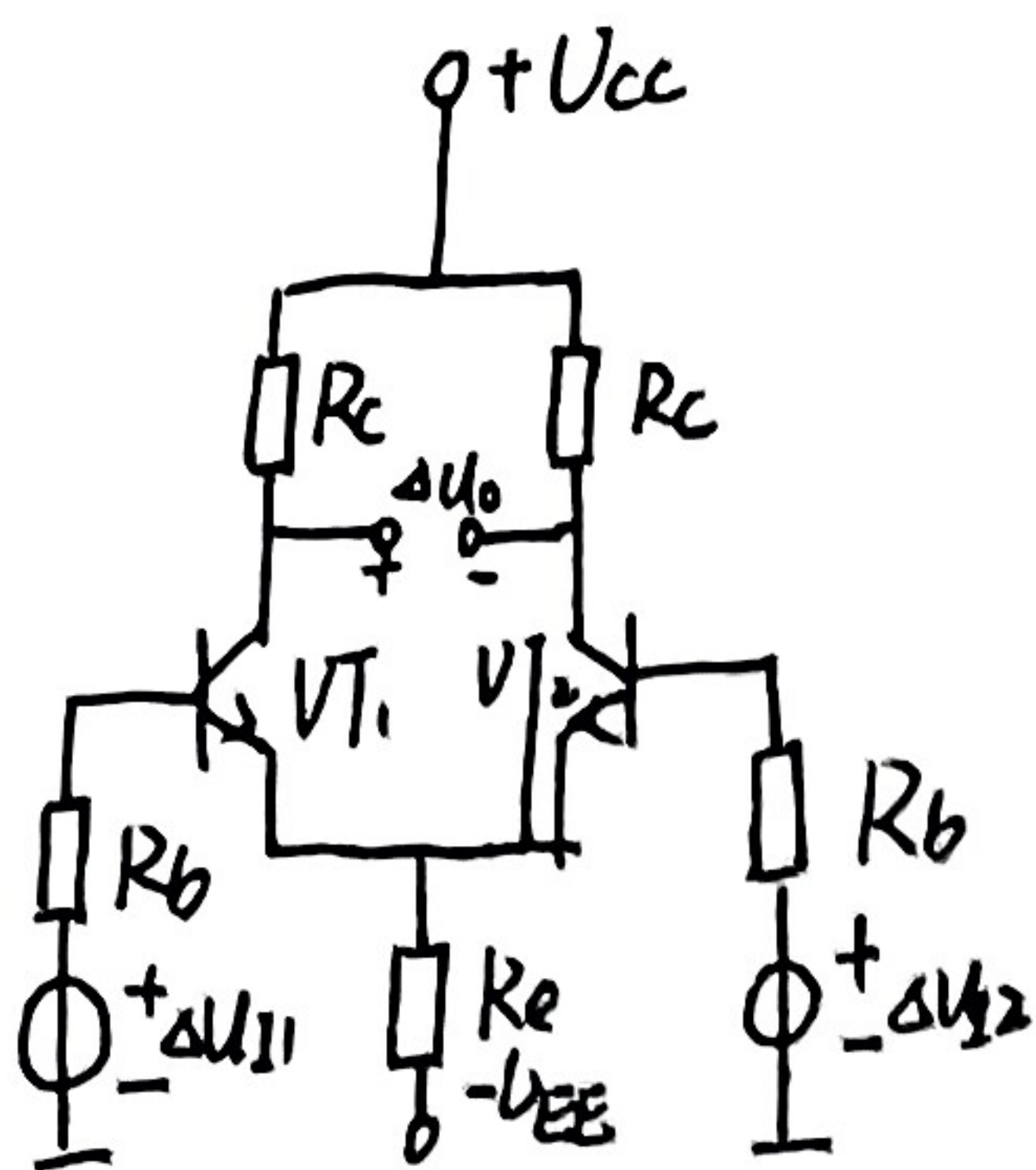
$$U_{gs} = U_i - I_d R_s$$

$$A_{um} = \frac{U_o}{U_i} = - \frac{g_m r_{ds} R_L'}{r_{ds} + R_L' + (1 + g_m r_{ds}) R_s}$$

$$R_i = R_g + R_1 || R_2$$

$$R_o = R_d || [r_{ds} + (1 + g_m r_{ds}) R_s]$$

1. 差分放大器 (零漂)



差模 $\Delta U_{1d} = \Delta U_{11} - \Delta U_{12}$

$\Delta U_{11} = -\Delta U_{12}$ 一对差模输入信号

共模 $\Delta U_{1c} = \frac{\Delta U_{11} + \Delta U_{12}}{2}$ ← 零漂引起

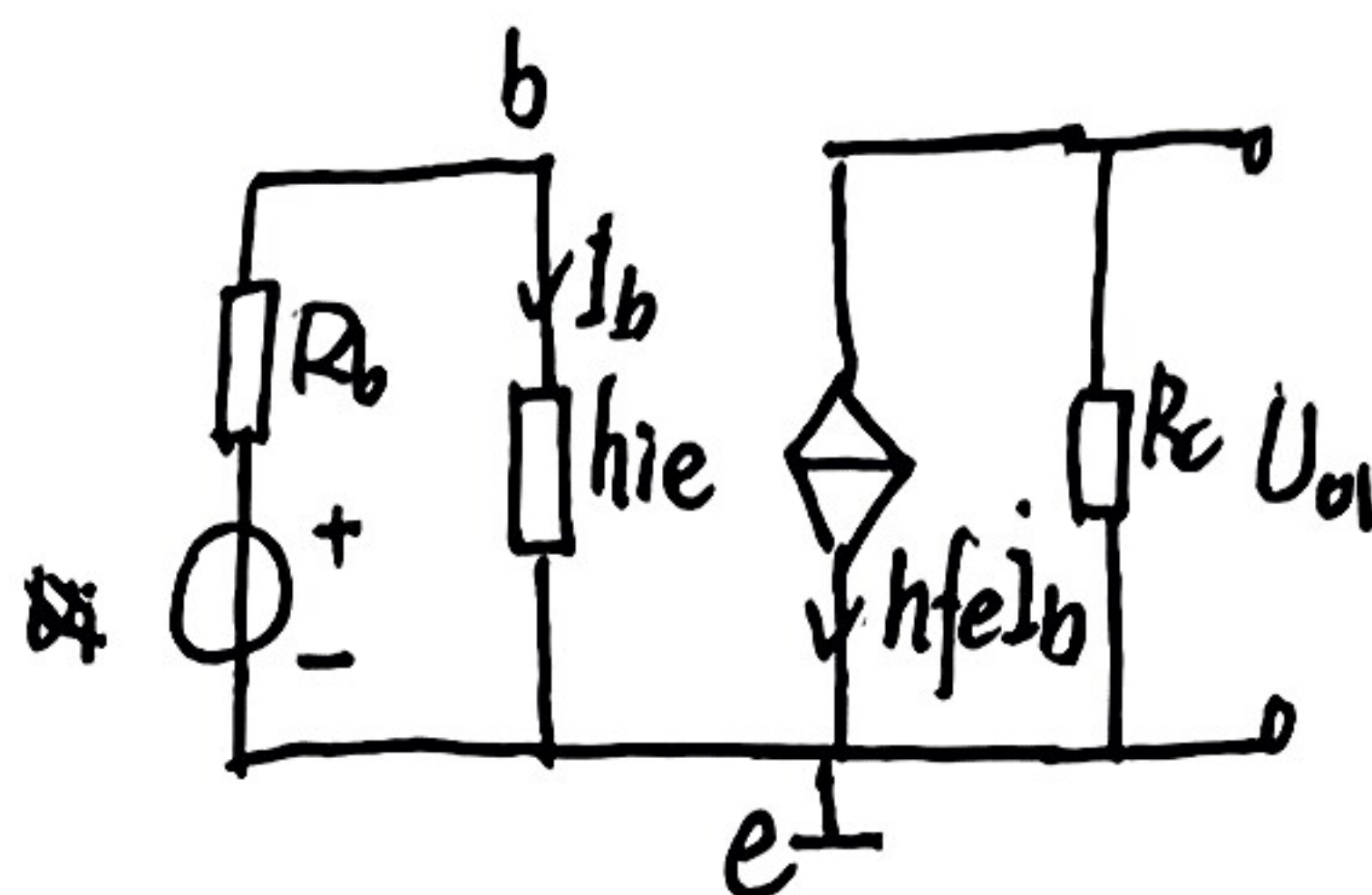
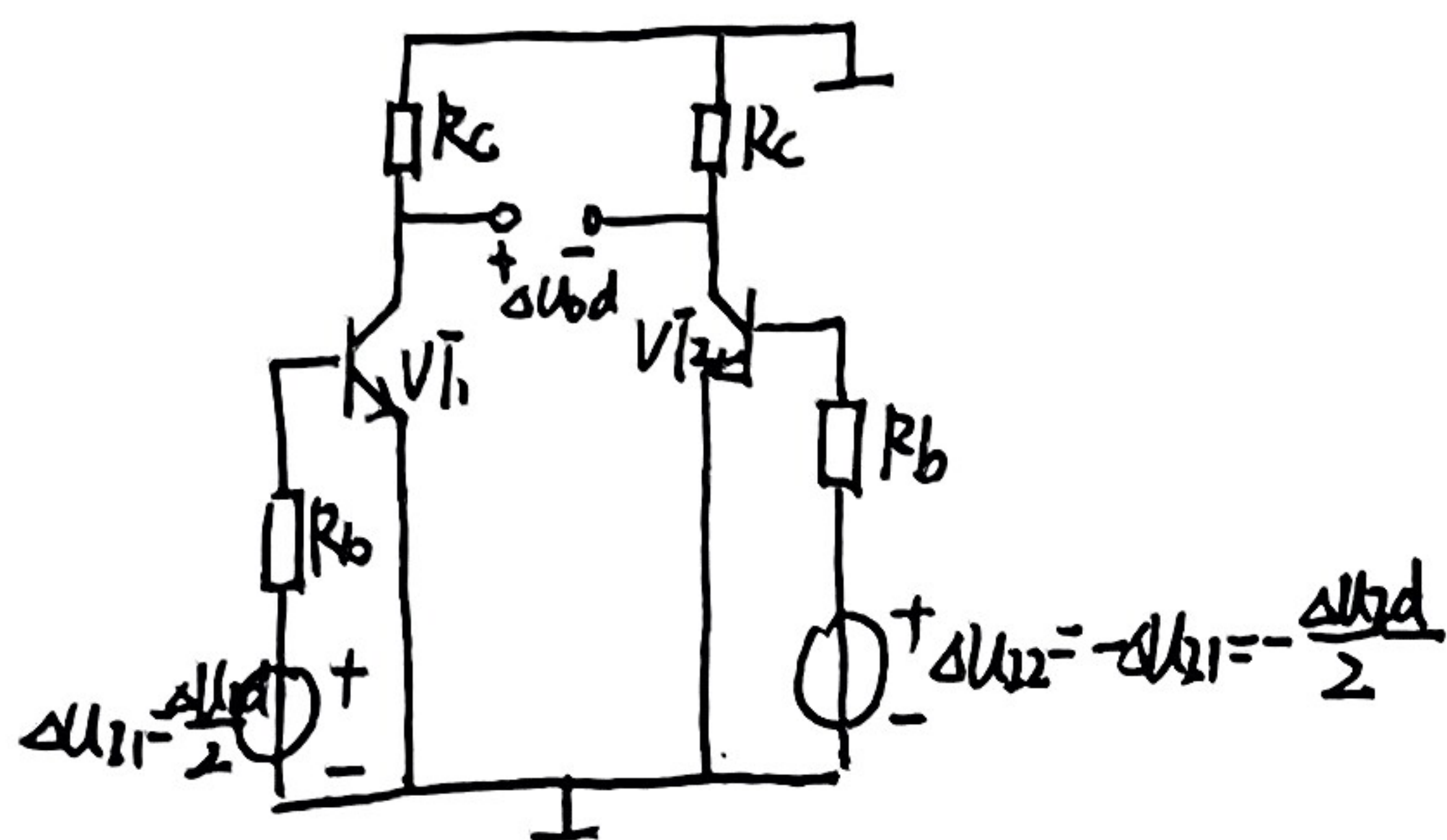
任意信号可分解成共模与差模信号之和/差

$$\Delta U_{11} = \frac{\Delta U_{1d}}{2} + \Delta U_{1c}$$

$$\Delta U_{12} = -\frac{\Delta U_{1d}}{2} + \Delta U_{1c}$$

2. 性能分析.

差模电压放大倍数 A_{ud} , A_{ud1} (A_{ud2})



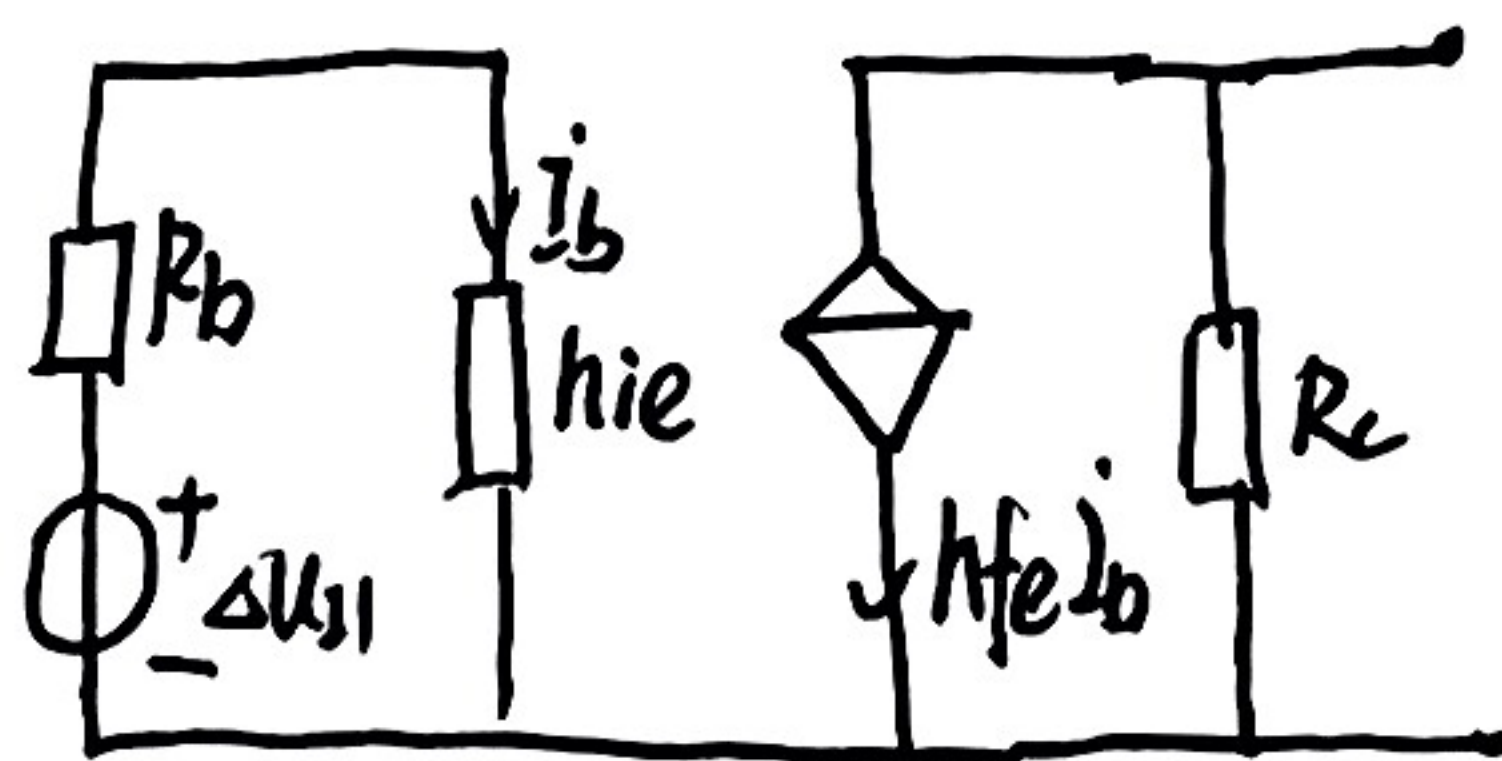
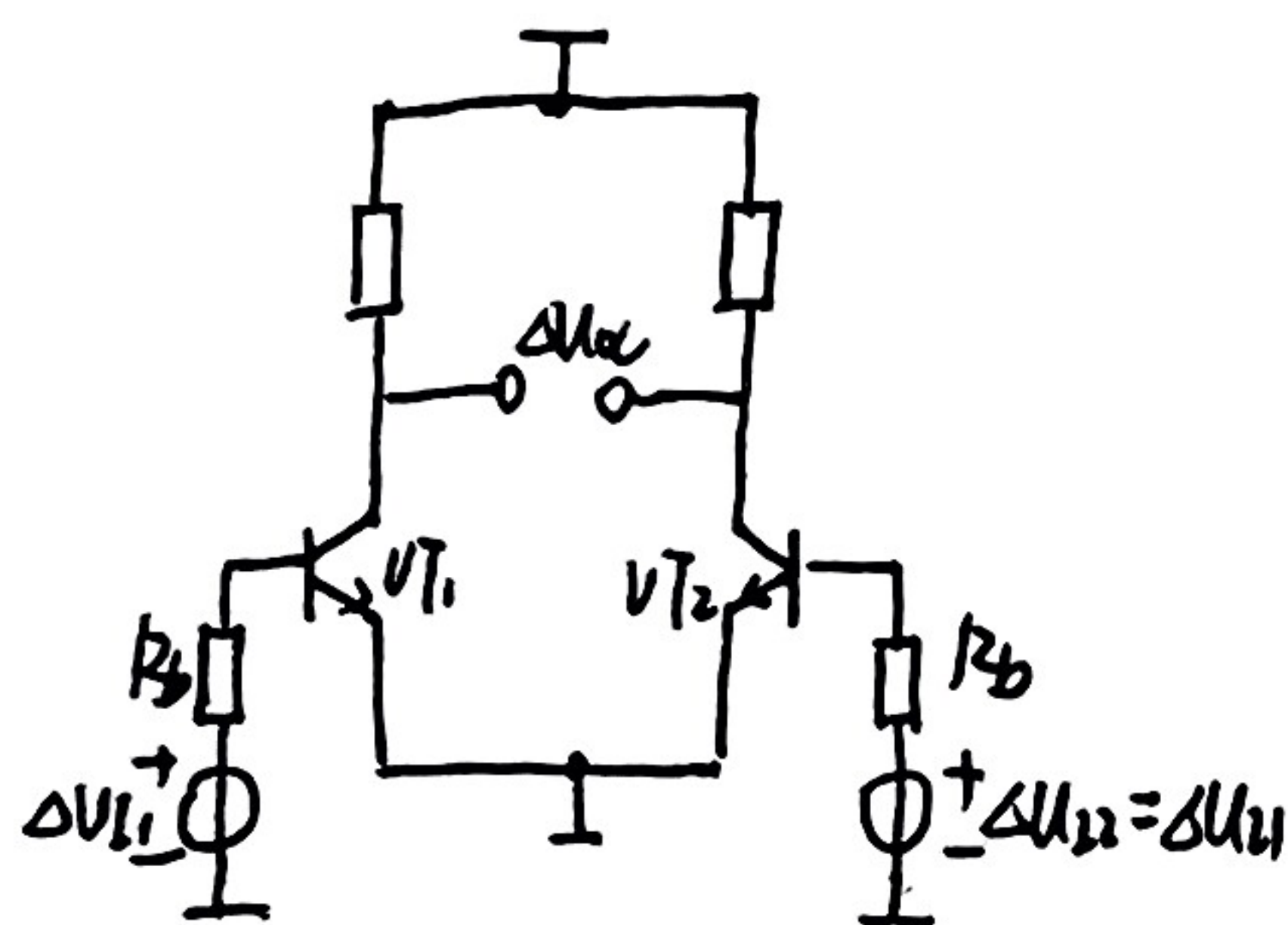
$$A_{ud1} = \frac{\Delta U_{o1}}{\Delta U_{1d}} = \frac{\Delta U_{o1}}{2\Delta U_{11}}$$

$$= \frac{-h_{fe} I_b R_c}{2 I_b (R_b + h_{ie})} = -\frac{h_{fe} R_c}{2(R_b + h_{ie})}$$

$$A_{ud2} = \frac{\Delta U_{o2}}{\Delta U_{1d}} = \frac{\Delta U_{o2}}{-2\Delta U_{11}} = \frac{h_{fe} R_c}{2(R_b + h_{ie})}$$

$$A_{ud} = \frac{\Delta U_o}{\Delta U_{1d}} = \frac{\Delta U_{o1} - \Delta U_{o2}}{\Delta U_{1d}} = -\frac{h_{fe} R_c}{R_b + h_{ie}}$$

共模电压放大倍数 A_{uc} , A_{uc1} (A_{uc2})



$$A_{uc1} = \frac{\Delta U_{o1}}{\Delta U_{1c}} = \frac{\Delta U_{o1}}{\Delta U_{11}} = -\frac{h_{fe} R_c}{R_b + h_{ie}}$$

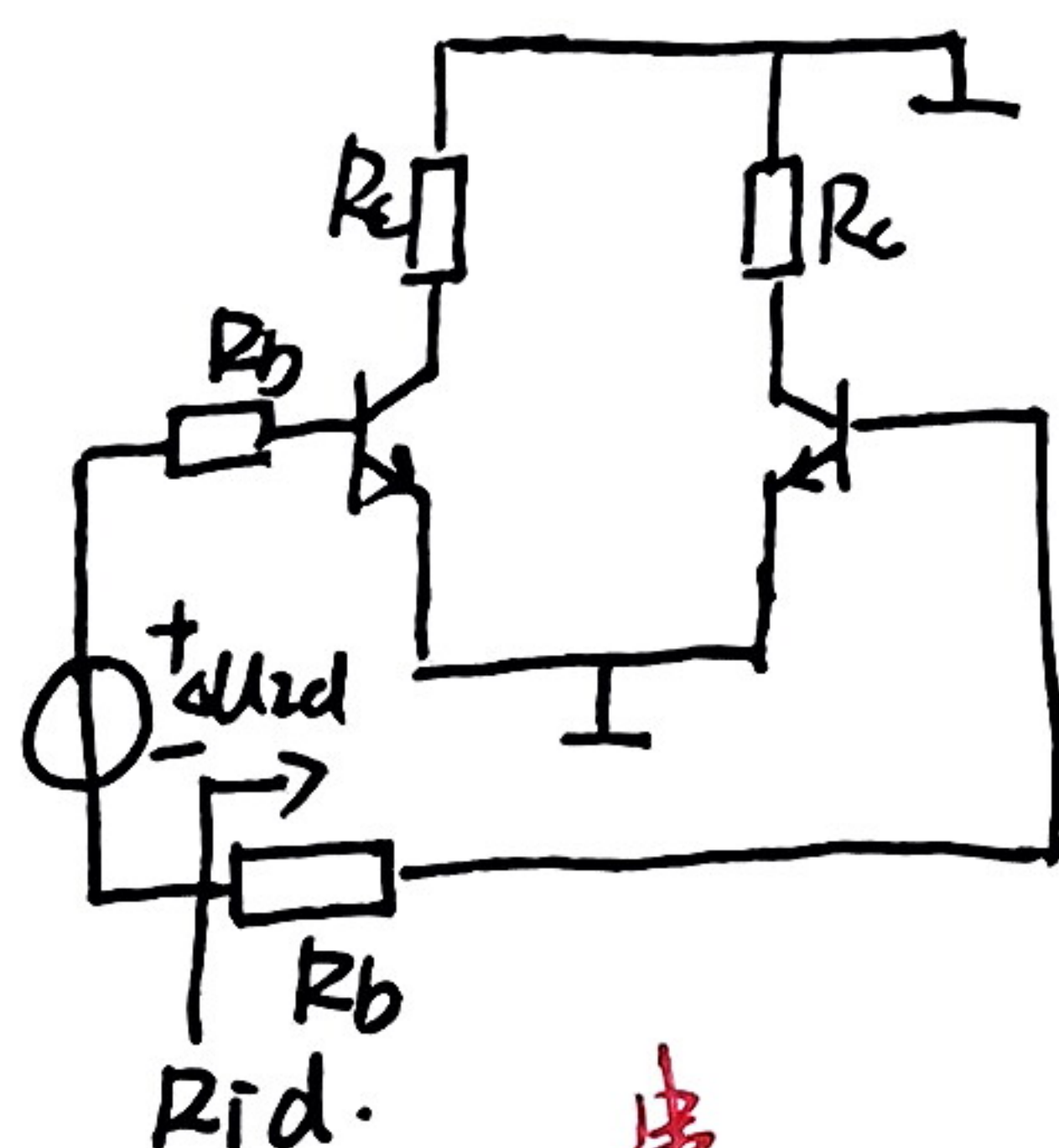
$$A_{uc2} = \frac{\Delta U_{o2}}{\Delta U_{1c}} = \frac{\Delta U_{o1}}{\Delta U_{1c}} = A_{uc1}$$

$$A_{uc} = \frac{\Delta U_{oc}}{\Delta U_{1c}} = \frac{\Delta U_{o1} - \Delta U_{o2}}{\Delta U_{1c}} = 0$$

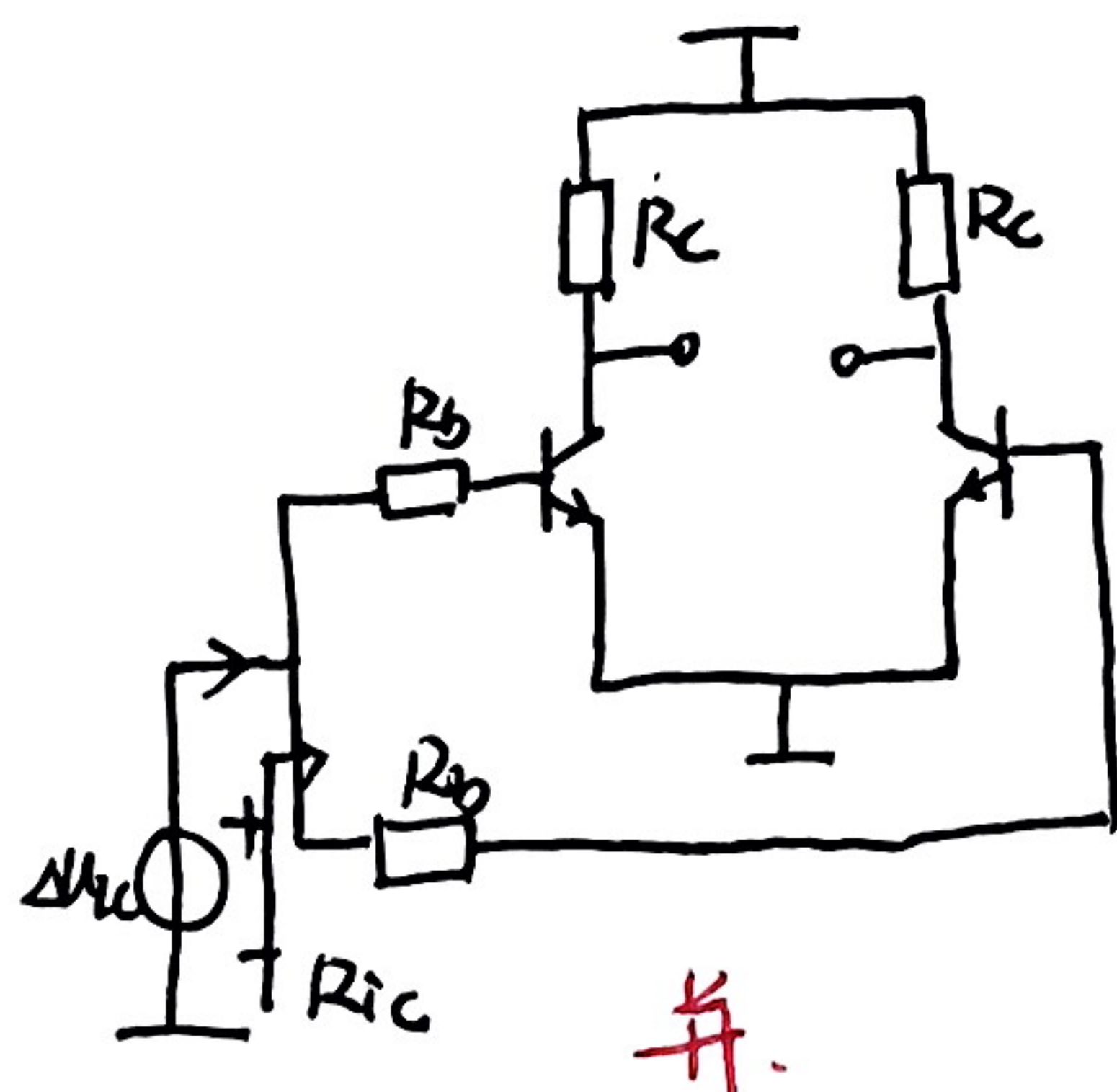
差模输入电阻

共模输入电阻

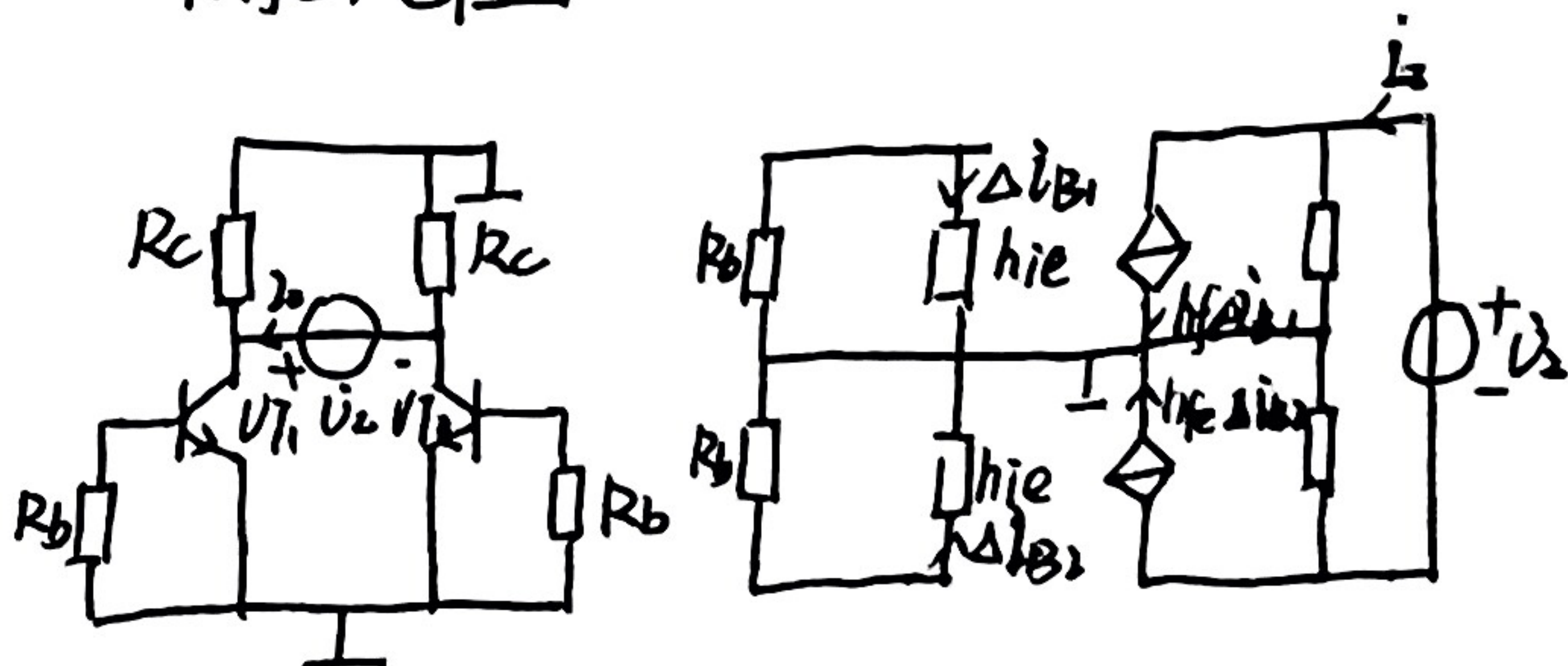
输出电阻



$$R_{id} = 2(R_b + h_{ie})$$



$$R_{ic} = \frac{1}{2}(R_b + h_{ie})$$



$$R_o = 2R_c$$

共模抑制比

$$K_{CMR} = \left| \frac{A_{ud}}{A_{uc}} \right|$$

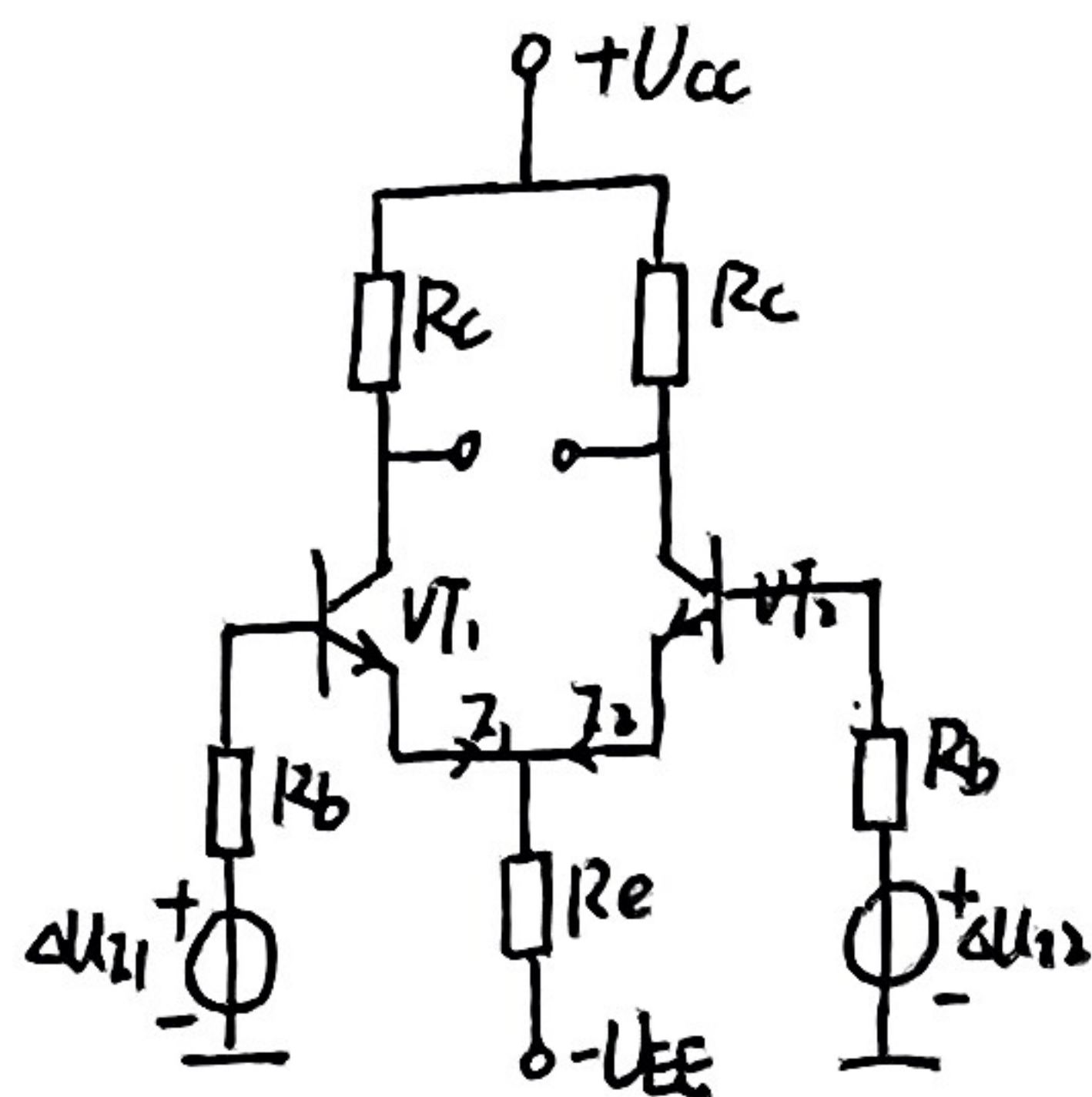
$$K_{CMR}(dB) = 20 \lg K_{CMR}$$

$$K_{CMR1,2} = \left| \frac{A_{ud1,2}}{A_{uc1,2}} \right|$$

双端输出 $K_{CMR} \rightarrow \infty$

单端输出 $K_{CMR} = 0.5$ 需提高

3. 双电源长尾式差分放大器 (提高单端输出性能)



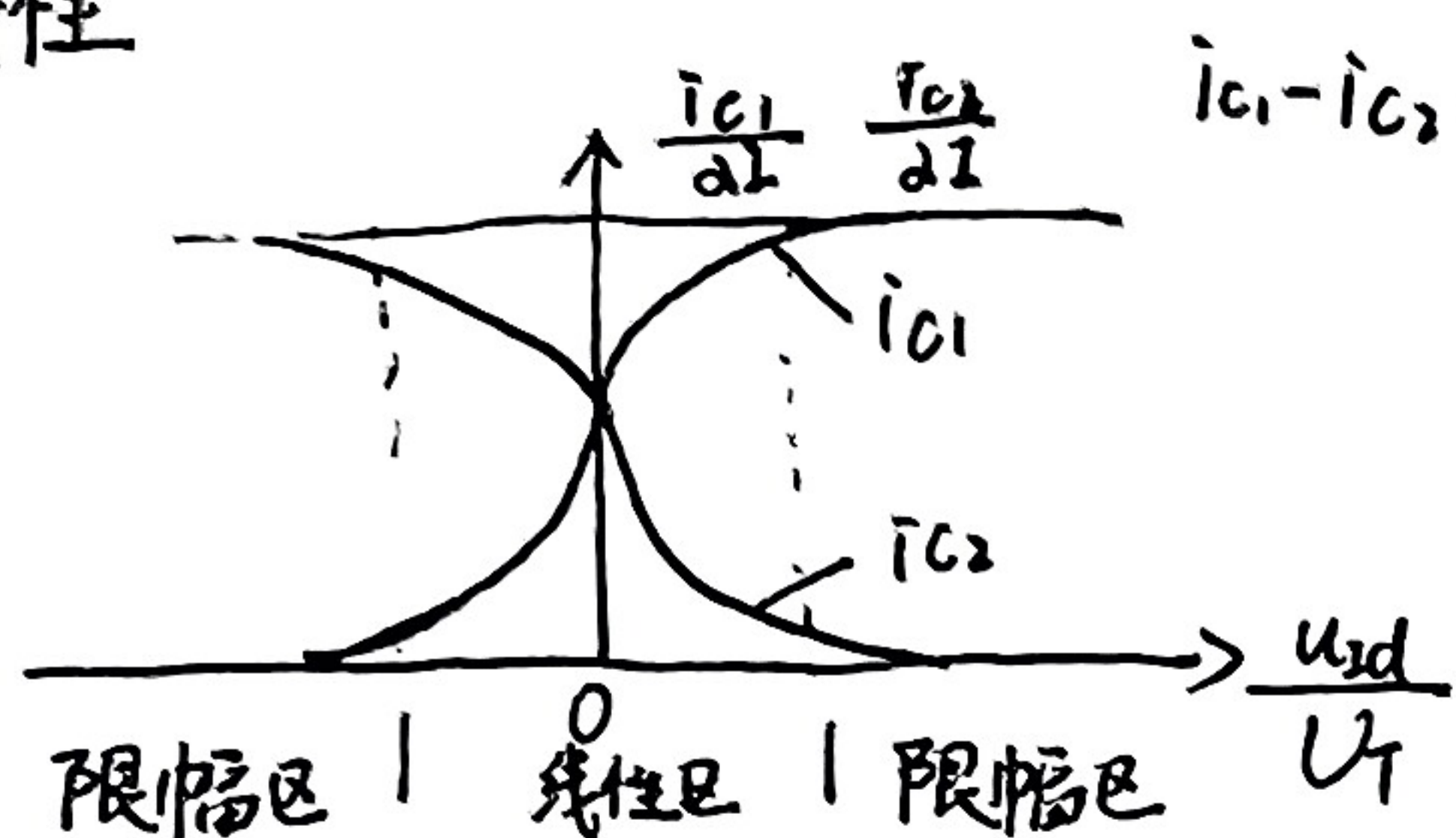
差模信号: $I_2 = -I_1$. R_e 上无电流 A_{ud} 不变

共模信号: $I_1 = I_2$ $U_e = 2I R_e = 2R_e \cdot I$

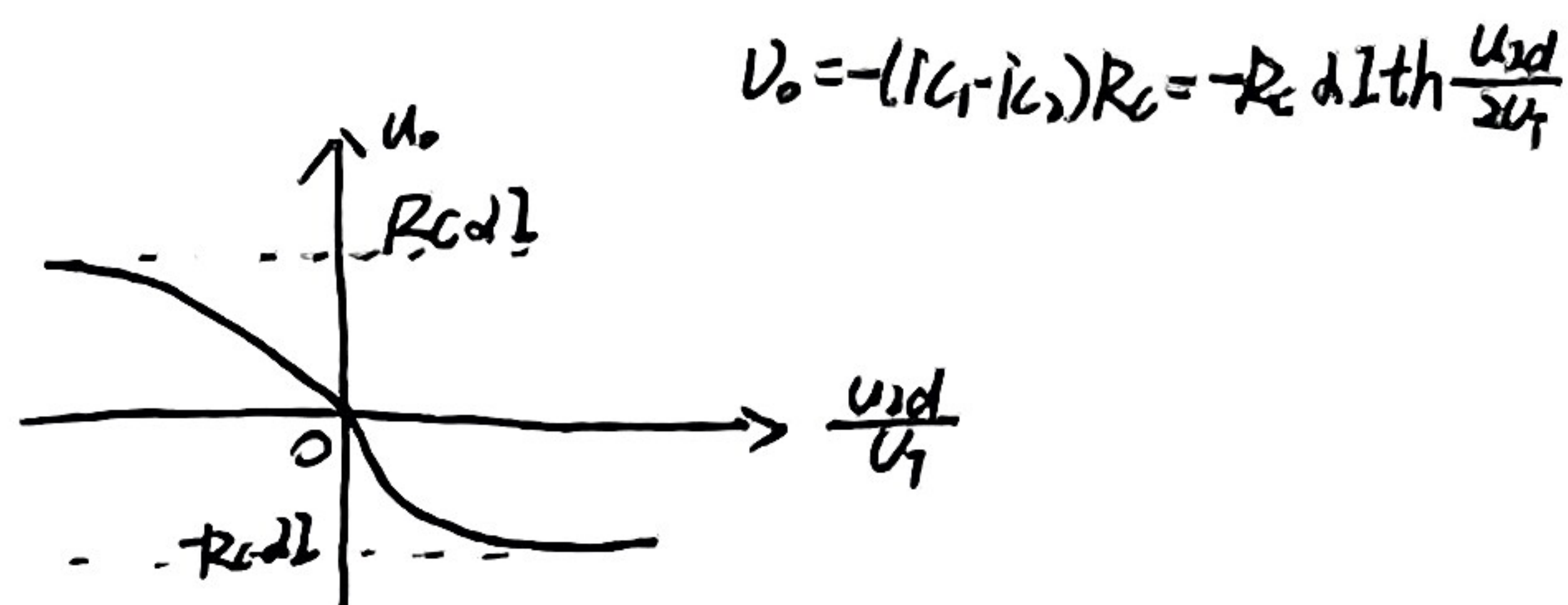
相当于两射极分别接 $2R_e$ 的电阻 ΔU_{be} 减小, A_{uc1} 减小

$$A_{uc1} = A_{uc2} \approx -\frac{R_c}{2R_e}$$

4. 传输特性



$$i_{C1} - i_{C2} = \alpha I_{th} \frac{u_{Id}}{2U_T}$$

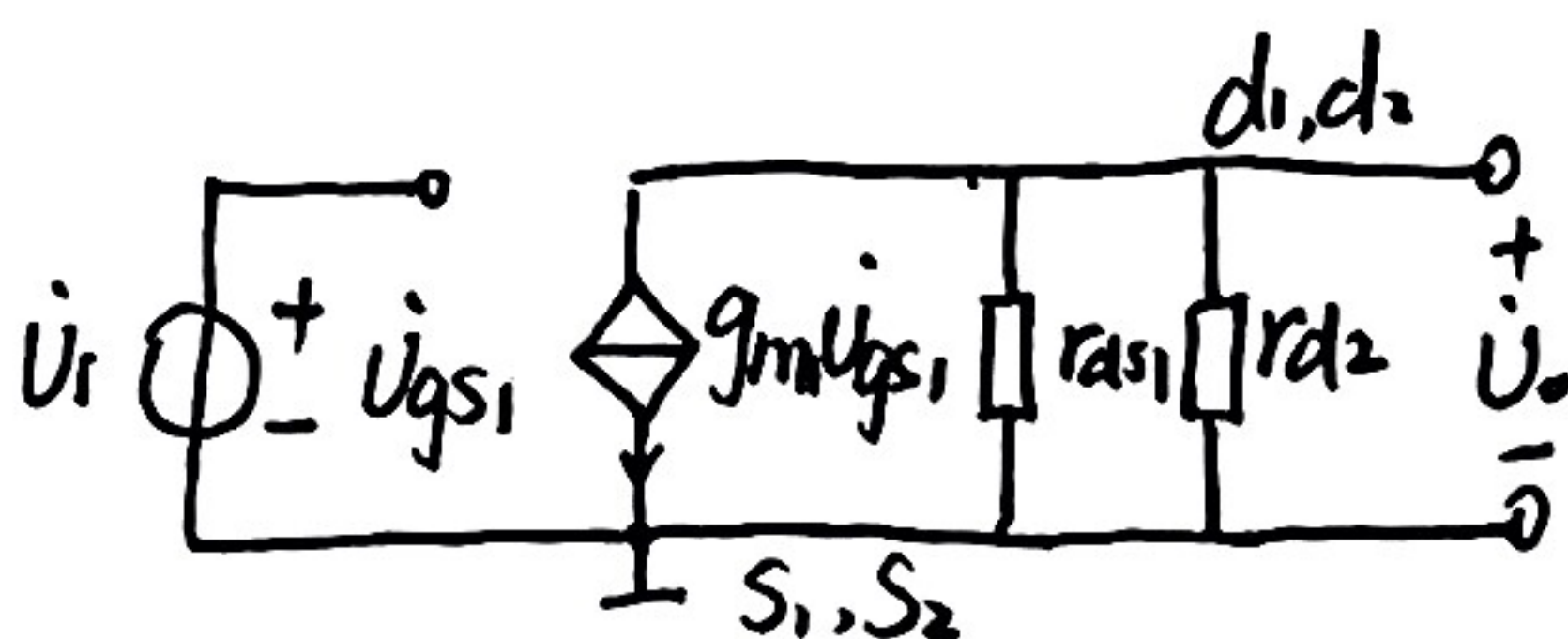
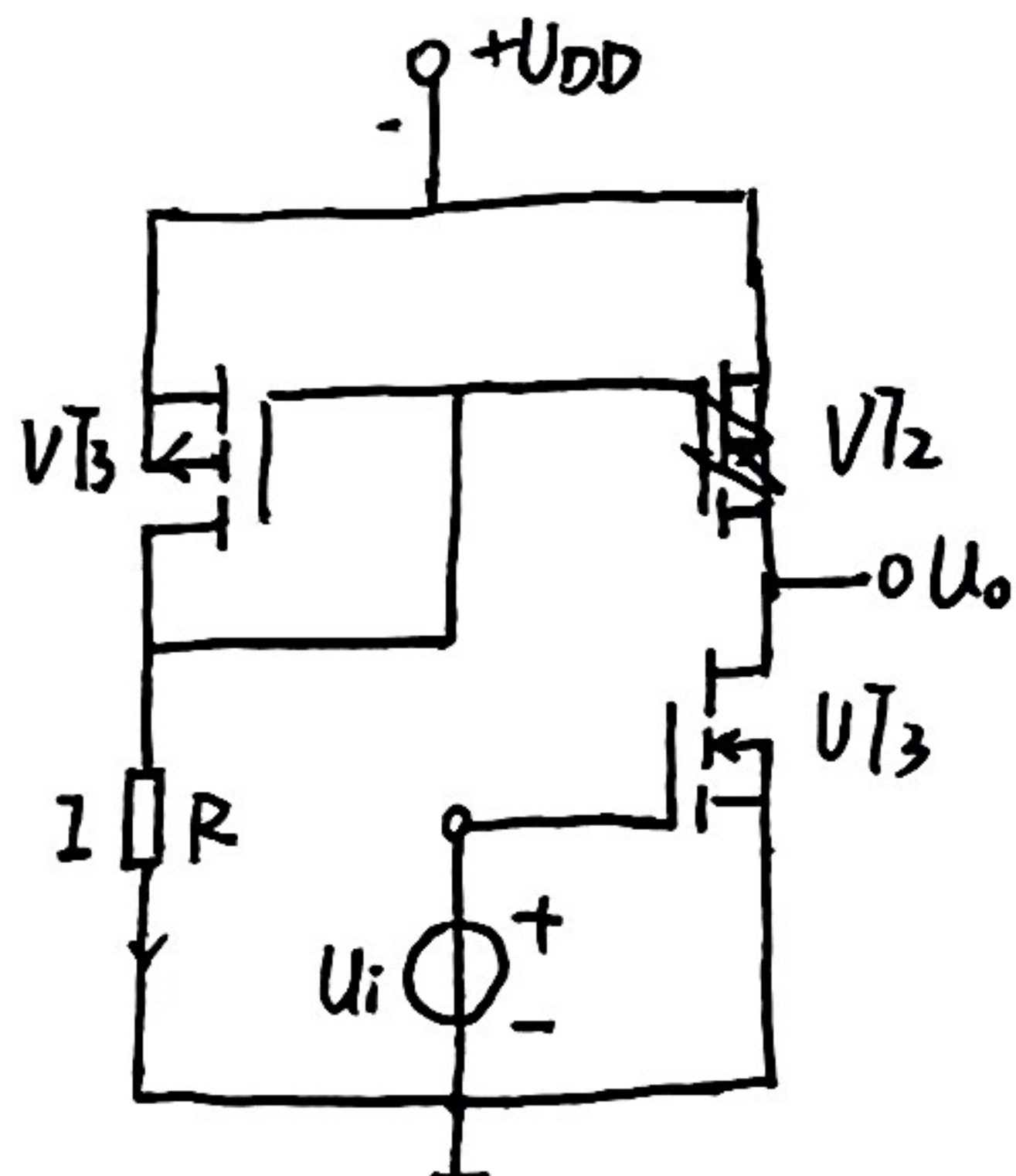


$$U_o = -(I_{C1} - I_{C2}) R_c = -R_c \alpha I_{th} \frac{u_{Id}}{2U_T}$$

扩大差分放大器的线性范围: 加入负反馈 (射极电阻)

5. 场效应管有源负载放大器

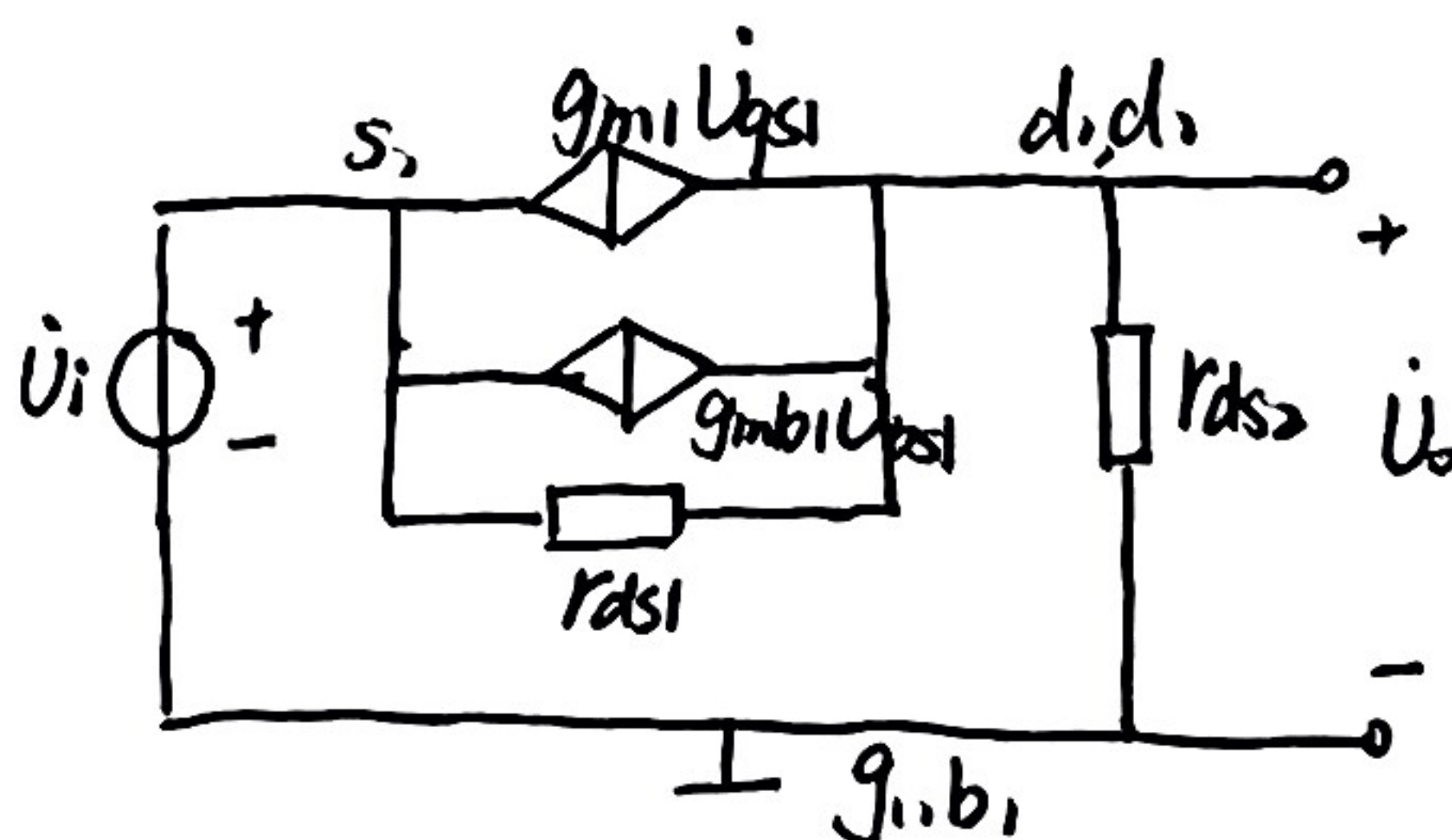
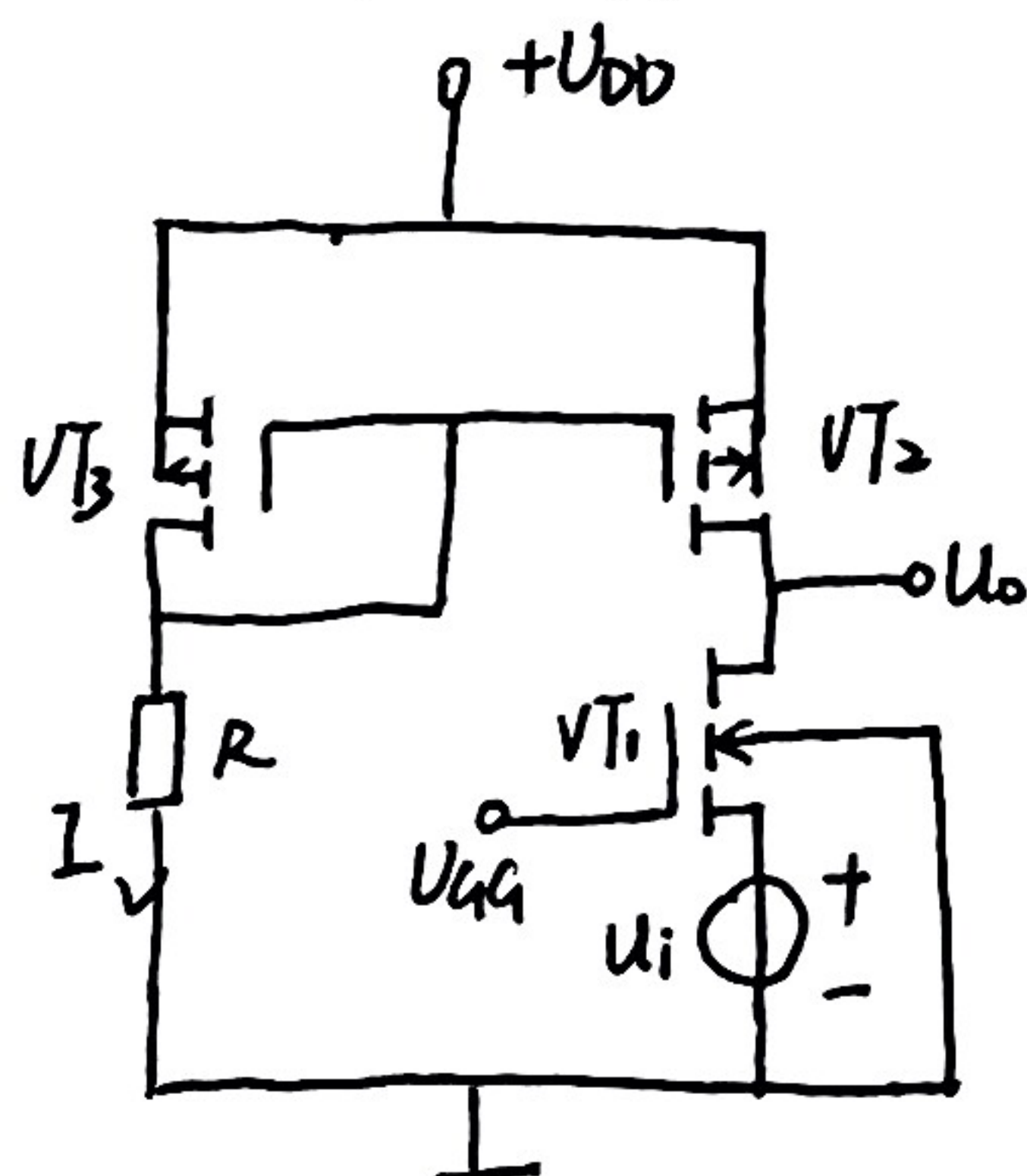
共源接法有源负载放大器



$$A_{um} = \frac{\dot{U}_o}{\dot{U}_i} = -g_m (r_{ds1} \parallel r_{ds2})$$

$$R_o = r_{ds1} \parallel r_{ds2}$$

共栅接法有源负载放大器



$$A_{um} = \frac{\dot{U}_o}{\dot{U}_i} = \frac{[1 + (g_{m1} + g_{mb1}) r_{ds1}] r_{ds2}}{r_{ds1} + r_{ds2}}$$

$$R_i = \frac{U_i}{I_i} = \frac{r_{ds1} + r_{ds2}}{1 + (g_{m1} + g_{mb1}) r_{ds1}}$$

$$R_o = r_{ds1} \parallel r_{ds2}$$

共漏接法有源负载放大器

第3章 放大器的频率特性

1. 线性失真 (电抗元件)

幅度失真: 与振幅频率特性有关 (放大器对不同频率分量的放大倍数不同) 通频带大于信号带宽
相位失真: 与相位 (滞后时间不相等)

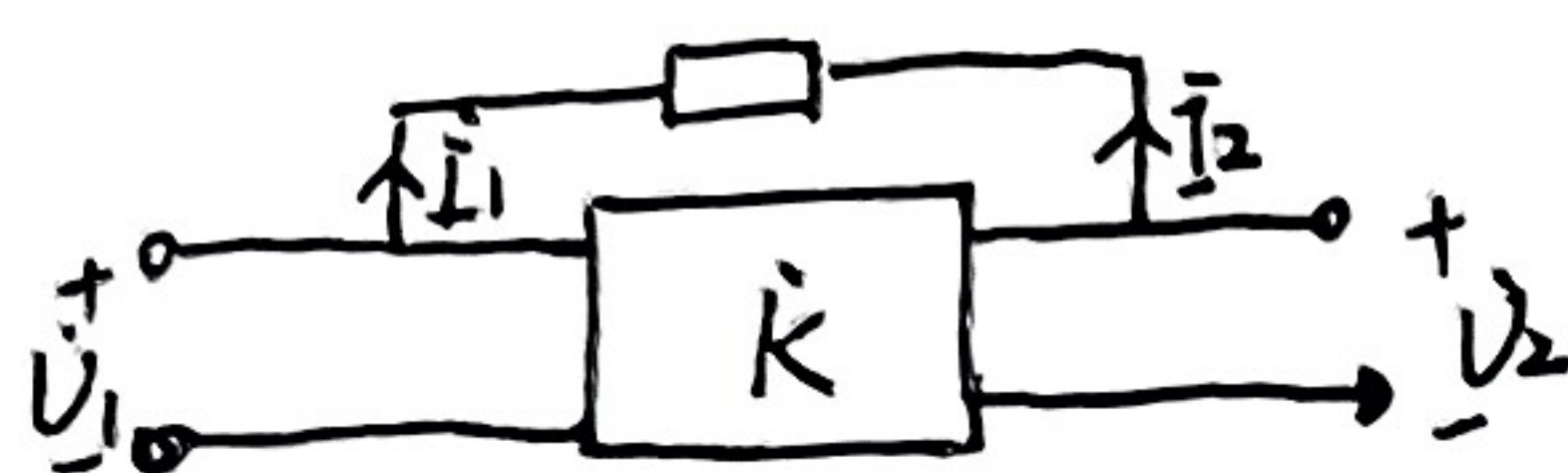
非线性失真: 非线性元件, 与信号幅度有关, 产生输入信号没有的新的频率成分

例: 判断失真类型

① 看最大不失真输出电压 $\hat{U}_o$ 是否大于输入电压 $\hat{U}_i$ 大于则发生非线性失真

② 看每个量的波的频率是否都在上、下限频率间, 不在则发生线性失真 (单频信号除外).

2. 密勒定理



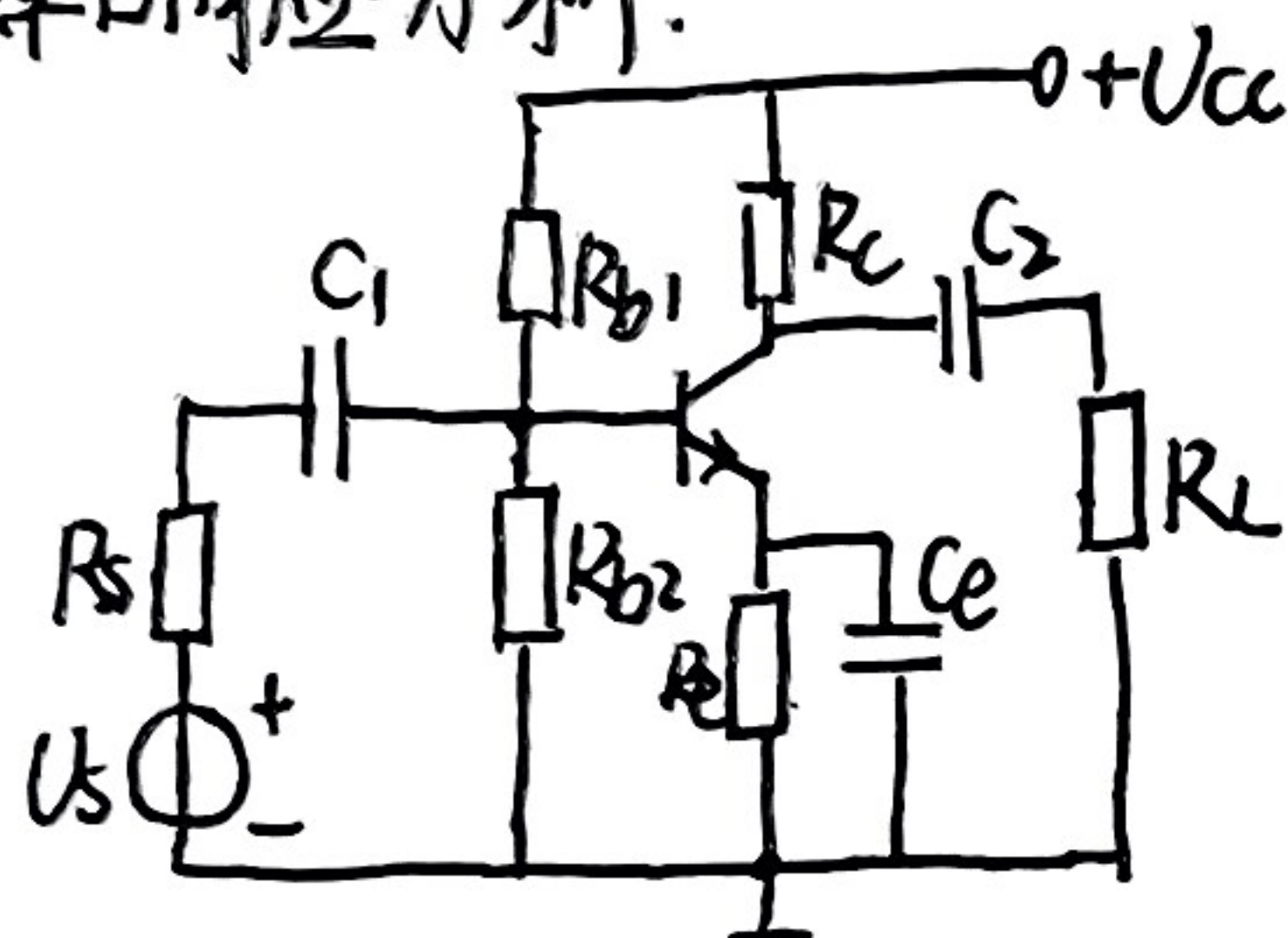
→



$$k = \frac{U_2}{U_1}$$

$$Z_1 = Z / (1 - k) \quad Z_2 = Z / (1 - 1/k)$$

3. 频率响应分析



C_1, C_2, C_e 容值较大

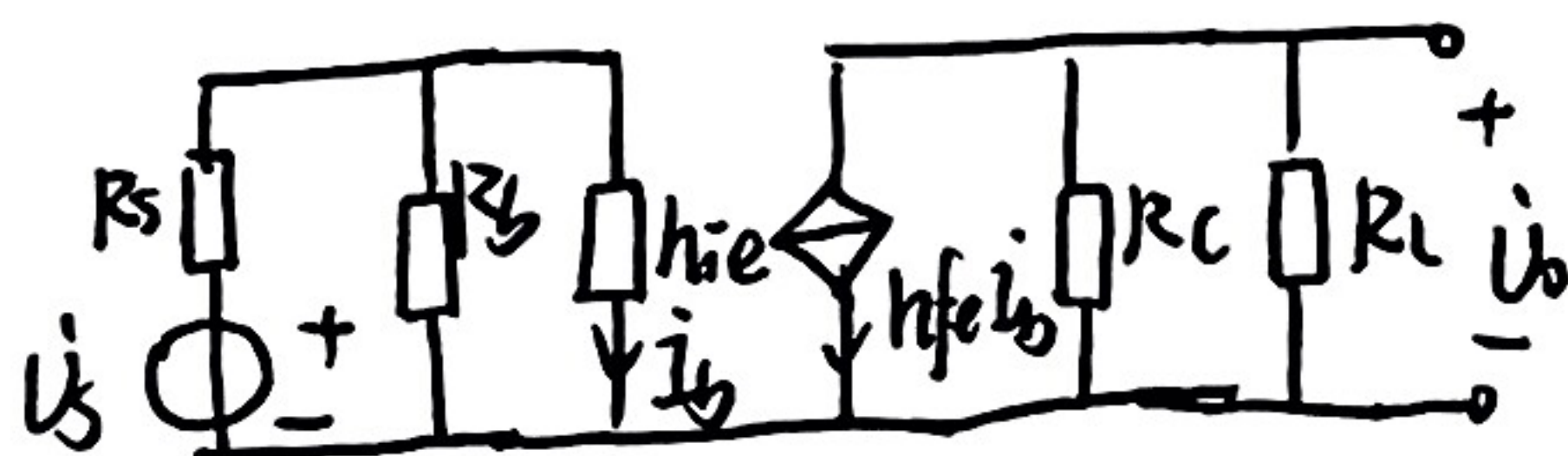
C_{be}, C_{bc} 容值较小

$$Z_c = \frac{1}{j\omega C}$$

中频

C_1, C_2, C_e 容量大 短路

C_{be}, C_{bc} 容量小 开路



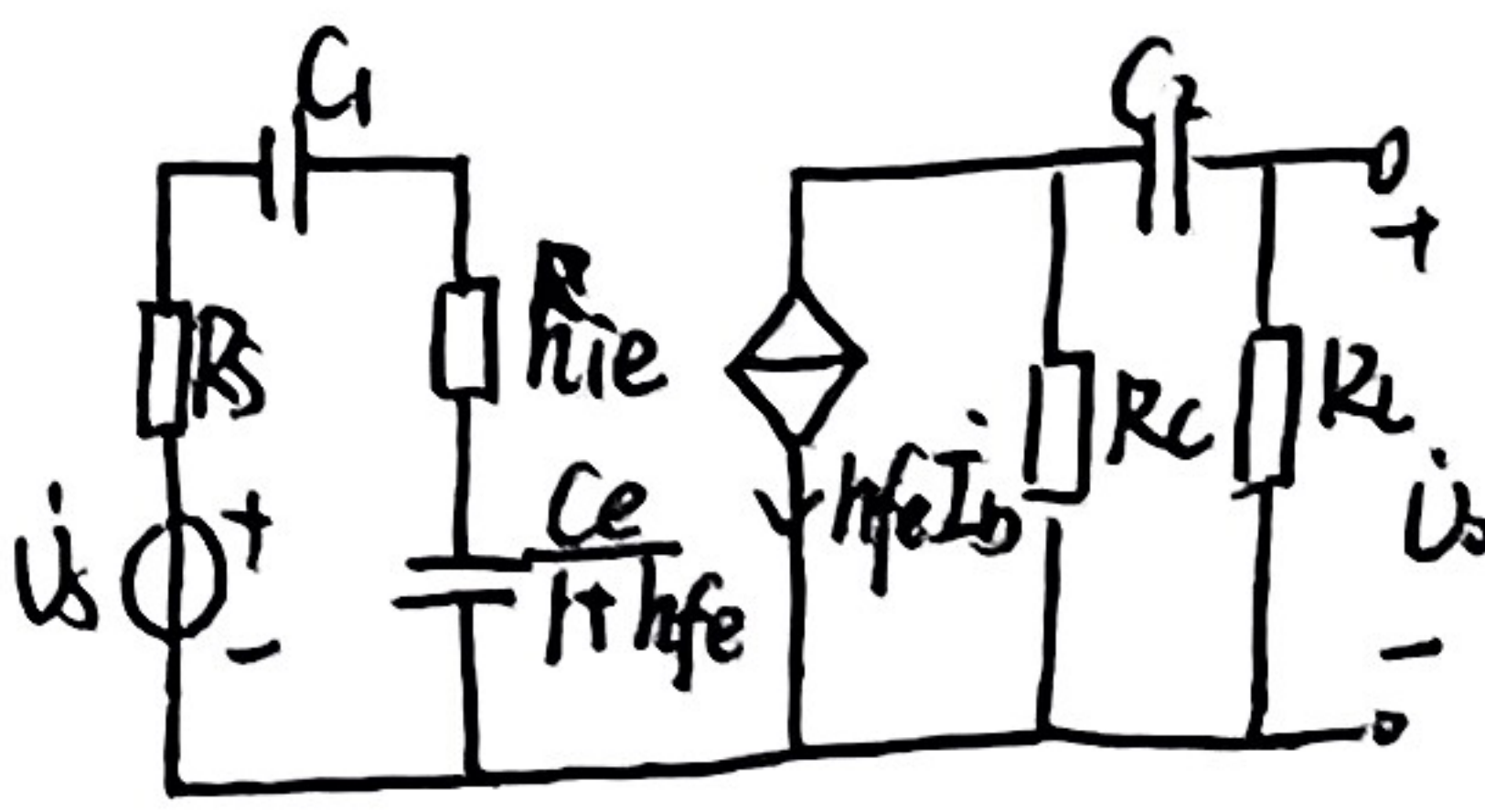
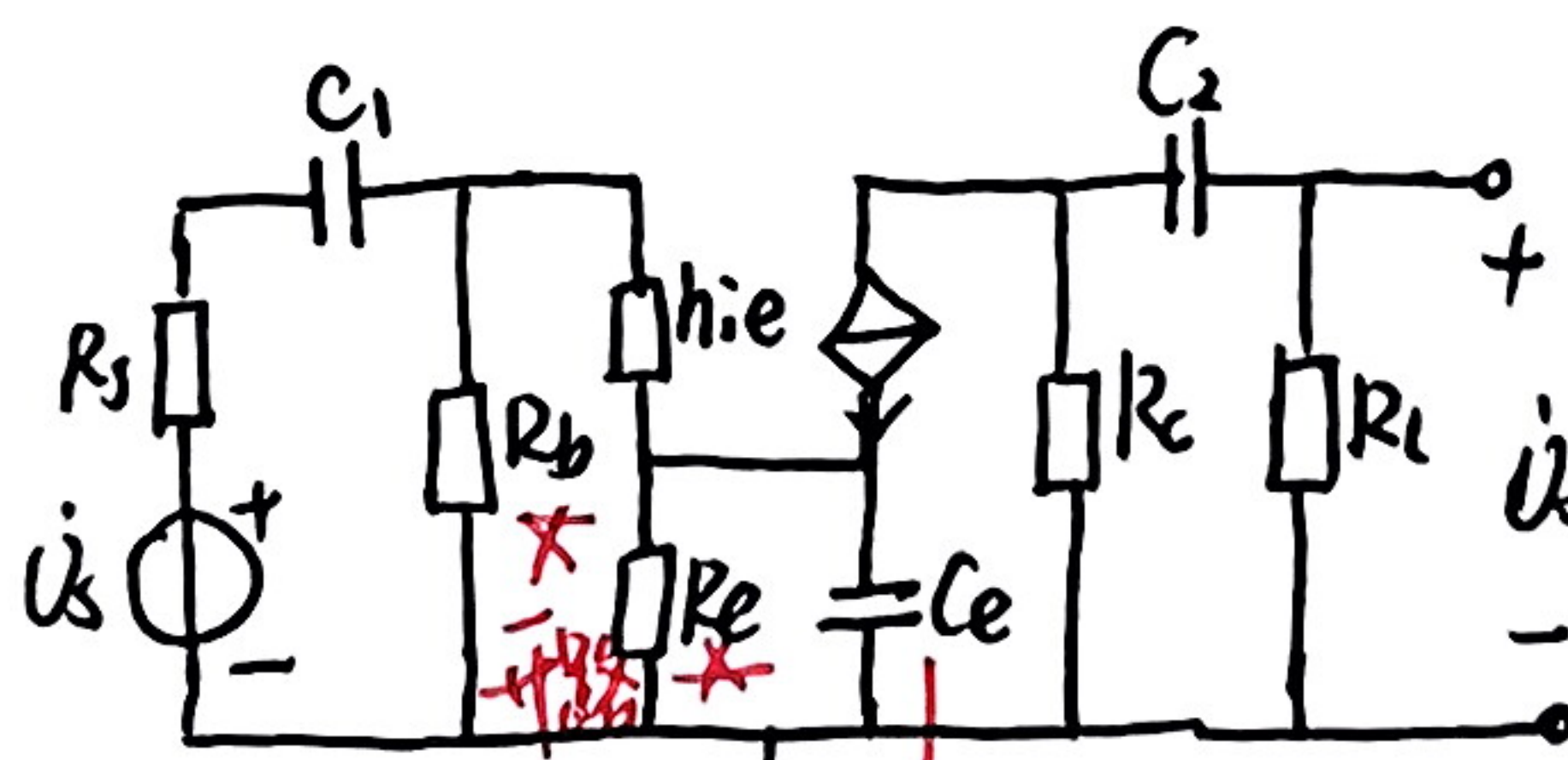
$$A_{usm} = \frac{U_o}{U_s} = \frac{U_o}{U_i} \frac{U_i}{U_s} = \frac{-h_{fe} R_c'}{h_{ie}} \frac{R_i}{R_s + R_i}$$

$$R_i = R_{b1} \parallel R_{b2} \parallel h_{ie}$$

低频 $\omega \downarrow$

对 C_{be}, C_{bc} $Z_c = \frac{1}{j\omega C}$ 开路

C_1, C_2, C_e 不可忽略.



$$\text{输入回路 } T_1 = (R_s + h_{ie}) C_1' \quad (C_1' = C_1 \parallel \frac{C_e}{1 + h_{fe}})$$

$$\text{输出回路 } T_2 = (R_c + R_L) C_2$$

$$A_{uL}(j\omega) = \frac{A_{usm}}{(1 + \frac{j\omega T_1}{1}) (1 + \frac{j\omega T_2}{1})}$$

$$C_e \rightarrow \infty \quad \omega_L = \frac{1}{(R_s + h_{ie}) C_1}$$

$$C_e, C_2 \rightarrow \infty \quad A_{uL}(j\omega) = \frac{A_{usm}}{(1 + \frac{j\omega T_1}{1})}$$

$$C_e, C_1 \rightarrow \infty \quad A_{uL}(j\omega) = \frac{A_{usm}}{1 + \frac{j\omega T_2}{1}}$$

求 f_L

① 先算 $\omega = \frac{1}{RC}$

② $f_L = \frac{1}{2\pi RC}$

③ 若 $T = \sqrt{T_1^2 + T_2^2}$

高频 $\omega \uparrow$

C_1, C_2, C_e 短路.

考虑 C_{be}, C_{bc}

4. 多级放大器的频率响应

画出增益函数, 分高频、低频写出增益的零点和极点, 化成标准形式

$$f_L = \frac{1}{2\pi \sqrt{\omega_{p1}^2 + \omega_{p2}^2 + \dots}} \quad f_H = \frac{1}{2\pi \sqrt{\omega_{z1}^2 + \omega_{z2}^2 + \dots}}$$

$$\text{例: 已知 } A_u(s) = \frac{2 \times 10^3 s(s+5)}{(s+10)(s+30)(s+10^5)(s+2 \times 10^6)} = \frac{10^2 (1 + \frac{s}{5})}{(1 + \frac{s}{10})(1 + \frac{s}{30})(1 + \frac{s}{10^5})(1 + \frac{s}{2 \times 10^6})}$$

低频 $1/s$ 高频 10^5 10^6

$$A_{uL}(j\omega) = \frac{100 (1 + \frac{j\omega}{5})}{(1 + \frac{j\omega}{10})(1 + \frac{j\omega}{30})}$$

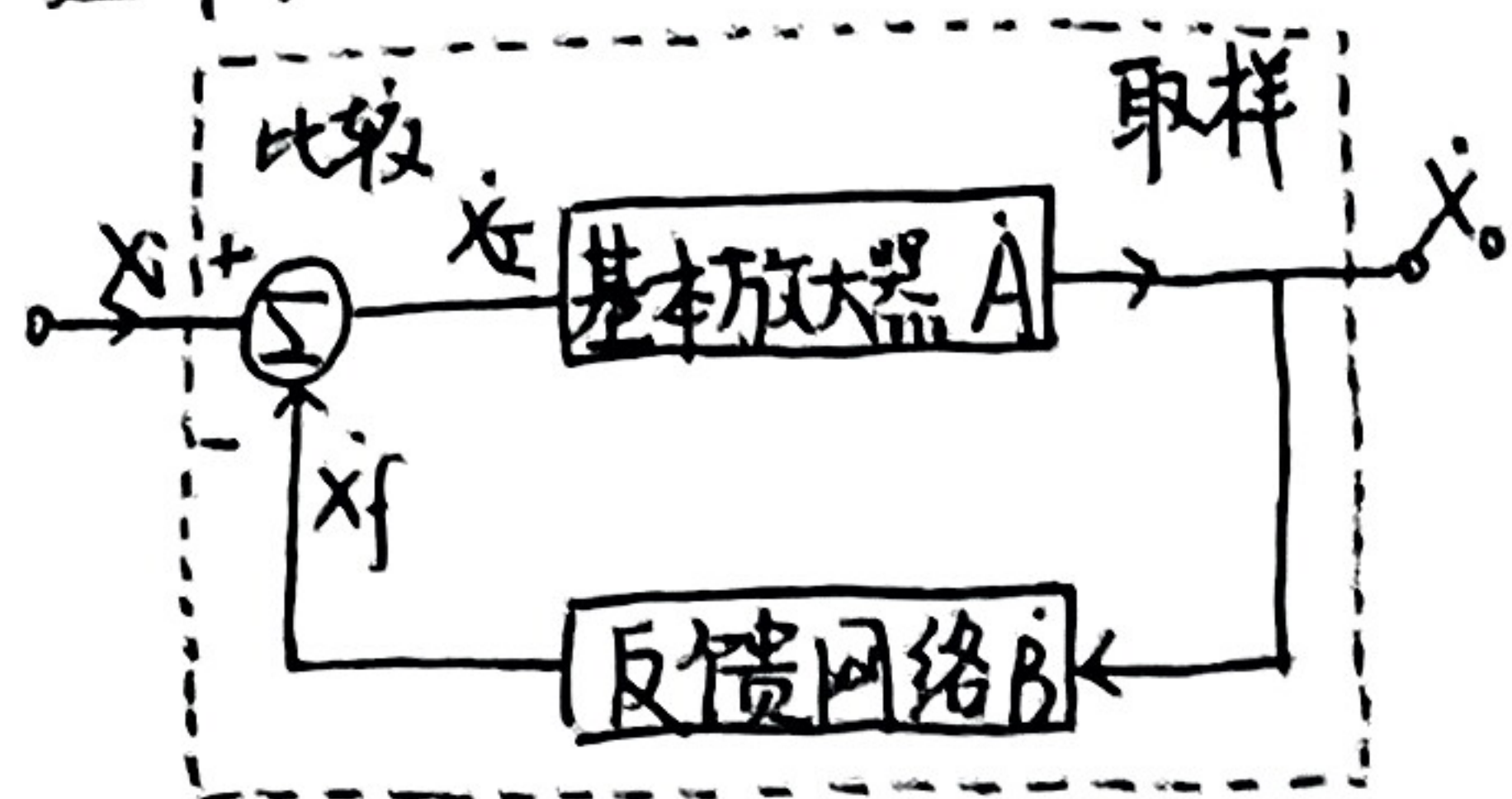
$$A_{uH}(j\omega) = \frac{100}{(1 + \frac{j\omega}{10^5})(1 + \frac{j\omega}{2 \times 10^6})}$$

$$A_{um} = 100 \quad f_L = \frac{1}{2\pi} \sqrt{10^2 + 30^2 - 2 \times 5^2}$$

$$f_H = \frac{1}{2\pi \sqrt{(10^5)^2 + (2 \times 10^6)^2}}$$

第4章 负反馈放大器

1. 基本概念



$$A = \frac{\dot{X}_o}{\dot{X}_e} \quad \text{开环增益}$$

$$\beta = \frac{\dot{X}_f}{\dot{X}_o} \quad \text{闭环...}$$

$$A_f = \frac{\dot{X}_o}{\dot{X}_i}$$

2. 影响

提高放大倍数的稳定性 反馈深度 $F = 1 + AB$

深负: $F \gg 10, A_f \approx \frac{1}{\beta}$

展宽了通频带, 减少线性失真

减小了非线性失真

抑制内部噪声和干扰

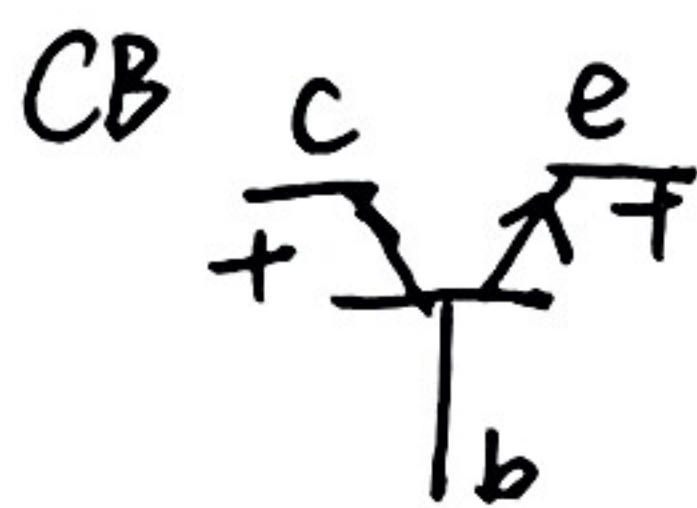
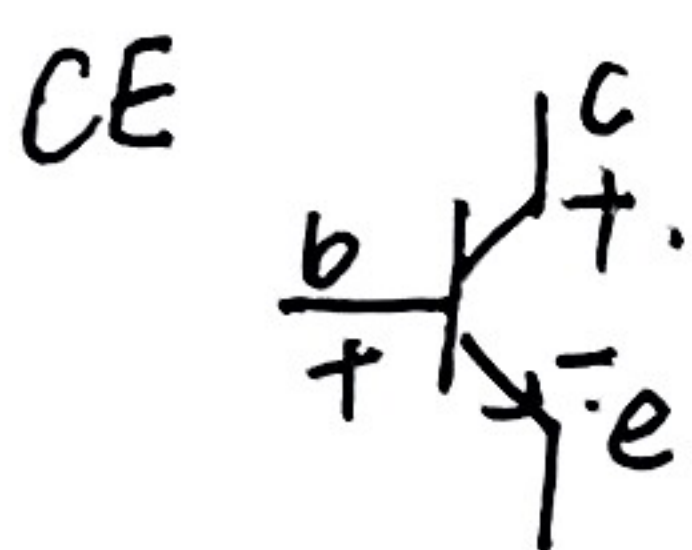


①. U_o

②. U_f 常数

③. $U_i = U_o + U_f$

3. 反馈的判断和引入



差分的两个输出-正-负

4. 负反馈放大器的分析方法

深负: $A_f \approx \frac{1}{\beta}$

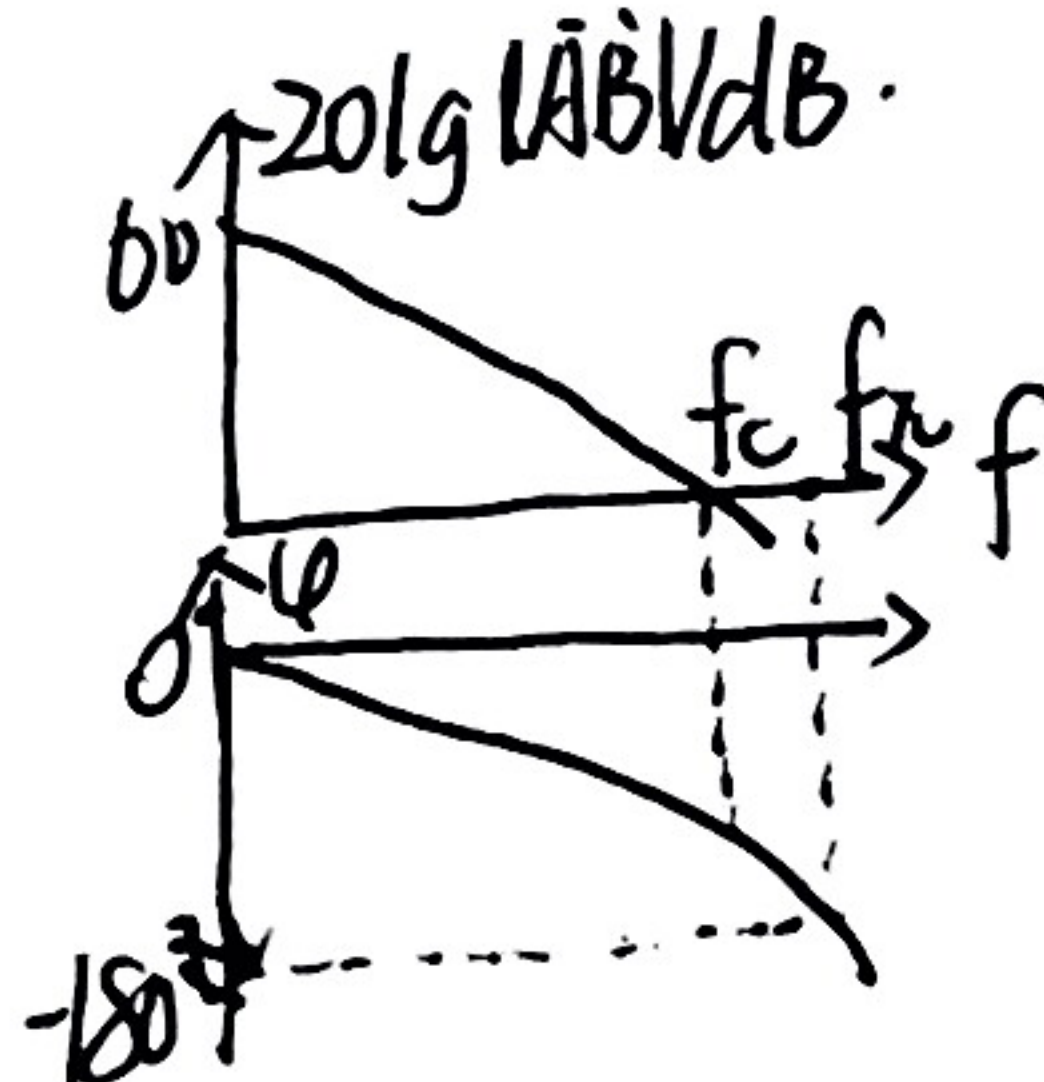
5. 稳定性

自激条件 $A_f = \frac{\dot{A}}{1 + \dot{A}\dot{B}}$

$$1 + \dot{A}\dot{B} = 0 \rightarrow A_f = \infty$$

$$\begin{cases} |\dot{A}\dot{B}| = 1 \\ \arg \dot{A}\dot{B} = \pm(2n+1)\pi \end{cases}$$

判定



增益交界频率

$f_c < f_p$ 稳定
相位交界频率

相位裕度 $\varphi_m = 180^\circ - |\varphi_{AB}(f_c)| \geq 45^\circ$

增益裕度 $G_m = 20\lg|\dot{A}(f_p)\dot{B}(f_p)| \leq 10\text{dB}$

第5章 低频功率放大器

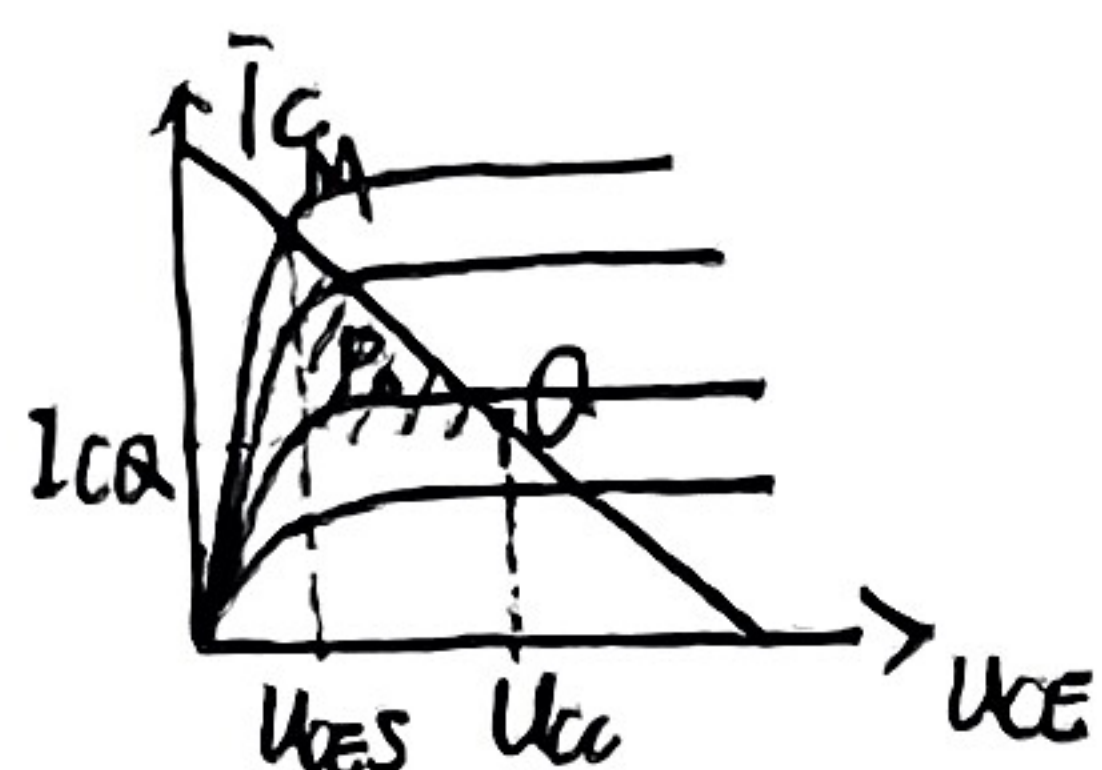
1. 分类: 导通时间

甲 乙 甲乙 丙 丁
- 直通 - 半

$$\eta_{甲} < \eta_{甲乙} < \eta_{乙} < \eta_{丙} < \eta_{丁}$$

50% 78.5%

2. 甲类工作原理

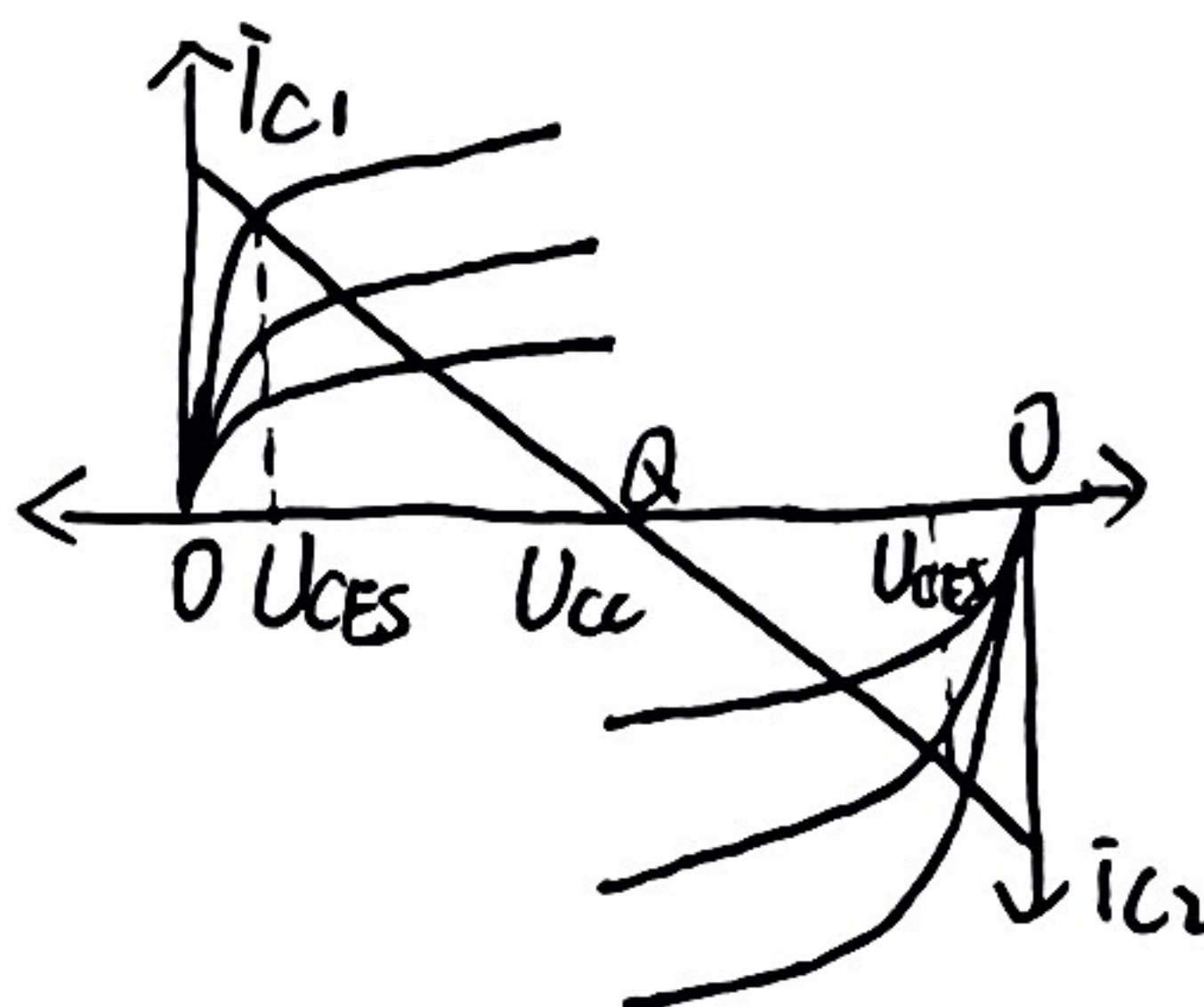
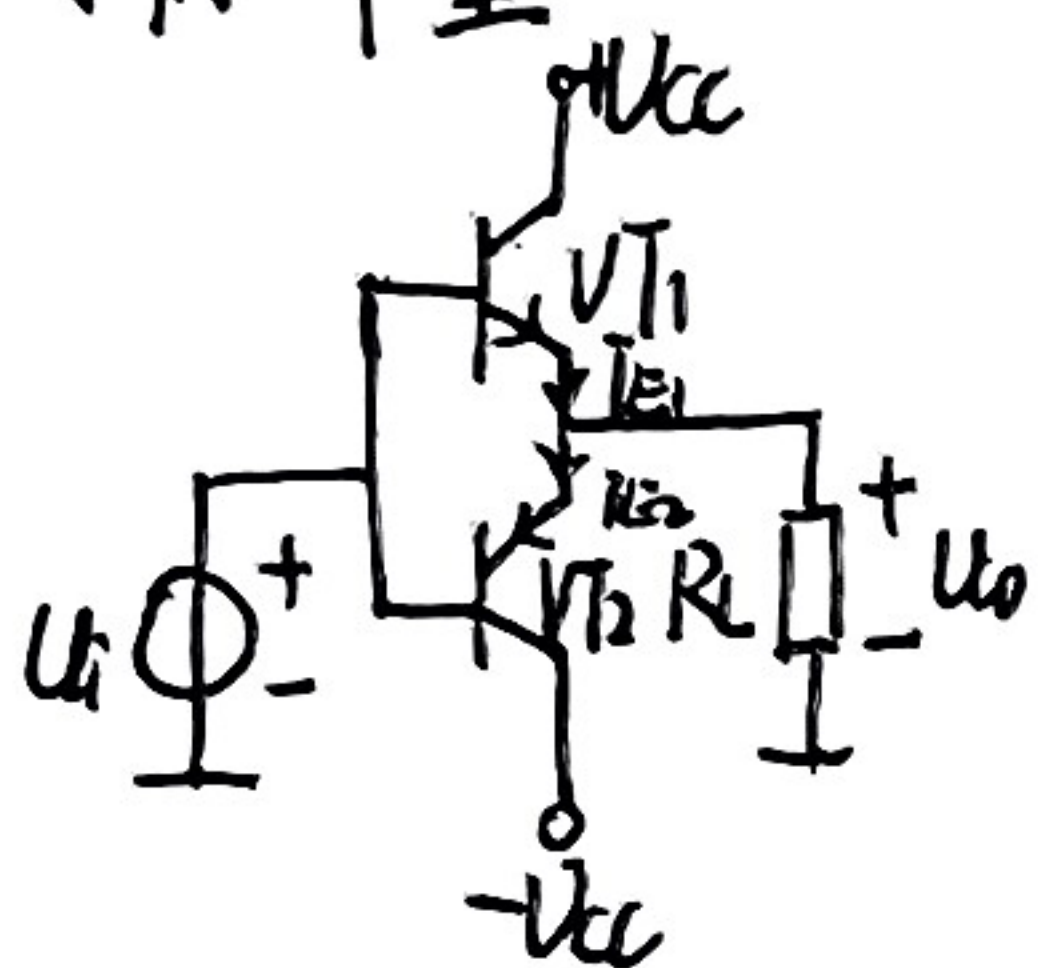


$$P_o = \frac{1}{2} I_{cem} U_{cem} = \frac{1}{2} \frac{U_{cem}^2}{R_L} = \frac{1}{2} \frac{(U_{cc} - U_{ces})^2}{R_L}$$

$$P_{omax} = \frac{1}{2} \frac{U_{cc}^2}{R_L}$$

3. 乙类推挽功率放大器

对称异型



$$P_o = \frac{1}{2} I_{cm} U_{cem} = \frac{1}{2} \frac{U_{cc}^2}{R_L} \eta^2$$

$$\eta = \frac{U_{cem}}{U_{cc}}$$

$$P_u = 2 I_o U_{cc} = \frac{2 I_{cm} U_{cc}}{\pi} = \frac{2}{\pi} \frac{U_{cc}^2}{R_L} \eta$$

$$\eta = \frac{P_o}{P_u} = \frac{\pi}{4} \eta \quad \eta_{max} = 78.5\%$$

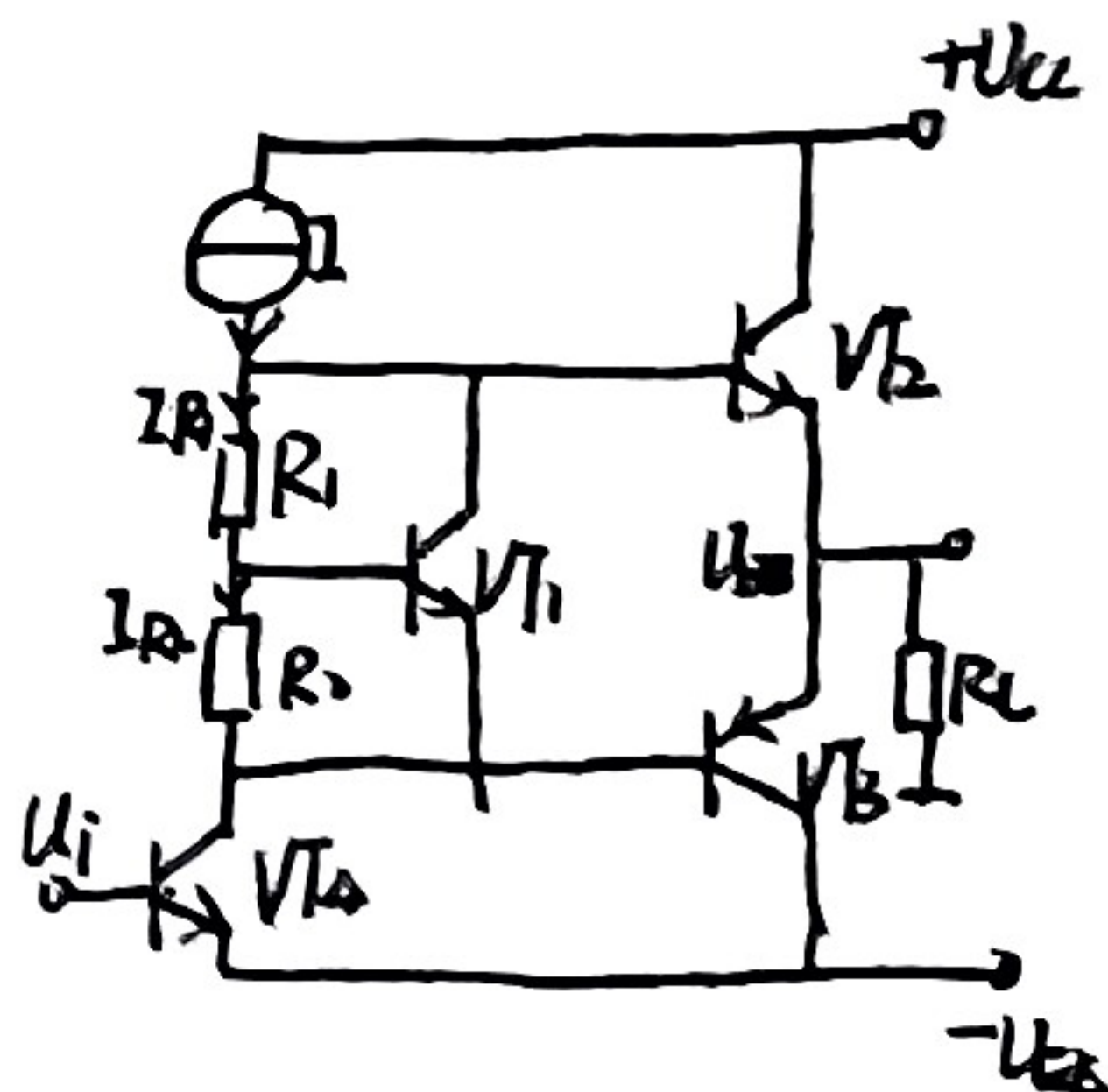
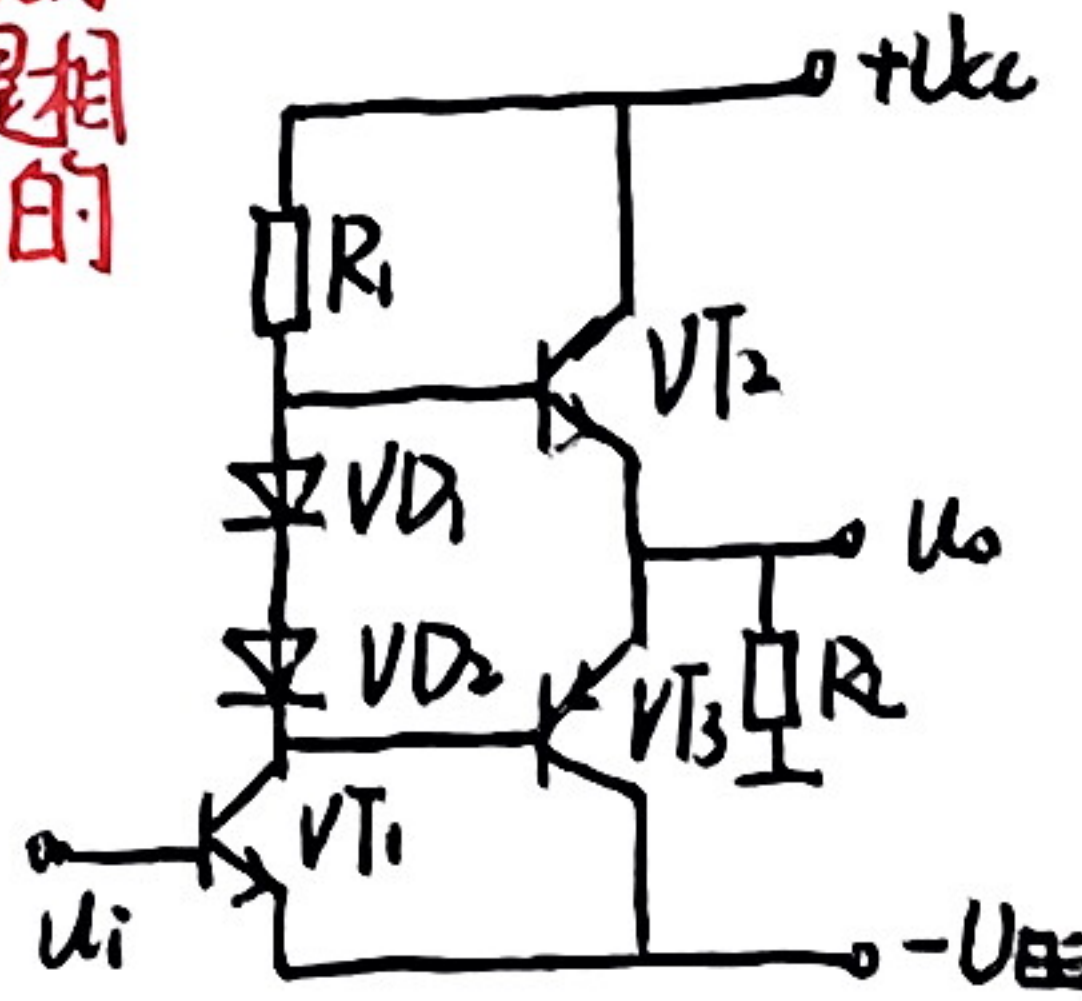
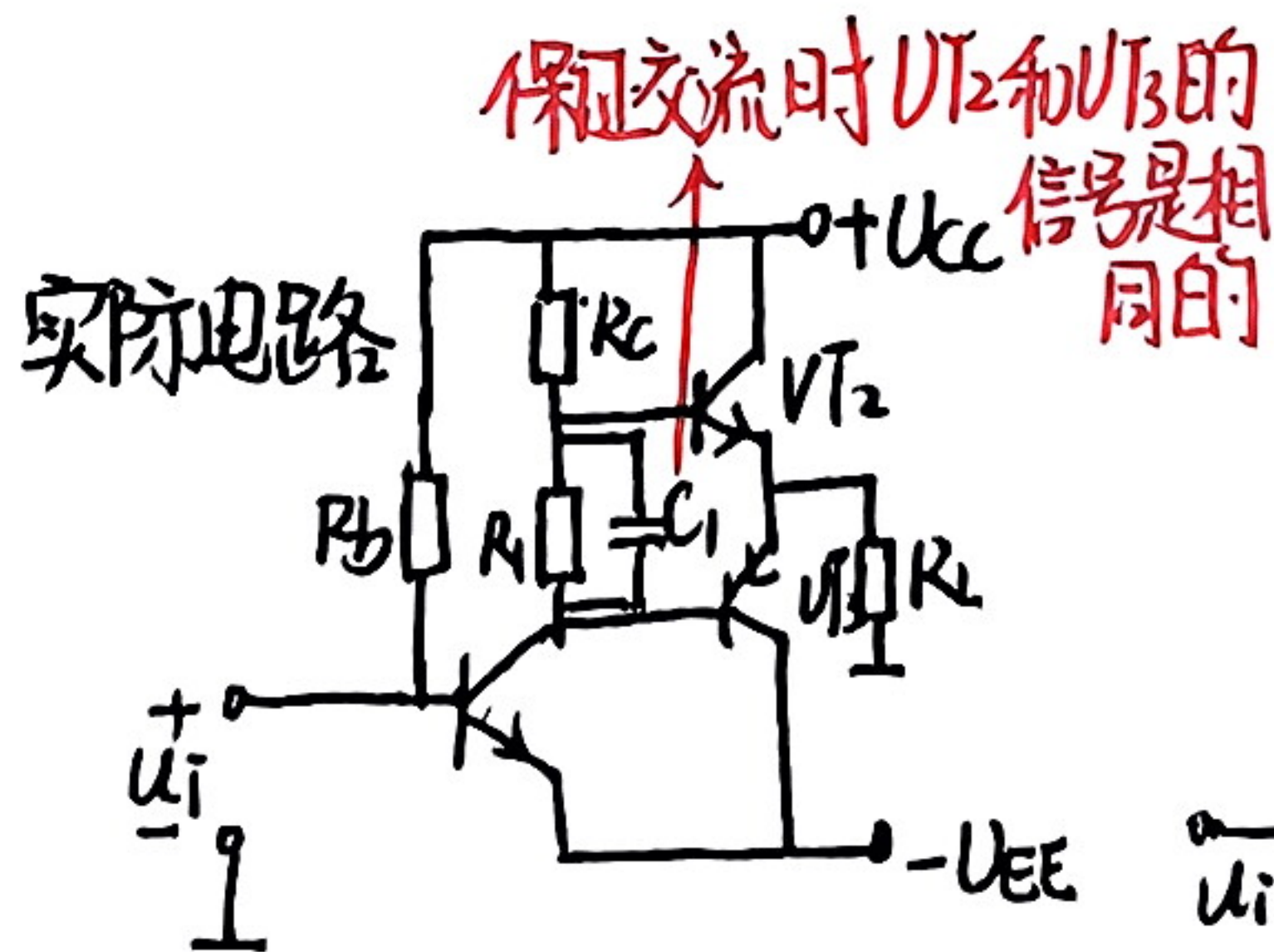
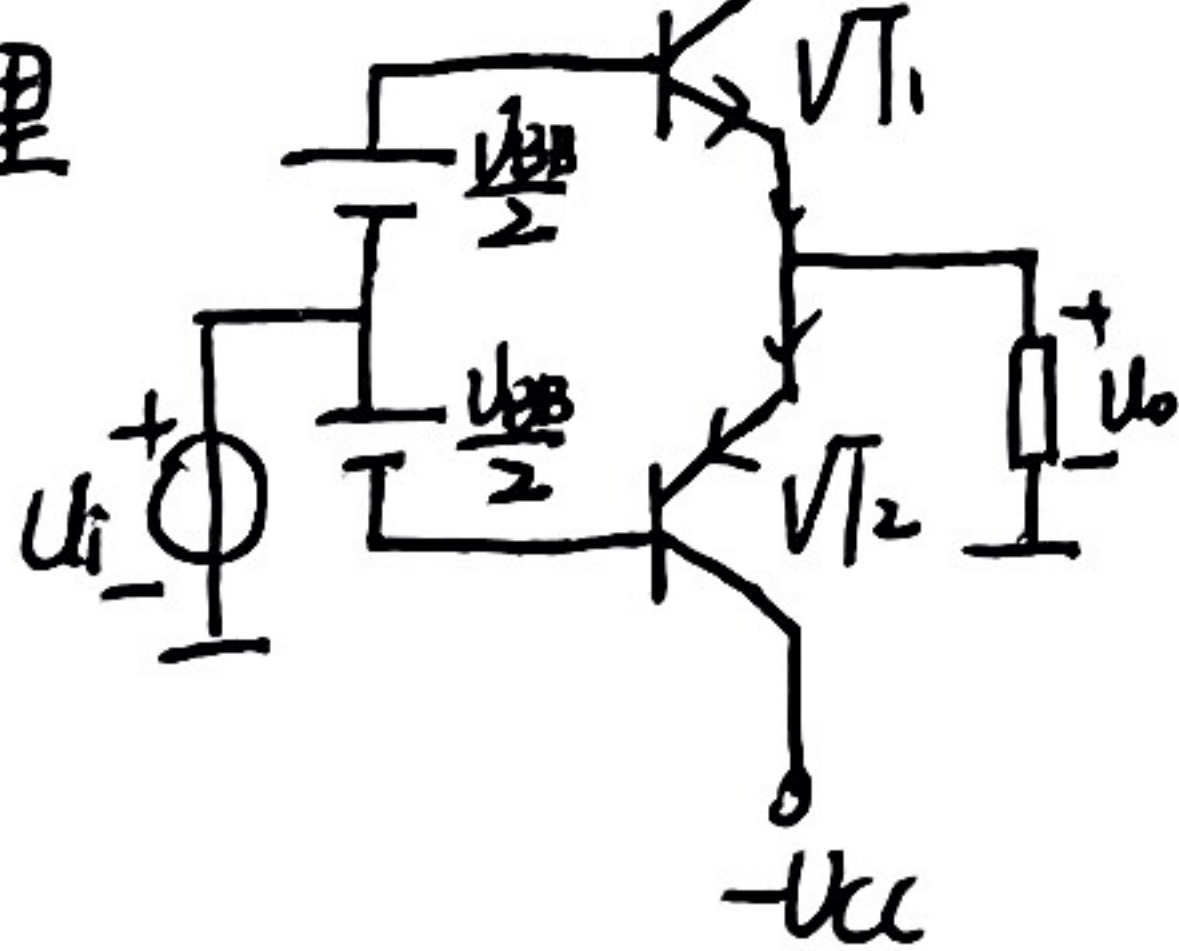
选择依据

- 集电极功耗 $P_{CM} > 0.2 P_{omax}$
- 反向击穿电压 $U_{BR}(CEO) > 2U_{cc}$
- 集电极最大允许电流 $I_{cm} > \frac{U_{cc}}{R_L}$

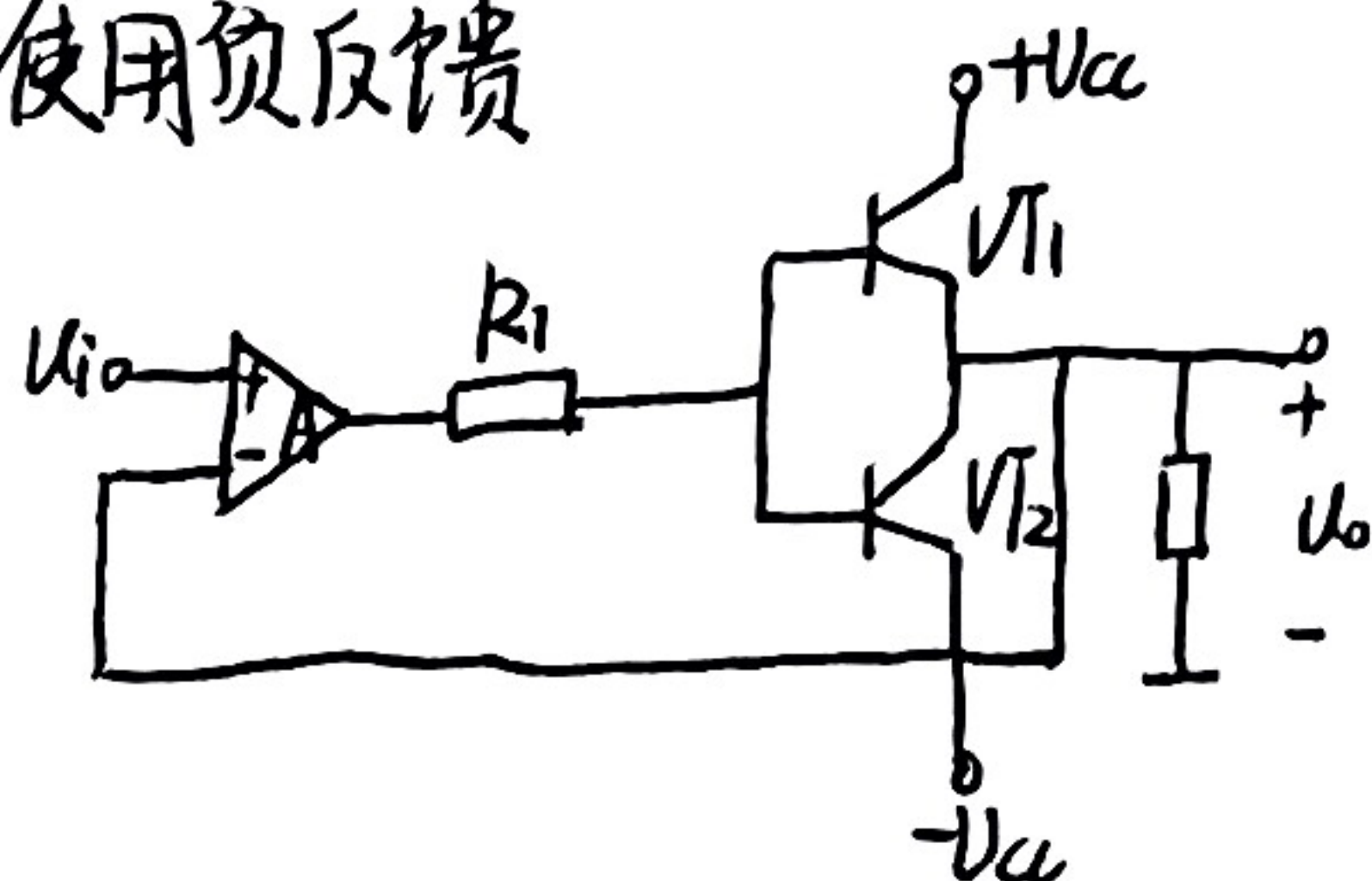
4. 消除交越失真

改为甲乙类工作状态

原理



使用负反馈



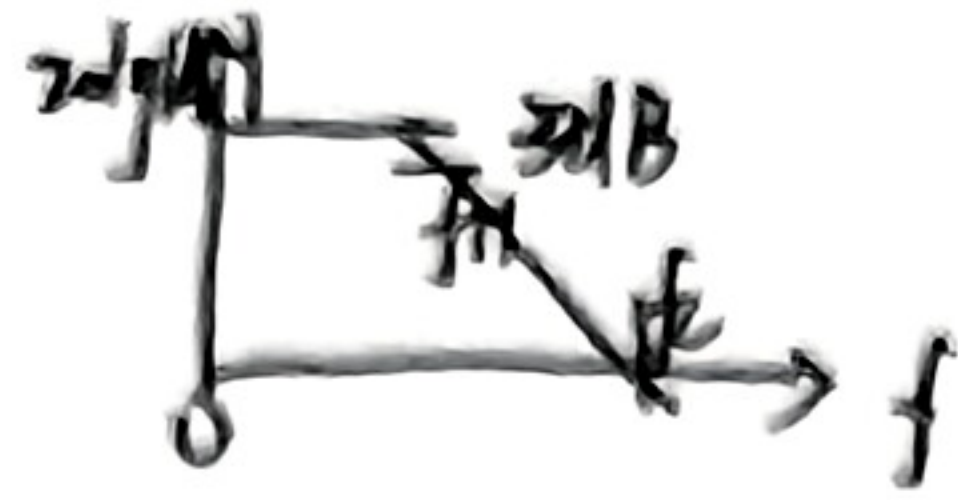
第6章 集成运算放大器原理及其应用

1. 性能参数

额定输出电压 U_{om} (最大不失真输出电压): 比电源电压低 1-2V $U_{om} = U_{cc} - (1/2)U$

-3dB 带宽 f_H

单位增益带宽 f_c



2. 基本特性和基本组态

特性: 线性放大

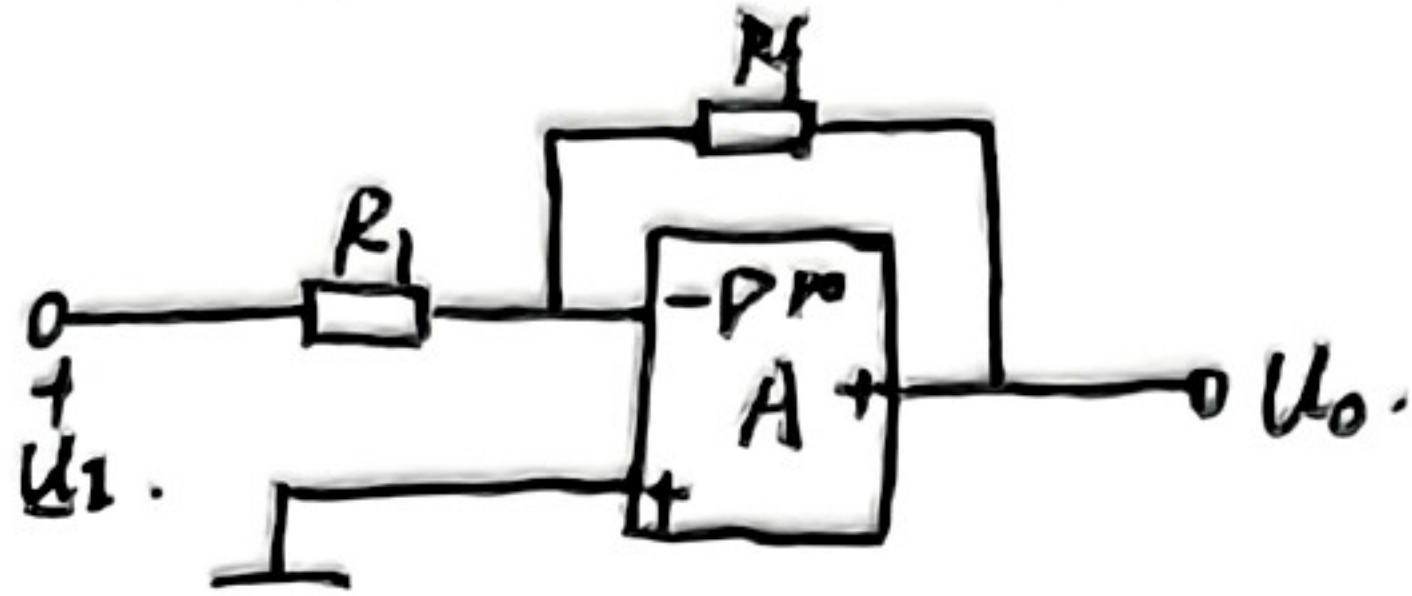
饱和(非线性)

虚断: $i_i = 0$ 两个输入端无电流, 开路 $\leftrightarrow i_i = 0$

虚短: $U_{id} = U_+ - U_- = 0 \quad U_+ = U_-$

$\leftrightarrow U_+ > U_- \quad U_o = U_{omax}; U_+ < U_- \quad U_o = U_{omin}$

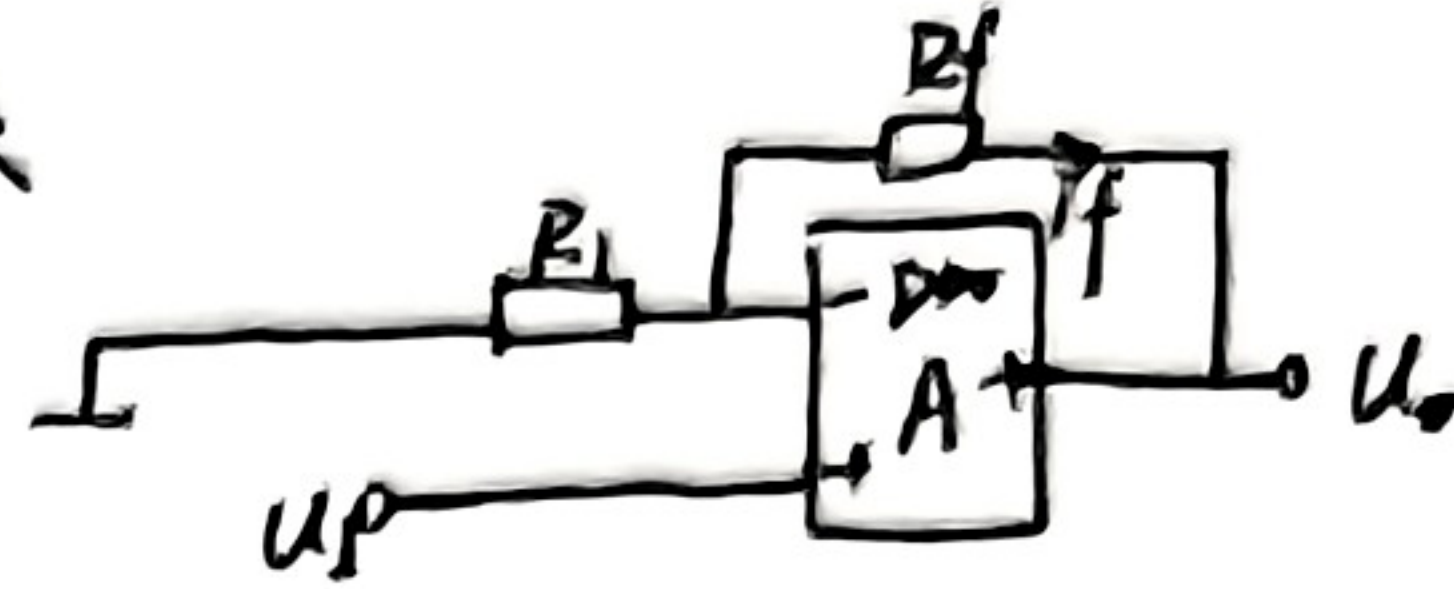
反相放大



输入 "-"

$$A_{uf} = -\frac{R_f}{R_1}$$

同相放大



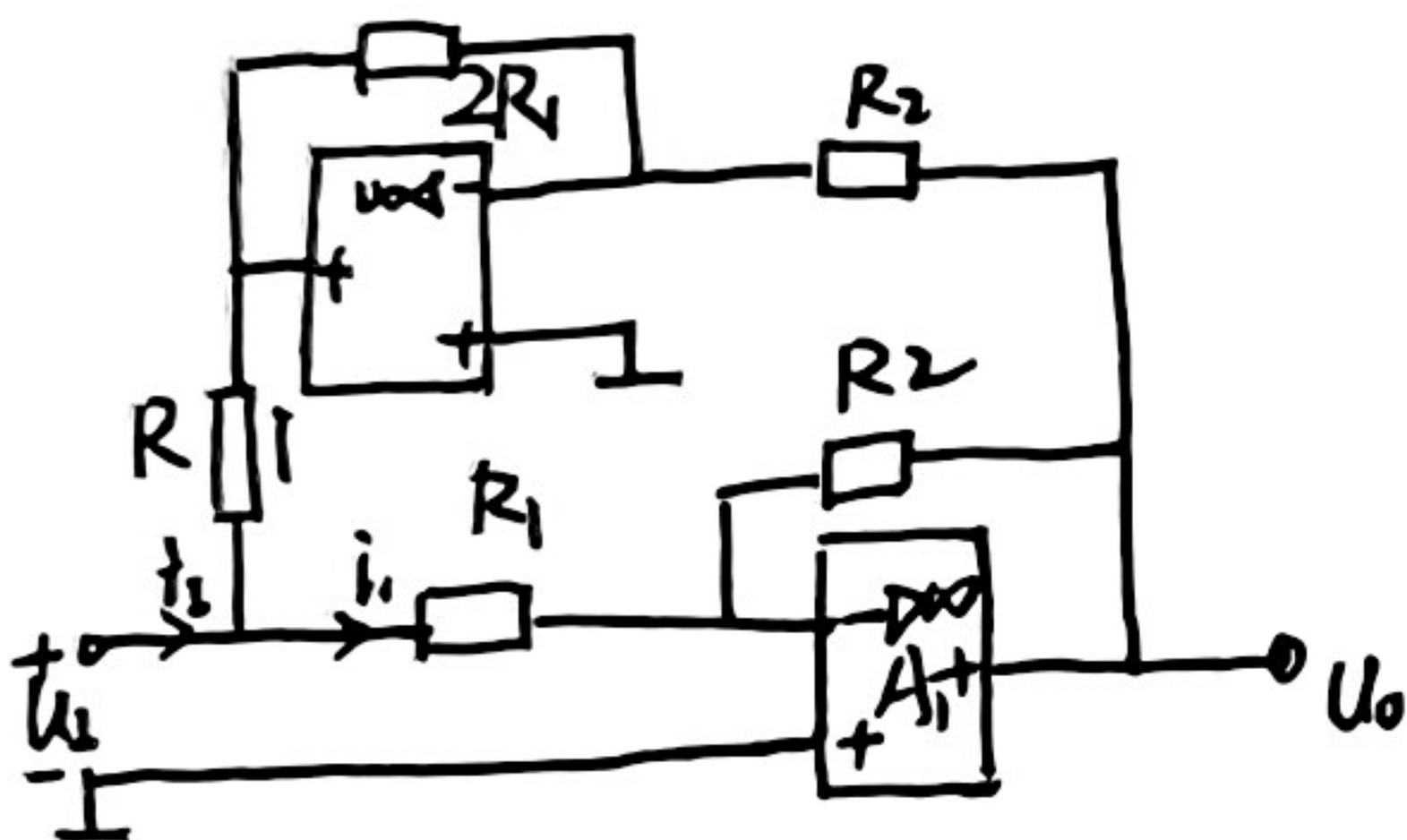
输入 "+"

$$A_{uf} = 1 + \frac{R_f}{R_2}$$

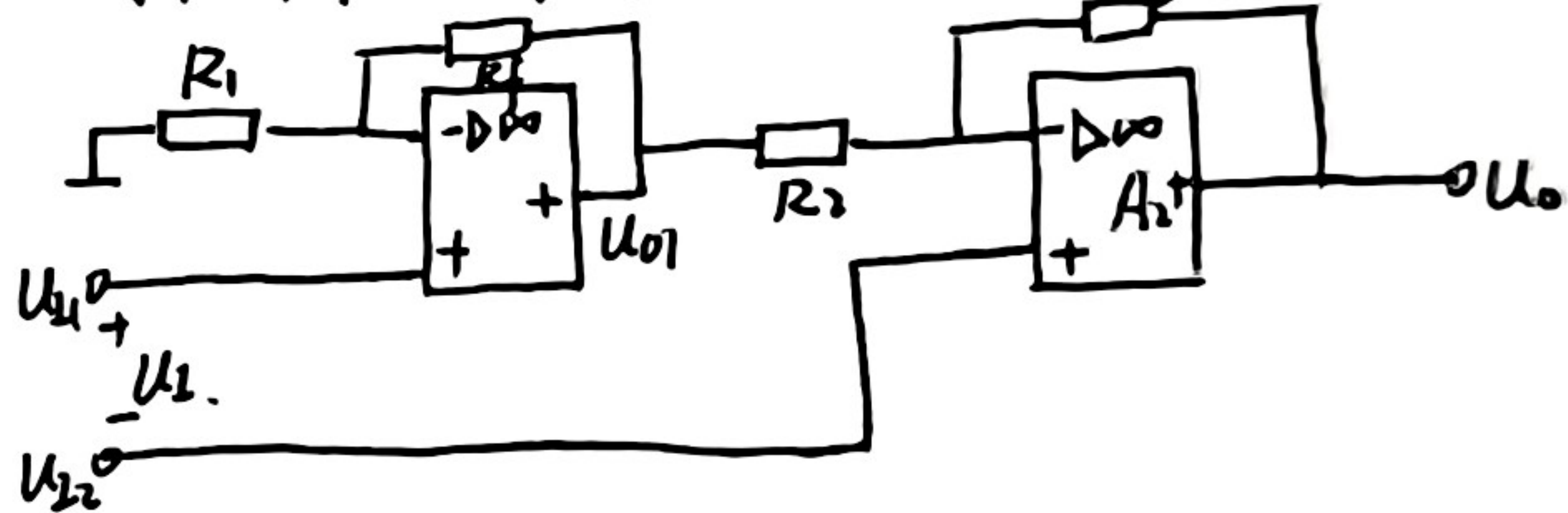
3. 应用

① 高输入阻抗

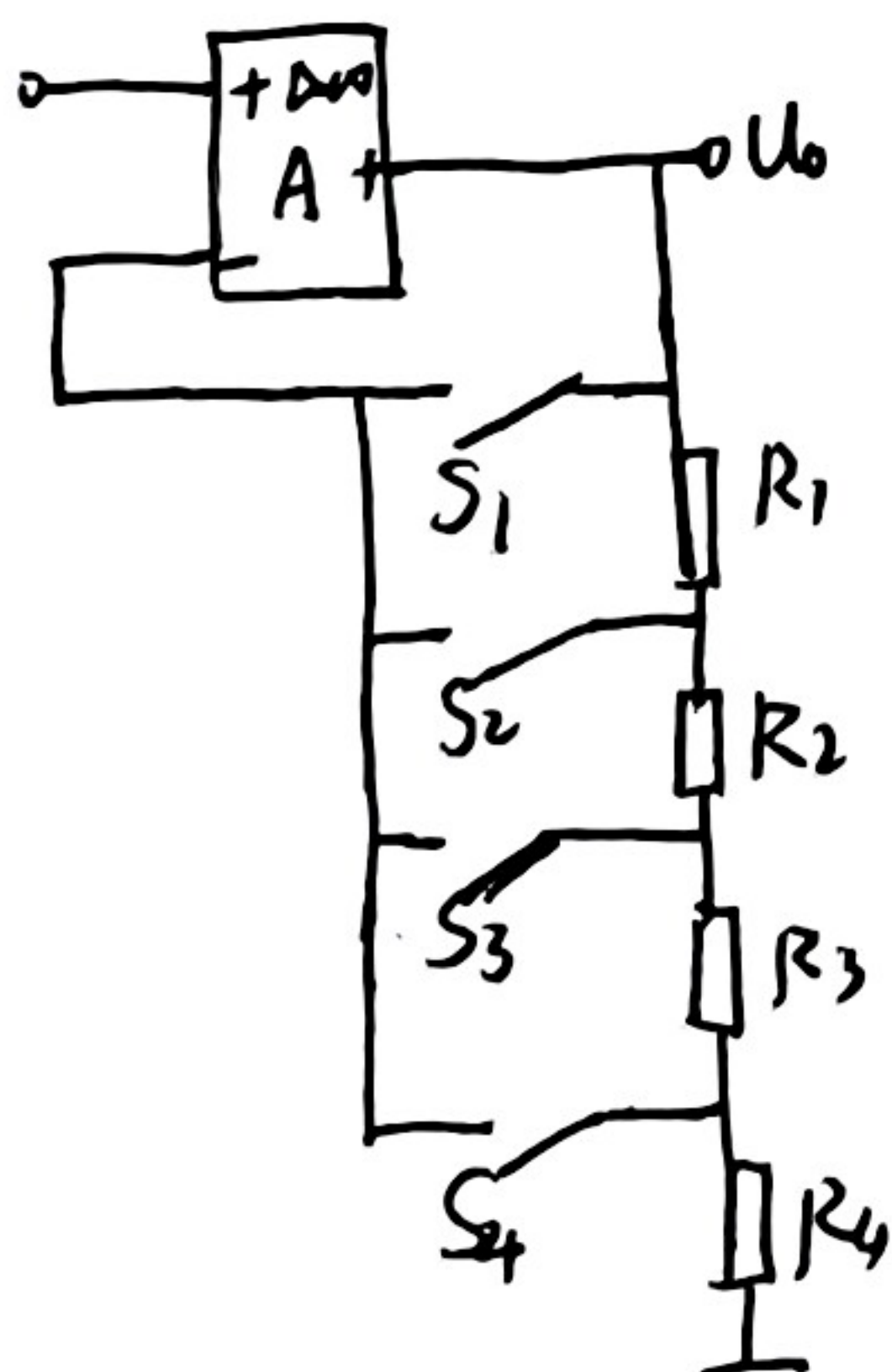
输入电阻扩展电路



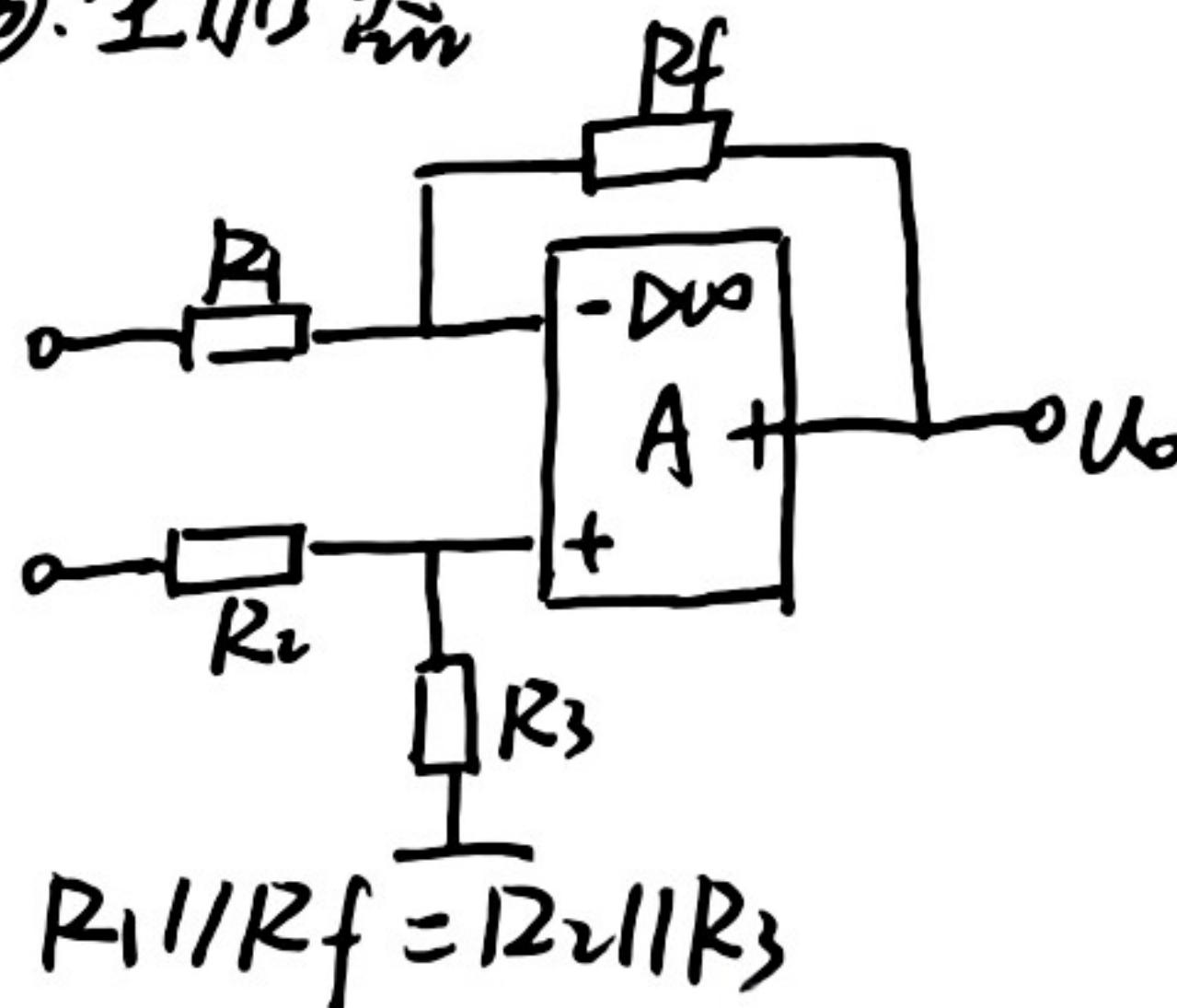
同相串联差分式高输入阻抗放大器



② 可编程增益放大器(同相)

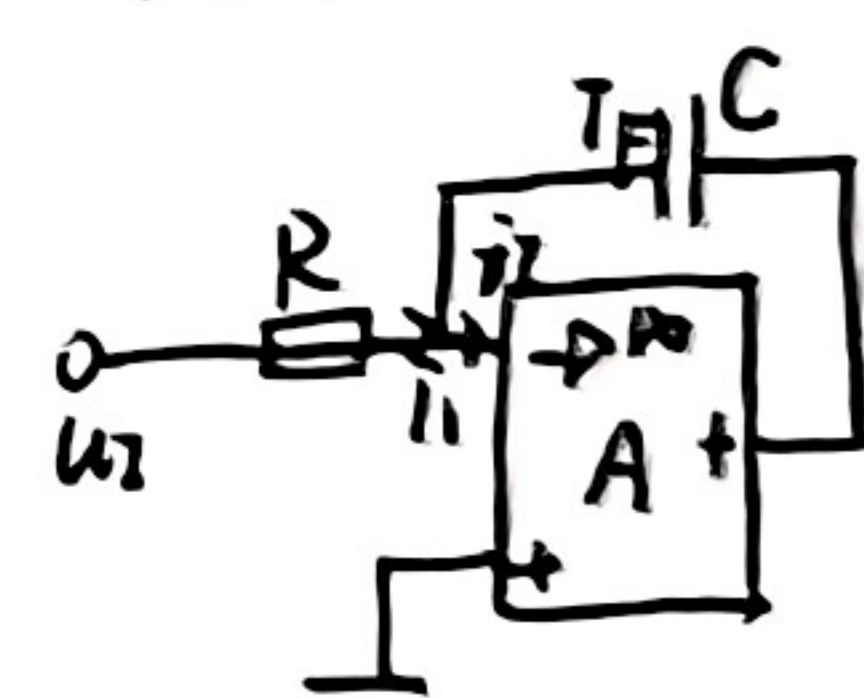


③ 全加器



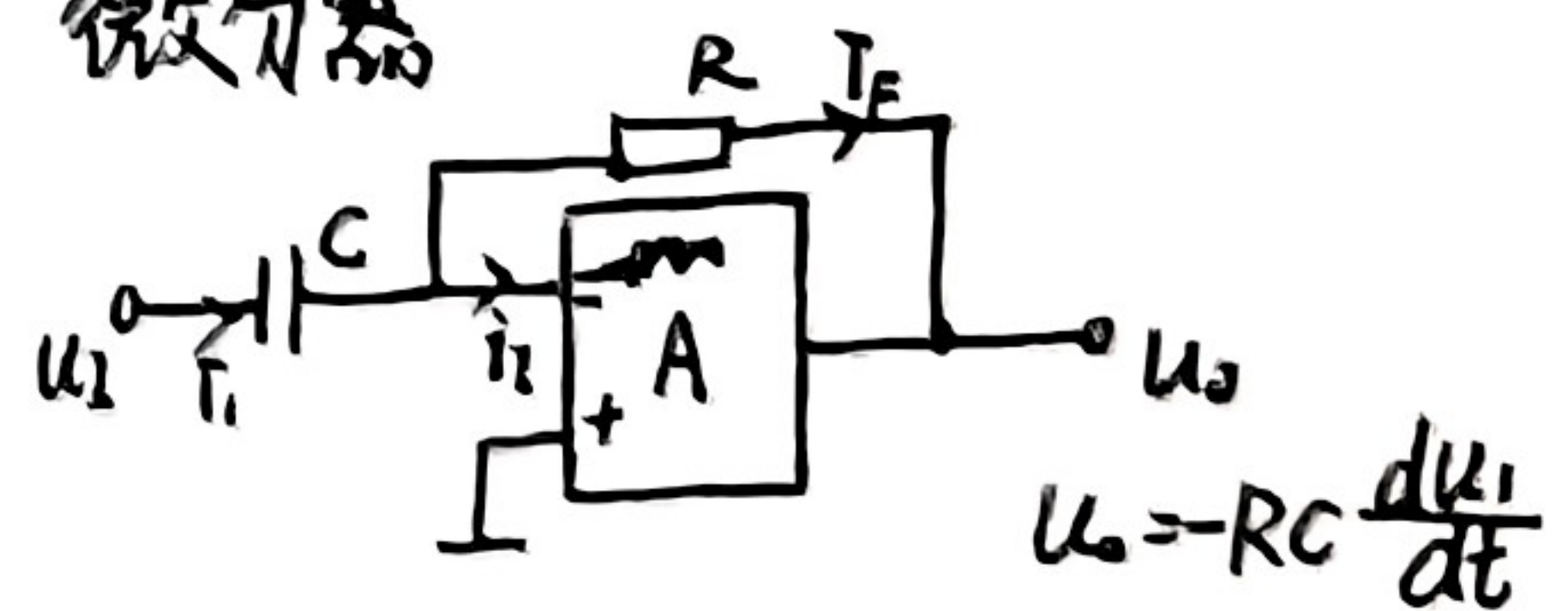
$$R_1 // R_f = R_2 // R_3$$

④ 积分器



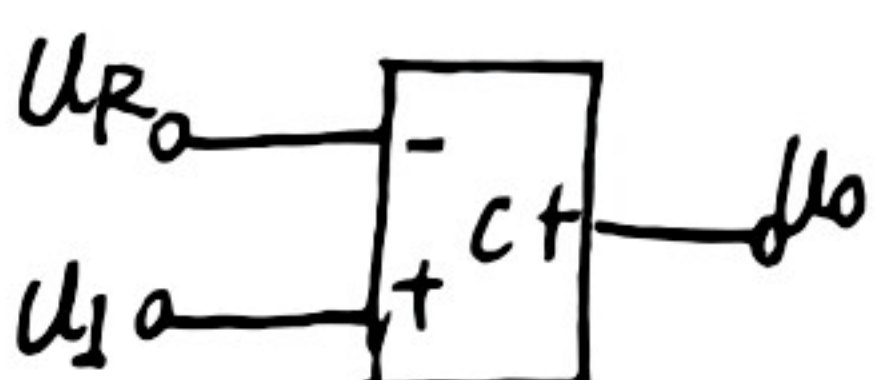
$$\begin{aligned} i_i &= 0 \\ U_- &= U_+ = 0 \\ i_i &= \frac{U_i}{R} = i_F \\ i_F &= -C \frac{dU_o}{dt} \\ U_o &= -\frac{1}{RC} \int U_i dt \end{aligned}$$

微分器



$$U_o = -RC \frac{dU_i}{dt}$$

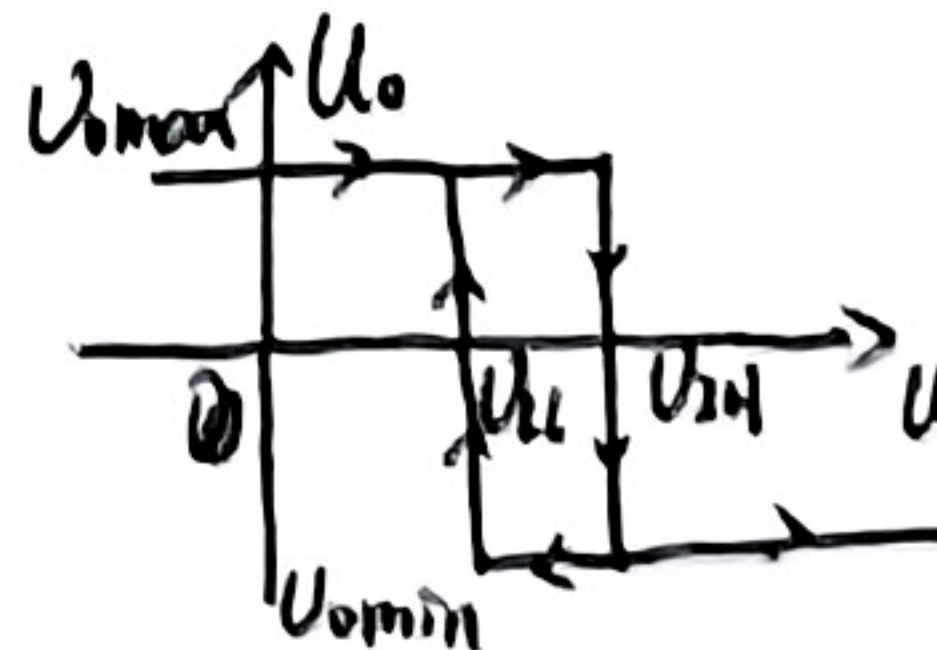
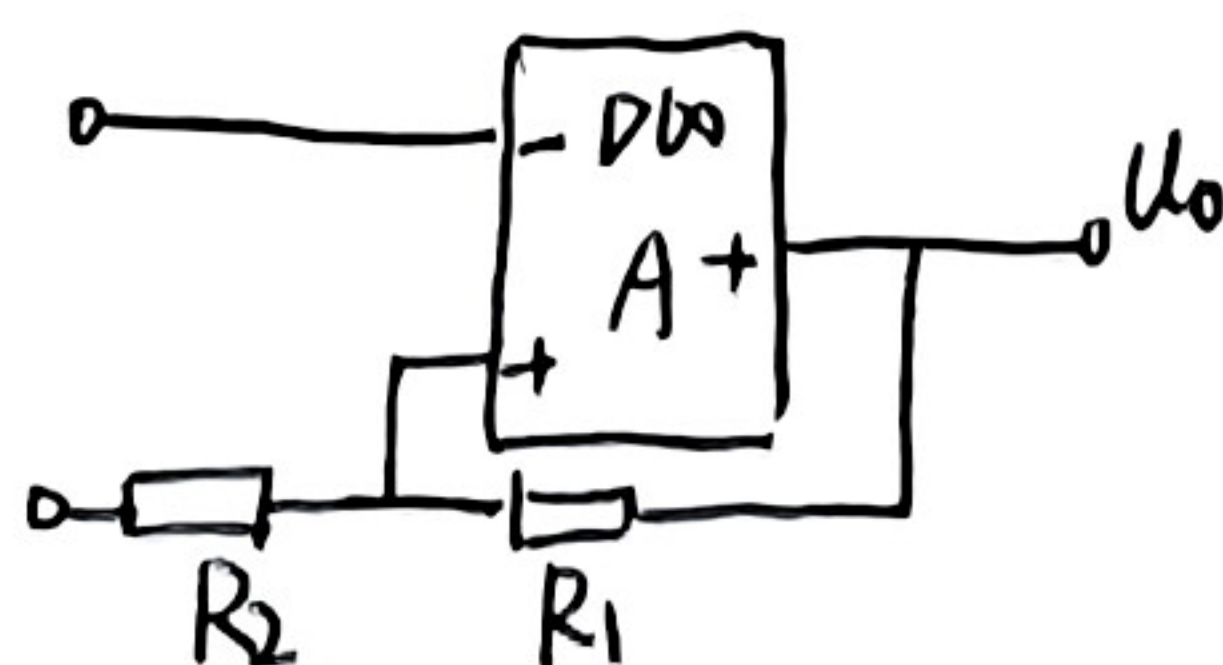
⑤ 电压比较器



$$U_i > U_R \rightarrow U_o = U_{omax}$$

$$U_i < U_R \rightarrow U_o = U_{omin}$$

⑥ 迟滞比较器



$$U_{IH} = \frac{R_1}{R_1 + R_2} U_R + \frac{R_1}{R_1 + R_2} U_{omax}$$

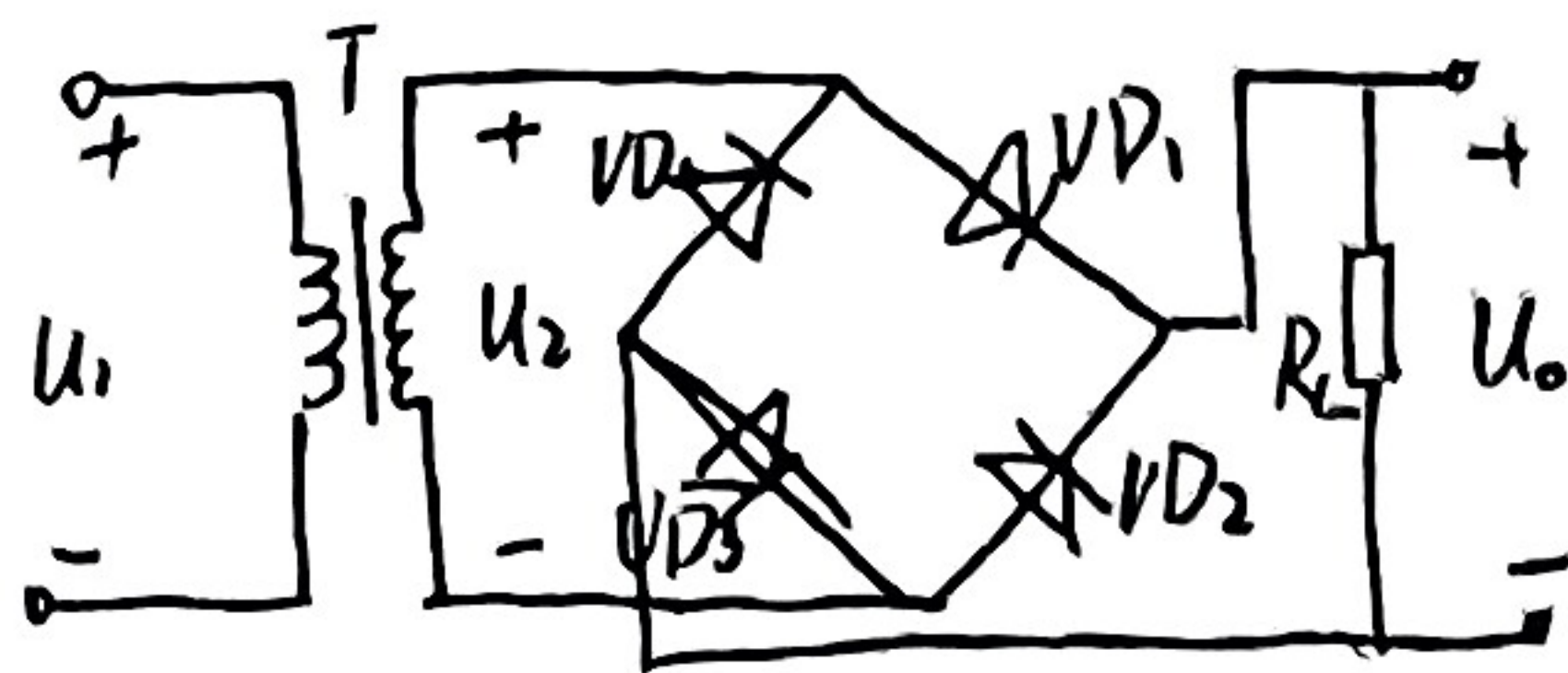
$$U_{IL} = \frac{R_1}{R_1 + R_2} U_R + \frac{R_1}{R_1 + R_2} U_{omin}$$

$$\Delta U_I = U_{IH} - U_{IL} = \frac{R_1}{R_1 + R_2} (U_{omax} - U_{omin})$$

第1章 直流稳压电源

220V 变压器 $\rightarrow$ 整流适当峰峰值 $\rightarrow$ 整流 负半转正 滤波 稳压 直流

1. 桥式整流电路



$$U_o = 0.9 U_2$$

二极管: ① 最大反向电压 $U_{DRmax} = U_{2m}$

② 平均电流 $I_D = \frac{1}{2} I_L = \frac{0.45 U_{2m}}{R_L}$

2. 平滑滤波器

输出端加一电容 $U_o \approx 1.2 U_2$